

Estimation of Linear Functionals in Multilayer Panels under Staggered Adoption

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September 10, 2026

Abstract

We study the estimation of bilinear forms from noisy, partially observed multilayer data. The signal follows a Tucker2 model, with shared unit and time factors across tensor layers and slice-specific cores. The missingness pattern is structured and motivated by staggered adoption designs, which are common in causal inference and related applications. We first analyze the four-block missingness pattern, the basic building block for general staggered adoption, and propose a spectral algorithm that pools information across layers and targets the functional directly. We prove a non-asymptotic mean squared error bound that exhibits a phase transition in the number of layers, showing when pooling improves estimation, and match it with a local minimax lower bound up to constants when ranks and logarithmic factors are treated as constant-order quantities. We then extend the construction to general staggered adoption designs via an anchored four-block reduction, and derive analogous theoretical guarantees. Finally, we validate our theoretical findings using synthetic and real-world data on Castle Doctrine laws and COVID-19 policies.

Keywords: Bilinear forms; Common-subspace models; Low-rank tensor models; Missing not at random; Spectral methods.

1 Introduction

In modern statistics, data often depart from the idealized i.i.d. setting and, in many applications, take the form of multiple incomplete panels. For instance, we may observe different responses for each individual over time, with data recorded at some time points but not others. Rather than recovering all missing values, practitioners may instead aim to estimate a simple functional of a target layer, such as a *bilinear form*, using its observed entries and possibly incorporating information from additional layers. This naturally raises two questions: how should information from auxiliary panels be used to estimate the target functional, and when does adding further panels cease to improve accuracy?

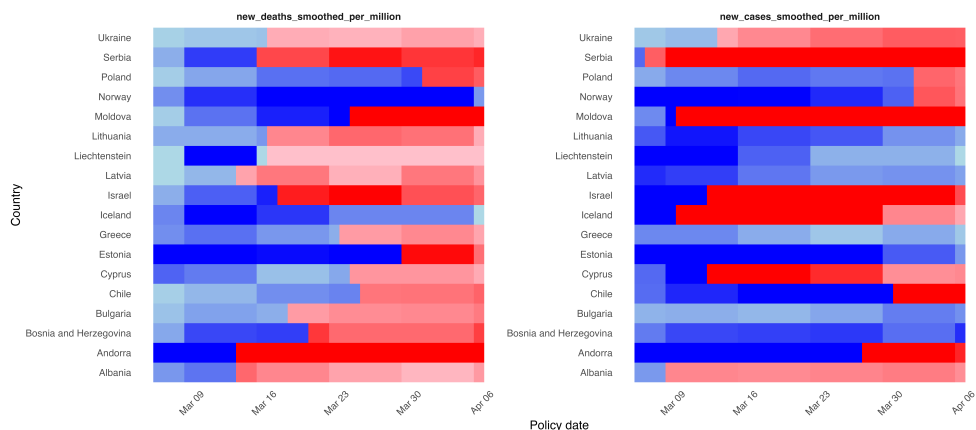


Figure 1: Outcome tensor from the merged OxCGRT–OWID panel. Rows are countries, columns are dates from March 5 to April 5, 2020, and panels are outcome–policy pairs. Outcomes are measured at a 28-day delay, so date t shows the value at $t + 28$. Blue denotes policy-off entries, red policy-on entries, and darker shades larger outcome values.

We study these questions under a *Tucker2 model* (Kolda and Bader, 2009, Section 4) in which the panels share unit and time factors but have layer-specific cores, and the missingness follows a structured missing-not-at-random (MNAR) pattern. In particular, we focus on *staggered adoption* designs (Athey and Imbens, 2022), which often arise in causal inference and related applications, where units are observed over multiple periods and, once treated, remain treated thereafter. From the perspective of untreated potential outcomes, observations are therefore available for all units before adoption and missing for treated units after adoption. The simplest instance of this structure is the four-block setting illustrated in Fig. 2 and analyzed in Section 2. A concrete example of a multilayer panel

with a more general staggered missingness is provided by the COVID-19 policy application studied in Section 5.3. Fig. 1 shows the two panels corresponding to stay-at-home policies and contact tracing, with red and blue indicating whether the policy is on or off, respectively, and color intensity representing the number of new deaths and new cases.

In applications of this kind, our contribution is a spectral method for estimating bilinear forms of a target layer under staggered missingness while borrowing information from related auxiliary layers. This class of functionals arises in many applications (e.g. Callaway and Sant’Anna, 2021, Section 3), and includes averages, contrasts, individual components, and temporal trends. Our theory shows when such pooling helps and when its benefits saturate. More precisely, since additional layers improve estimation of the shared row and column spaces but do not reduce uncertainty in the layer-specific core of the target panel, the estimation error initially decreases as more auxiliary information becomes available, but eventually reaches a target-specific floor. This *phase transition* is illustrated in Fig. 3.

Related work. An alternative way to estimate a functional of a partially observed panel is to impute the missing entries and use the completed layer to construct plug-in estimators of these quantities. Low-rank matrix completion provides a natural framework for this task. Although such methodologies were developed mainly for missing completely at random observation patterns (e.g., Candès and Recht, 2009; Keshavan et al., 2010; Negahban and Wainwright, 2012; Koltchinskii et al., 2011; Klopp, 2014; Chi et al., 2019), they can be repurposed in causal panels by treating unobserved untreated outcomes as missing entries of an approximately low-rank matrix. Athey et al. (2021) formalized this connection by relating low-rank matrix completion to unconfoundedness-based methods (e.g., Imbens and Rubin, 2015) and synthetic-control methods (e.g., Abadie, 2021), and proposed estimating the missing counterfactual entries through nuclear-norm penalized least squares.

A subsequent literature has developed matrix-completion methods for MNAR settings with structured missingness. Choi and Yuan (2026) study staggered adoption designs

and extend the nuclear-norm approach of [Athey et al. \(2021\)](#) by partitioning the missing entries into groups and applying convex relaxation within each group. They prove ℓ_∞ estimation error bounds that improve on the Frobenius-norm bound obtained in [Athey et al. \(2021\)](#). Similarly, [Agarwal et al. \(2026\)](#) study a related problem with row and column side information, providing Frobenius-norm guarantees for an estimator based on sieve projection and nuclear-norm penalization. Alongside optimization-based approaches, [Yan and Wainwright \(2024\)](#) consider panels with staggered adoption and propose a spectral algorithm based on singular value decomposition, with non-asymptotic entrywise guarantees and Gaussian approximations. Related factor-based approaches, including [Bai and Ng \(2021\)](#) and [Cahan et al. \(2023\)](#), exploit tall and wide observed blocks to estimate latent factors and impute missing entries. Finally, [Agarwal et al. \(2023\)](#) developed a completion method based on synthetic nearest neighbors for a broad class of MNAR patterns, with ℓ_∞ error bounds and asymptotic normality.

When the data are naturally organized as multiple related panels, as in our COVID-19 application, this matrix-completion perspective extends to tensors, allowing dependence across layers to be exploited. Recent work has begun to develop tensor-completion methods under structured missingness, but the theory remains less developed than in the matrix case. [Auerbach et al. \(2022\)](#) arrange multivariate longitudinal outcomes as a unit-by-time-by-outcome tensor and use nuclear-norm penalization to impute the missing entries and study COVID-19 mandates. [Agarwal et al. \(2025\)](#) extend synthetic-control ideas to multiple treatments using a low-rank tensor factor model. [Mandal and Parkes \(2019\)](#) and [Gao et al. \(2025\)](#) consider tensor formulations for longitudinal causal problems, where treatment histories are stacked along an additional tensor mode. In particular, [Gao et al. \(2025\)](#) estimate the latent tensor using an inverse-probability-weighted low-rank Tucker formulation, implemented by projected gradient descent. Their main guarantee is a non-asymptotic Frobenius-norm bound for tensor recovery (Theorem 1). In Remark 2, they relate their

framework to [Athey et al. \(2021\)](#), noting that a special case reduces to a staggered-adoption panel setting with two potential-outcome matrices.

The primary target in much of this literature, however, remains recovery of the missing matrix or tensor, either as a whole or entry by entry. The problem of estimating general bilinear forms is briefly mentioned by [Xia et al. \(2024, Appendix C\)](#), but no theoretical guarantees are provided for this target. Instead, existing error bounds are typically stated for full recovery, for instance in Frobenius norm or entrywise ℓ_∞ norm, even when the ultimate object of interest is only a particular functional of the missing entries. Compared with these plug-in approaches, our method estimates the requested bilinear form without first materializing the entire missing block, thereby reducing computational cost and allowing a functional-specific analysis.

Moreover, pooling information across panel slices can improve statistical estimation. In the COVID-19 application in [Section 5.3](#), our method is more accurate than approaches using only the target layer. This is likely due to the limited information available in the first layer, which contains just two fully observed rows ([Fig. 1](#)), making estimation particularly challenging and highlighting the importance of exploiting shared structure across slices.

Notation. For $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and index sequences $I^{(t)} = \{i_1^{(t)}, \dots, i_{|I^{(t)}|}^{(t)}\} \subseteq [n_t]$, $t \in \{1, 2, 3\}$, let $\mathcal{X}_{I^{(1)}, I^{(2)}, I^{(3)}} \in \mathbb{R}^{|I^{(1)}| \times |I^{(2)}| \times |I^{(3)}|}$ denote the subtensor with $(\mathcal{X}_{I^{(1)}, I^{(2)}, I^{(3)}})_{k_1, k_2, k_3} = \mathcal{X}_{i_{k_1}^{(1)}, i_{k_2}^{(2)}, i_{k_3}^{(3)}}$. The symbol $\bullet$ denotes the full index set in the corresponding mode, e.g., $\mathcal{X}_{\bullet, I^{(2)}, \bullet} := \mathcal{X}_{[n_1], I^{(2)}, [n_3]}$; singleton sets are written without braces, and the same convention applies to matrices and vectors. We write $\mathbf{0}_d$, $\mathbf{1}_d$, $\mathbf{I}_d$, and $\mathbf{e}_j^{(d)}$ for the zero vector, all-one vector, identity matrix, and j th canonical basis vector in $\mathbb{R}^d$, omitting d when clear, and set $\mathbf{O}_{d_1 \times d_2} := \mathbf{0}_{d_1} \mathbf{0}_{d_2}^\top$ and $\mathbf{1}_{d_1 \times d_2} := \mathbf{1}_{d_1} \mathbf{1}_{d_2}^\top$. For symmetric $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{d \times d}$, $\mathbf{A} \succeq 0$ means that $\mathbf{A}$ is positive semidefinite, and $\mathbf{A} \succeq \mathbf{B}$ means $\mathbf{A} - \mathbf{B} \succeq 0$. We write $\text{tr}(\mathbf{A})$ for the trace and $\text{diag}(\mathbf{v})$ for the diagonal matrix with diagonal $\mathbf{v}$. The minimum, maximum, and j th largest eigenvalues of a symmetric matrix $\mathbf{A}$ are denoted by $\lambda_{\min}(\mathbf{A})$, $\lambda_{\max}(\mathbf{A})$, and $\lambda_j(\mathbf{A})$; for a

general matrix $\mathbf{B}$, the analogous singular values are $\sigma_{\min}(\mathbf{B})$, $\sigma_{\max}(\mathbf{B})$, and $\sigma_j(\mathbf{B})$. We define $P_{\Omega}(\mathbf{M}) := \Omega \odot \mathbf{M}$, where $\odot$ denotes the Hadamard product. Moreover, $\text{SVD}_r(\mathbf{A}) = (\mathbf{U}, \mathbf{\Sigma}, \mathbf{V})$ denotes the rank- r truncated SVD, where $\mathbf{U} \in \mathbb{R}^{n_1 \times r}$ and $\mathbf{V} \in \mathbb{R}^{n_2 \times r}$ contain the top- r left and right singular vectors and $\mathbf{\Sigma} \in \mathbb{R}^{r \times r}$ contains the corresponding singular values. If $\mathbf{A} = \mathbf{U} \text{diag}(\sigma_1, \dots, \sigma_r) \mathbf{V}^\top$ with $\sigma_i > 0$, then $\mathbf{A}^\dagger := \mathbf{V} \text{diag}(\sigma_1^{-1}, \dots, \sigma_r^{-1}) \mathbf{U}^\top$ is its Moore–Penrose pseudoinverse. We use $\|\cdot\|_p$ for vector ℓ_p -norms, $\|\cdot\|_{\text{op}}$ and $\|\cdot\|_F$ for matrix spectral and Frobenius norms, and $\langle \cdot, \cdot \rangle$ for the Euclidean inner product. Also, $\mathcal{S}^{d-1} := \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 = 1\}$ is the unit sphere, and $\mathcal{O}(x)$ denotes a quantity whose absolute value is bounded by Cx for some constant $C > 0$.

2 Bilinear form estimation with 4-block missingness

2.1 Statistical setting and main result

In this section and the next, we study bilinear-form estimation in the four-block setting of Fig. 2, the simplest nontrivial building block for the staggered adoption design in Section 4. For fixed dimensions $N, T, K \geq 1$, we define the deterministic missingness-pattern tensor $\Omega \in \{0, 1\}^{N \times T \times K}$ where, for all $j \in [K]$,

$$\Omega_{\bullet, \bullet, j} = \begin{pmatrix} \mathbf{1}_{N_{1j} \times T_{1j}} & \mathbf{1}_{N_{1j} \times T_{2j}} \\ \mathbf{1}_{N_{2j} \times T_{1j}} & \mathbf{0}_{N_{2j} \times T_{2j}} \end{pmatrix} \in \mathbb{R}^{N \times T}, \quad (1)$$

with $N = N_{1j} + N_{2j}$ and $T = T_{1j} + T_{2j}$. The tensor Ω will be fixed throughout this section.

We observe $\mathcal{Y} \in \mathbb{R}^{N \times T \times K}$ with

$$\mathcal{Y}_{\bullet, \bullet, j} := P_{\Omega_{\bullet, \bullet, j}}(\mathcal{M}_{\bullet, \bullet, j} + \mathcal{E}_{\bullet, \bullet, j}) = \begin{pmatrix} \mathcal{M}_{\bullet, \bullet, j}^{(a)} + \mathcal{E}_{\bullet, \bullet, j}^{(a)} & \mathcal{M}_{\bullet, \bullet, j}^{(b)} + \mathcal{E}_{\bullet, \bullet, j}^{(b)} \\ \mathcal{M}_{\bullet, \bullet, j}^{(c)} + \mathcal{E}_{\bullet, \bullet, j}^{(c)} & \text{NA} \end{pmatrix}, \quad (2)$$

where a, b, c refer to the observed blocks in (1), and $\mathcal{E} \in \mathbb{R}^{N \times T \times K}$ is such that $\mathcal{E}_{i,t,j} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ for all $(i, t, j) \in [N] \times [T] \times [K]$. Fig. 2 illustrates the observed tensor $\mathcal{Y}$ for $K = 3$. The bottom-right block need not be missing in every layer. It suffices that each row is either fully observed or missing from some slice-specific time onward. We nevertheless use the four-block design in (1) to simplify the exposition.

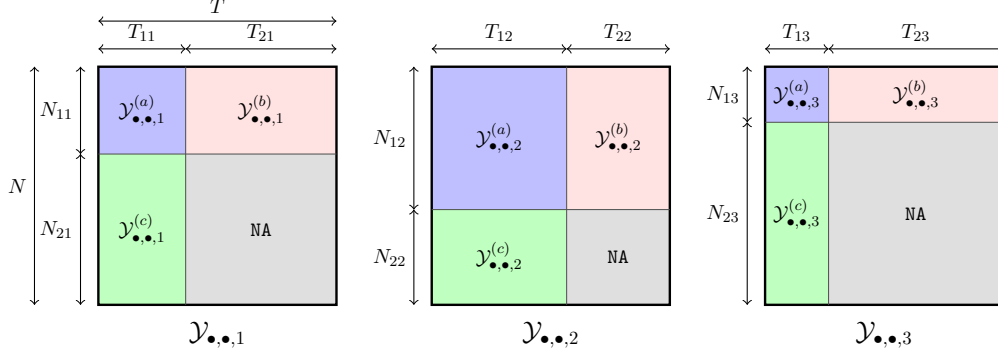


Figure 2: Layer-specific four-block structure for $K = 3$. In slice $\mathcal{Y}_{\bullet,\bullet,j}$, N_{1j} rows are fully observed, while N_{2j} rows are observed only in the first T_{1j} columns. Here, N and T denote the total numbers of rows and columns; a, b, c are observed blocks and d is missing.

As for the signal tensor $\mathcal{M} \in \mathbb{R}^{N \times T \times K}$, for $r \geq 1$ with $r \leq \min(N, T)$, we assume that $\mathcal{M}$ admits a Tucker2 decomposition of rank (r, r, K) , meaning that there exist factor matrices $\mathbf{U} \in \mathbb{R}^{N \times r}$ and $\mathbf{V} \in \mathbb{R}^{T \times r}$ satisfying $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_r$, and a core tensor $\mathcal{C} \in \mathbb{R}^{r \times r \times K}$ such that $\mathcal{M} = \mathcal{C} \times_1 \mathbf{U} \times_2 \mathbf{V} \times_3 \mathbf{I}_K$. Equivalently, each layer-specific signal matrix has rank at most r and admits the factorization $\mathcal{M}_{\bullet,\bullet,j} = \mathbf{U} \mathcal{C}_{\bullet,\bullet,j} \mathbf{V}^\top \in \mathbb{R}^{N \times T}$; a proof of the equivalence between these two formulations is given in Section S5.3 in the Supplement. This condition permits heterogeneity across slices while borrowing strength through shared latent row and column spaces. Similar modeling assumptions, often referred to as *common-subspace models*, have been studied in statistical settings under complete observation (Agterberg, 2026; Arroyo et al., 2021) and are motivated by biological applications, including neuroscience and RNA sequencing (Semedo et al., 2019; Ma and Ma, 2026).

It will also be convenient to partition $\mathbf{U} = (\mathbf{U}_{1j}; \mathbf{U}_{2j})$ and $\mathbf{V} = (\mathbf{V}_{1j}; \mathbf{V}_{2j})$ according to the (N_{1j}, N_{2j}) and (T_{1j}, T_{2j}) splits induced by slice j , where the semicolon denotes vertical stacking. Formally, for each $j \in [K]$ we define $\mathbf{U}_{1j} = \mathbf{U}_{[N_{1j}], \bullet} \in \mathbb{R}^{N_{1j} \times r}$ and $\mathbf{U}_{2j} = \mathbf{U}_{\{N_{1j}+1, \dots, N\}, \bullet} \in \mathbb{R}^{N_{2j} \times r}$; we define $\mathbf{V}_{1j}$ and $\mathbf{V}_{2j}$ analogously. Under this notation,

the unobserved bottom-right block satisfies $\mathcal{M}_{\bullet,\bullet,j}^{(d)} = \mathbf{U}_{2j} \mathcal{C}_{\bullet,\bullet,j} \mathbf{V}_{2j}^\top$.

Our focus is on estimating general bilinear forms of the missing d -blocks. Formally, fix $k \in [K]$ and unit vectors $\mathbf{x} \in \mathcal{S}^{N_{2k}-1}$, $\mathbf{y} \in \mathcal{S}^{T_{2k}-1}$, and define

$$\mu_{\mathbf{x}\mathbf{y}}^{(k)} := \mathbf{x}^\top \mathcal{M}_{\bullet,\bullet,k}^{(d)} \mathbf{y}. \quad (3)$$

This target class includes both average and individual counterfactual components, depending on whether $\mathbf{x}$, $\mathbf{y}$ are constant or canonical basis vectors. Our proposed estimator, given in Algorithm 1 in Section 2.2, predicts the missing d -block from the b -block by regressing the c -block on the a -block, exploiting the shared-subspace structure of the Tucker2 model. Its mean squared error is illustrated in Fig. 3.

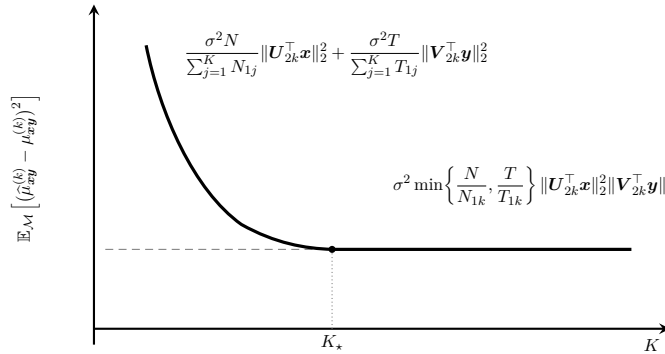


Figure 3: Pictorial illustration of the MSE of the estimator in Algorithm 1 based on Theorem 1.

In particular, under certain assumptions discussed in Section 3.2, Theorem 1 in Section 3.1 gives a precise upper bound on the estimation error that reveals the effect of pooling across layers. Treating ranks and logarithmic factors as constant-order quantities, the mean squared error is bounded up to constants by $\sigma^2 N \|\mathbf{U}_{2k}^\top \mathbf{x}\|_2^2 / \sum_{j=1}^K N_{1j} + \sigma^2 T \|\mathbf{V}_{2k}^\top \mathbf{y}\|_2^2 / \sum_{j=1}^K T_{1j}$ in the small- K regime, and by $\sigma^2 \min\left(\frac{N}{N_{1k}}, \frac{T}{T_{1k}}\right) \|\mathbf{U}_{2k}^\top \mathbf{x}\|_2^2 \|\mathbf{V}_{2k}^\top \mathbf{y}\|_2^2$ in the large- K regime. The first two terms arise from estimating the shared factors $\mathbf{U}$ and $\mathbf{V}$, and therefore decrease with K . The third is an irreducible layer-specific error that does not decrease with K , reflecting the difficulty of estimating $\mathcal{C}_{\bullet,\bullet,k}$. This phase transition in the rate as a function of K is illustrated by the simulation study in Fig. 5, and complemented by a matching local minimax lower bound in Theorem 2 in Section 3.3.

2.2 Proposed methodology for four-block missingness

We now introduce Algorithm 1 to estimate quantities of the form (3). To simplify the presentation of the method and its analysis, we introduce some notation, summarized in Table 1. We recall that parentheses with a space denote horizontal concatenation; semicolons denote vertical concatenation.

Notation	Definition	Notation	Definition
$\mathbf{M}_{\text{up}}^{(j)}$	$(\mathcal{M}_{\bullet,\bullet,j}^{(a)} \mathcal{M}_{\bullet,\bullet,j}^{(b)}) \in \mathbb{R}^{N_{1j} \times T}$	$\mathbf{M}_{\text{left}}^{(j)}$	$(\mathcal{M}_{\bullet,\bullet,j}^{(a)} ; \mathcal{M}_{\bullet,\bullet,j}^{(c)}) \in \mathbb{R}^{N \times T_{1j}}$
$\mathbf{E}_{\text{up}}^{(j)}$	$(\mathcal{E}_{\bullet,\bullet,j}^{(a)} \mathcal{E}_{\bullet,\bullet,j}^{(b)}) \in \mathbb{R}^{N_{1j} \times T}$	$\mathbf{E}_{\text{left}}^{(j)}$	$(\mathcal{E}_{\bullet,\bullet,j}^{(a)} ; \mathcal{E}_{\bullet,\bullet,j}^{(c)}) \in \mathbb{R}^{N \times T_{1j}}$
$\mathbf{Y}_{\text{up}}^{(j)}$	$(\mathcal{Y}_{\bullet,\bullet,j}^{(a)} \mathcal{Y}_{\bullet,\bullet,j}^{(b)}) \in \mathbb{R}^{N_{1j} \times T}$	$\mathbf{Y}_{\text{left}}^{(j)}$	$(\mathcal{Y}_{\bullet,\bullet,j}^{(a)} ; \mathcal{Y}_{\bullet,\bullet,j}^{(c)}) \in \mathbb{R}^{N \times T_{1j}}$
$\mathbf{M}_{\text{up}}^{\text{p}}$	$(\mathbf{M}_{\text{up}}^{(1)} ; \dots ; \mathbf{M}_{\text{up}}^{(K)}) \in \mathbb{R}^{N_{1,p} \times T}$	$\mathbf{M}_{\text{left}}^{\text{p}}$	$(\mathbf{M}_{\text{left}}^{(1)} \dots \mathbf{M}_{\text{left}}^{(K)}) \in \mathbb{R}^{N \times T_{1,p}}$
$\mathbf{E}_{\text{up}}^{\text{p}}$	$(\mathbf{E}_{\text{up}}^{(1)} ; \dots ; \mathbf{E}_{\text{up}}^{(K)}) \in \mathbb{R}^{N_{1,p} \times T}$	$\mathbf{E}_{\text{left}}^{\text{p}}$	$(\mathbf{E}_{\text{left}}^{(1)} \dots \mathbf{E}_{\text{left}}^{(K)}) \in \mathbb{R}^{N \times T_{1,p}}$
$\mathbf{Y}_{\text{up}}^{\text{p}}$	$(\mathbf{Y}_{\text{up}}^{(1)} ; \dots ; \mathbf{Y}_{\text{up}}^{(K)}) \in \mathbb{R}^{N_{1,p} \times T}$	$\mathbf{Y}_{\text{left}}^{\text{p}}$	$(\mathbf{Y}_{\text{left}}^{(1)} \dots \mathbf{Y}_{\text{left}}^{(K)}) \in \mathbb{R}^{N \times T_{1,p}}$
$\mathbf{W}_{\text{up}}$	$(\mathbf{U}_{11} \mathbf{C}_{\bullet,\bullet,1} ; \dots ; \mathbf{U}_{1K} \mathbf{C}_{\bullet,\bullet,K}) \in \mathbb{R}^{N_{1,p} \times r}$	$\mathbf{W}_{\text{left}}$	$(\mathbf{V}_{11} \mathbf{C}_{\bullet,\bullet,1}^{\top} ; \dots ; \mathbf{V}_{1K} \mathbf{C}_{\bullet,\bullet,K}^{\top}) \in \mathbb{R}^{T_{1,p} \times r}$
$(\mathbf{U}_{\text{up}}, \mathbf{\Sigma}_{\text{up}}, \mathbf{V}_{\text{up}})$	$\text{SVD}_r(\mathbf{M}_{\text{up}}^{\text{p}})$	$(\mathbf{U}_{\text{left}}, \mathbf{\Sigma}_{\text{left}}, \mathbf{V}_{\text{left}})$	$\text{SVD}_r(\mathbf{M}_{\text{left}}^{\text{p}})$
$(\widehat{\mathbf{U}}_{\text{up}}, \widehat{\mathbf{\Sigma}}_{\text{up}}, \widehat{\mathbf{V}}_{\text{up}})$	$\text{SVD}_r(\mathbf{Y}_{\text{up}}^{\text{p}})$	$(\widehat{\mathbf{U}}_{\text{left}}, \widehat{\mathbf{\Sigma}}_{\text{left}}, \widehat{\mathbf{V}}_{\text{left}})$	$\text{SVD}_r(\mathbf{Y}_{\text{left}}^{\text{p}})$
$N_{1,p}$	$\sum_{j=1}^K N_{1j}$	$T_{1,p}$	$\sum_{j=1}^K T_{1j}$
ρ_N	$N_{1,p}/N$	ρ_T	$T_{1,p}/T$
p_N	$\max\{N_{1,p}, T\}$	p_T	$\max\{N, T_{1,p}\}$
$\widehat{\mathbf{U}}_{1k}$	$(\widehat{\mathbf{U}}_{\text{left}})_{[N_{1k}], \bullet} \in \mathbb{R}^{N_{1k} \times r}$	$\widehat{\mathbf{U}}_{2k}$	$(\widehat{\mathbf{U}}_{\text{left}})_{\{N_{1k}+1, \dots, N\}, \bullet} \in \mathbb{R}^{N_{2k} \times r}$
$\widehat{\mathbf{U}}_{\text{up}}^{(k)}$	$(\widehat{\mathbf{U}}_{\text{up}})_{\{s_k+1, \dots, s_k+N_{1k}\}, \bullet} \in \mathbb{R}^{N_{1k} \times r}$	$\widehat{\mathbf{V}}_{2k}$	$(\widehat{\mathbf{V}}_{\text{up}})_{\{T_{1k}+1, \dots, T\}, \bullet} \in \mathbb{R}^{T_{2k} \times r}$
ζ_N	$\log(N_{1,p} + T)$	ζ_T	$\log(N + T_{1,p})$
s_k	$\sum_{j=1}^{k-1} N_{1j}$	ζ	$\max(\zeta_N, \zeta_T)$

Table 1: Notation used throughout the paper.

We now provide some intuition by considering the noiseless case. The key observation is that $\mathcal{M}_{\bullet,\bullet,k}^{(d)} = \mathbf{U}_{2k}(\mathbf{U}_{1k}^{\top} \mathbf{U}_{1k})^{-1} \mathbf{U}_{1k}^{\top} \mathcal{M}_{\bullet,\bullet,k}^{(b)}$, so the missing d -block can be recovered by mapping the observed b -block through the linear operator $\mathbf{U}_{2k}(\mathbf{U}_{1k}^{\top} \mathbf{U}_{1k})^{-1} \mathbf{U}_{1k}^{\top}$. For arbitrary unit vectors $\mathbf{x}$ and $\mathbf{y}$, this yields $\mu_{\mathbf{x}\mathbf{y}}^{(k)} = \mathbf{x}^{\top} \mathcal{M}_{\bullet,\bullet,k}^{(d)} \mathbf{y} = \langle \mathbf{U}_{2k}^{\top} \mathbf{x}, (\mathbf{U}_{1k}^{\top} \mathbf{U}_{1k})^{-1} \mathbf{U}_{1k}^{\top} \mathcal{M}_{\bullet,\bullet,k}^{(b)} \mathbf{y} \rangle$. This motivates a two-step procedure that exploits the shared-subspace assumption by first using the pooled left matrix $\mathbf{M}_{\text{left}}^{\text{p}} = \mathbf{U} \mathbf{W}_{\text{left}}^{\top}$ to estimate the left singular subspace and associated least-squares map, and then using the pooled upper matrix $\mathbf{M}_{\text{up}}^{\text{p}} = \mathbf{W}_{\text{up}} \mathbf{V}^{\top}$ to obtain a low-rank denoised estimate of the b -block, but only through its action on $\mathbf{y}$.

More precisely, if $\mathbf{W}_{\text{left}}$ has rank r , we have $\text{SVD}_r(\mathbf{M}_{\text{left}}^{\text{p}}) = (\mathbf{U}_{\text{left}}, \mathbf{\Sigma}_{\text{left}}, \mathbf{V}_{\text{left}})$ with $\mathbf{U}_{\text{left}} = \mathbf{U} \mathbf{Q}_{\text{left}}$ for some orthogonal $\mathbf{Q}_{\text{left}} \in \mathbb{R}^{r \times r}$. Importantly, the operator $\mathbf{U}_{2k}(\mathbf{U}_{1k}^{\top} \mathbf{U}_{1k})^{-1} \mathbf{U}_{1k}^{\top}$ is rotationally invariant. On the other hand, if $\mathbf{W}_{\text{up}}$ has rank

r , for the pooled upper matrix we have $\text{SVD}_r(\mathbf{M}_{\text{up}}^{\text{p}}) = (\mathbf{U}_{\text{up}}, \mathbf{\Sigma}_{\text{up}}, \mathbf{V}_{\text{up}})$, where $\mathbf{V}_{\text{up}} = \mathbf{V}\mathbf{Q}_{\text{up}}$ for some orthogonal $\mathbf{Q}_{\text{up}} \in \mathbb{R}^{r \times r}$. Writing $\mathbf{U}_{\text{up}} = (\mathbf{U}_{\text{up}}^{(1)}; \dots; \mathbf{U}_{\text{up}}^{(K)})$ with $\mathbf{U}_{\text{up}}^{(j)} \in \mathbb{R}^{N_{1j} \times r}$, we have $\mathbf{U}_{\text{up}}\mathbf{\Sigma}_{\text{up}} = \mathbf{W}_{\text{up}}\mathbf{Q}_{\text{up}}$ and $\mathbf{U}_{\text{up}}^{(k)}\mathbf{\Sigma}_{\text{up}} = \mathbf{U}_{1k}\mathcal{C}_{\bullet,\bullet,k}\mathbf{Q}_{\text{up}}$, hence $\mathbf{U}_{\text{up}}^{(k)}\mathbf{\Sigma}_{\text{up}}(\mathbf{V}_{\text{up}})_{\{T_{1k}+1,\dots,T\},\bullet}^{\top} = \mathbf{U}_{1k}\mathcal{C}_{\bullet,\bullet,k}\mathbf{Q}_{\text{up}}\mathbf{Q}_{\text{up}}^{\top}\mathbf{V}_{2k}^{\top} = \mathbf{U}_{1k}\mathcal{C}_{\bullet,\bullet,k}\mathbf{V}_{2k}^{\top} = \mathcal{M}_{\bullet,\bullet,k}^{(b)}$. This also shows that this quantity is rotationally invariant, and further ensures that no cross-alignment between the two SVDs is required.

Algorithm 1 BILINEAR_TENSOR4BLOCK for estimating $\mu_{\mathbf{x}\mathbf{y}}^{(k)} = \mathbf{x}^{\top}\mathcal{M}_{\bullet,\bullet,k}^{(d)}\mathbf{y}$ in slice k of a tensor with four-block missingness

Require: integer $k \in [K]$, rank r , unit vectors $\mathbf{x} \in \mathcal{S}^{N_{2k}-1}$, $\mathbf{y} \in \mathcal{S}^{T_{2k}-1}$, data $\mathcal{Y}$, block sizes $\{(N_{1j}, N_{2j}, T_{1j}, T_{2j})\}_{j=1}^K$ satisfying $N = N_{1j} + N_{2j}$ and $T = T_{1j} + T_{2j}$ for all $j \in [K]$, parameter $\tau > 0$.

- 1: Form pooled left matrix $\mathbf{Y}_{\text{left}}^{\text{p}} \leftarrow (\mathbf{Y}_{\text{left}}^{(1)} \dots \mathbf{Y}_{\text{left}}^{(K)}) \in \mathbb{R}^{N \times T_{1,\text{p}}}$.
 - 2: Compute rank- r truncated singular value decomposition $(\hat{\mathbf{U}}_{\text{left}}, \hat{\mathbf{\Sigma}}_{\text{left}}, \hat{\mathbf{V}}_{\text{left}}) \leftarrow \text{SVD}_r(\mathbf{Y}_{\text{left}}^{\text{p}})$.
 - 3: Set $\hat{\mathbf{U}}_{1k} \leftarrow (\hat{\mathbf{U}}_{\text{left}})_{[N_{1k}],\bullet}$ and $\hat{\mathbf{U}}_{2k} \leftarrow (\hat{\mathbf{U}}_{\text{left}})_{\{N_{1k}+1,\dots,N\},\bullet}$.
 - 4: Compute $\hat{\mathbf{H}}_k \leftarrow \hat{\mathbf{U}}_{1k}^{\top}\hat{\mathbf{U}}_{2k} \in \mathbb{R}^{r \times r}$, take the eigendecomposition $\hat{\mathbf{H}}_k = \mathbf{Q} \text{diag}(\lambda_1, \dots, \lambda_r)\mathbf{Q}^{\top}$, and set
$$\hat{\mathbf{H}}_{k,\tau}^{\text{inv}} \leftarrow \mathbf{Q} \text{diag}\left(\left\{\frac{1}{\max[\lambda_i, \tau]}\right\}_{i=1}^r\right)\mathbf{Q}^{\top}.$$
 - 5: Compute $\hat{\mathbf{\alpha}}_x^{(k)} \leftarrow \hat{\mathbf{U}}_{2k}^{\top}\mathbf{x} \in \mathbb{R}^r$.
 - 6: Form pooled upper matrix $\mathbf{Y}_{\text{up}}^{\text{p}} \leftarrow (\mathbf{Y}_{\text{up}}^{(1)}; \dots; \mathbf{Y}_{\text{up}}^{(K)}) \in \mathbb{R}^{N_{1,\text{p}} \times T}$.
 - 7: Compute rank- r truncated singular value decomposition $(\hat{\mathbf{U}}_{\text{up}}, \hat{\mathbf{\Sigma}}_{\text{up}}, \hat{\mathbf{V}}_{\text{up}}) \leftarrow \text{SVD}_r(\mathbf{Y}_{\text{up}}^{\text{p}})$.
 - 8: Compute $s_k \leftarrow \sum_{j=1}^{k-1} N_{1j}$, and extract $\hat{\mathbf{U}}_{\text{up}}^{(k)} \leftarrow (\hat{\mathbf{U}}_{\text{up}})_{\{s_k+1,\dots,s_k+N_{1k}\},\bullet}$, $\hat{\mathbf{V}}_{2k} \leftarrow (\hat{\mathbf{V}}_{\text{up}})_{\{T_{1k}+1,\dots,T\},\bullet}$.
 - 9: Compute $\mathbf{T}_y \leftarrow \hat{\mathbf{V}}_{2k}^{\top}\mathbf{y} \in \mathbb{R}^r$, $\mathbf{W}_y \leftarrow \hat{\mathbf{\Sigma}}_{\text{up}}\mathbf{T}_y \in \mathbb{R}^r$, and $\mathbf{X}_y \leftarrow \hat{\mathbf{U}}_{\text{up}}^{(k)}\mathbf{W}_y \in \mathbb{R}^{N_{1k}}$.
 - 10: Compute $\hat{\beta}_y^{(k)} \leftarrow \hat{\mathbf{H}}_{k,\tau}^{\text{inv}}\hat{\mathbf{U}}_{1k}^{\top}\mathbf{X}_y \in \mathbb{R}^r$.
 - 11: **return** $\hat{\mu}_{\mathbf{x}\mathbf{y}}^{(k)} \leftarrow \langle \hat{\mathbf{\alpha}}_x^{(k)}, \hat{\beta}_y^{(k)} \rangle$.
-

In the noisy setting, Algorithm 1 follows the same principle, but applied to the observed tensor $\mathcal{Y}$ rather than the signal tensor $\mathcal{M}$. Our method computes the rank- r truncated SVDs of $\mathbf{Y}_{\text{left}}^{\text{p}}$ and $\mathbf{Y}_{\text{up}}^{\text{p}}$, formed by horizontally stacking the blue/green blocks and vertically stacking the blue/pink blocks in Fig. 2, respectively. This step, often referred to as *Stack-SVD*, exploits the singular subspaces shared across panels in order to improve subspace estimation; see Ma and Ma (2026); Baharav et al. (2025) for theoretical guarantees and comparisons with alternative aggregation schemes.

We also notice that our denoise-then-regress procedure targets the linear functional directly rather than reconstructing the entire missing d -block. Specifically, instead of reconstructing the whole missing block, in Steps 9-10 we compute only $\hat{\beta}_y^{(k)} \in \mathbb{R}^r$, and then

return its inner product with $\hat{\alpha}_x^{(k)} \in \mathbb{R}^r$ calculated in Step 5. We further elaborate on the computational complexity of our procedure in Appendix S3.

Finally, we use the clipped inverse $\hat{\mathbf{H}}_{k,\tau}^{\text{inv}}$ to estimate $(\mathbf{U}_{1k}^\top \mathbf{U}_{1k})^{-1}$ and safeguard against ill-conditioning. For sufficiently small τ , clipping is inactive with high probability, while stabilizing the inverse on the complementary event. In the numerical experiments of Section 5, the procedure appears stable for values of τ below 10^{-2} , with little need for careful tuning; see Fig. 6(a) for further details.

3 Theoretical analysis

3.1 Upper bound

We now present the theoretical analysis of Algorithm 1. Because the masks $\Omega_{\bullet,\bullet,j}$ may be adversarial and induce pathological MNAR patterns, assumptions are required to ensure that $\mu_{xy}^{(k)}$ is identifiable from the observed data. Building on prior work on MNAR matrix completion (Bai and Ng, 2021; Yan and Wainwright, 2024; Choi and Yuan, 2026; Agarwal et al., 2026), we impose a condition controlling the spectra of the restricted Gram matrices $\mathbf{U}_{1j}^\top \mathbf{U}_{1j}$ and $\mathbf{V}_{1j}^\top \mathbf{V}_{1j}$.

Assumption A1. *There exist $0 < c_\ell \leq c_u$ such that for all $j \in [K]$ we have*

$$c_\ell \frac{N_{1j}}{N} I_r \preceq \mathbf{U}_{1j}^\top \mathbf{U}_{1j} \preceq c_u \frac{N_{1j}}{N} I_r, \quad c_\ell \frac{T_{1j}}{T} I_r \preceq \mathbf{V}_{1j}^\top \mathbf{V}_{1j} \preceq c_u \frac{T_{1j}}{T} I_r.$$

Our theoretical result also requires the following three conditions, where $c_0 > 0$ and $c_{\text{dim}} > 0$ are fixed, sufficiently small absolute constants, while $c_{\text{blk}} > 0$ is a fixed absolute constant. We use the additional notation in Table 1.

Assumption A2. *We have $r + \zeta \leq c_{\text{dim}} \min(N - r, T - r, N_{1k}, T_{1k})$ and $c_{\text{blk}} \min(\zeta_N, \zeta_T) \leq r$.*

Assumption A3. *Define $\theta := \sigma \gamma_{\min}^{-1} \max(\sqrt{N}, \sqrt{T}, \sqrt{N/\rho_T}, \sqrt{NT/N_{1k}})$, and assume*

this signal-to-noise ratio satisfies $\theta \leq c_0$.

Assumption A4. We define the incoherence parameters $\nu_x := \sqrt{N/r} \|\mathbf{U}_{2k}^\top \mathbf{x}\|_2$ and $\nu_y := \sqrt{T/r} \|\mathbf{V}_{2k}^\top \mathbf{y}\|_2$, and assume they are of constant order.

A detailed discussion of these assumptions is deferred to Section 3.2. We collect all admissible signal tensors into the following class. For fixed $r, N, T, K, \gamma_{\min}, \gamma_{\max}, \Omega$ and $0 < c_\ell \leq c_u$, we define

$$\begin{aligned} \mathcal{F}(c_\ell, c_u) &:= \left\{ \mathcal{M} \in \mathbb{R}^{N \times T \times K} : \mathcal{M} = \mathcal{C} \times_1 \mathbf{U} \times_2 \mathbf{V} \times_3 \mathbf{I}_K, \right. \\ &\quad \mathcal{C} \in \mathbb{R}^{r \times r \times K}, \quad \mathbf{U} \in \mathbb{R}^{N \times r}, \quad \mathbf{V} \in \mathbb{R}^{T \times r}, \mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_r, \\ &\quad 0 < \gamma_{\min} \leq \sigma_{\min}(\mathcal{C}_{\bullet, \bullet, j}) \leq \sigma_{\max}(\mathcal{C}_{\bullet, \bullet, j}) \leq \gamma_{\max} < \infty \quad \text{for all } j \in [K], \\ &\quad \left. \text{Assumption (A1) holds with constants } c_\ell, c_u \text{ for the fixed design } \Omega \right\}. \end{aligned}$$

We now state our main result. We write $\kappa := \gamma_{\max}/\gamma_{\min}$ for the condition number of $\mathcal{C}$.

Theorem 1. Fix absolute constants $0 < c_\ell \leq c_u$, a tensor $\mathcal{M} \in \mathcal{F}(c_\ell, c_u)$, an index $k \in [K]$, and unit vectors $\mathbf{x} \in \mathcal{S}^{N_{2k}-1}, \mathbf{y} \in \mathcal{S}^{T_{2k}-1}$. Let $\mu_{\mathbf{x}\mathbf{y}}^{(k)}$ be as in (3), and define $\hat{\mu}_{\mathbf{x}\mathbf{y}}^{(k)}$ to be the output of Algorithm 1 run with $0 < \tau \leq \frac{c_\ell N_{1k}}{2N}$. Assume (A2), (A3), and (A4) with $\nu_x \neq 0, \nu_y \neq 0$. Let

$$\Upsilon_{xy} := \frac{\sigma^2(r + \zeta_N)}{\rho_N} \|\mathbf{U}_{2k}^\top \mathbf{x}\|_2^2 + \frac{\sigma^2(r + \zeta_T)}{\rho_T} \|\mathbf{V}_{2k}^\top \mathbf{y}\|_2^2 + \frac{\sigma^2 N}{N_{1k}} \|\mathbf{U}_{2k}^\top \mathbf{x}\|_2^2 \|\mathbf{V}_{2k}^\top \mathbf{y}\|_2^2,$$

and further suppose that

$$\frac{\gamma_{\max}^2}{\tau} \frac{N_{1k}}{N} \left(p_N^{-10} + p_T^{-10} \right) + \frac{\sigma^2}{\tau} (N_{1k} + T) \left(p_N^{-5} + p_T^{-5} \right) \leq c_0 \Upsilon_{xy}. \quad (4)$$

There exists $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\dim}, c_{\text{blk}}, \kappa, \nu_x, \nu_y) > 0$ such that $\mathbb{E}_{\mathcal{M}}[\{\hat{\mu}_{\mathbf{x}\mathbf{y}}^{(k)} - \mu_{\mathbf{x}\mathbf{y}}^{(k)}\}^2] \leq c_1 \Upsilon_{xy}$.

We note that the term $\sigma^2 \frac{N}{N_{1k}} \|\mathbf{U}_{2k}^\top \mathbf{x}\|_2^2 \|\mathbf{V}_{2k}^\top \mathbf{y}\|_2^2$ is asymmetric because Algorithm 1 uses vertical regression to predict the missing d -block from the observed b -block; see

Section 2.2. However, applying the same construction to the transposed tensor gives $\frac{T}{T_{1k}}$ in place of $\frac{N}{N_{1k}}$, so choosing the better orientation yields the symmetric quantity $\sigma^2 \min\left(\frac{N}{N_{1k}}, \frac{T}{T_{1k}}\right) \|\mathbf{U}_{2k}^\top \mathbf{x}\|_2^2 \|\mathbf{V}_{2k}^\top \mathbf{y}\|_2^2$.

The interpretation of this result is illustrated in Fig. 3, and discussed immediately after. A useful comparison is with Yan and Wainwright (2024), who study entrywise inference for causal panel data under staggered adoption in the special case $K = 1$, $\mathbf{x} = \mathbf{e}_i$, and $\mathbf{y} = \mathbf{e}_t$. Both methods build on the spectral approach of Bai and Ng (2021), but differ in their inferential target and estimation procedure. Yan and Wainwright (2024, Algorithm 1) estimate missing entries of $\mathbf{M}_d$ by completing the full missing d -block, whereas we directly estimate a general bilinear form. More generally, however, our setting also allows $K \geq 1$ and extends this direct regress-then-denoise approach to tensor data with four-block missingness.

The theoretical comparison is also transparent in the matrix entrywise case. In this case, the upper bound in Theorem 1 matches their Equation 4.7 up to constants, apart from the term proportional to $\|\mathbf{U}_{2k}^\top \mathbf{e}_i\|_2^2 \|\mathbf{V}_{2k}^\top \mathbf{e}_t\|_2^2$. Since $\|\mathbf{V}_{2k}^\top \mathbf{e}_t\|_2^2 \leq 1$, this term is lower order and can be absorbed into the first one. In our tensor setting, however, it is important to keep this term explicit, as it captures the error component that does not decrease under pooling across layers and thereby characterizes the phase transition in the rate as K grows.

At the proof level, both the leave-one-block-out method of Yan and Wainwright (2024) and ours expand $\hat{\mu}_{xy}^{(k)} - \mu_{xy}^{(k)}$ into Gaussian terms and remainders. The key difference is how the remainders are controlled. Their approach could be adapted to our tensor setting, but yields negligible remainders only under signal-to-noise (SNR) conditions that worsen with K . The proof of Theorem 1 instead applies the Haar compression bounds from Section S4 of the Supplement to a centered version of $\mathbf{Y}_{\text{left}}^p (\mathbf{Y}_{\text{left}}^p)^\top$, following the opening paragraph of the proof of Lemma S4, and obtains the same type of expansion under weaker assumptions. Moreover, beyond allowing general unit vectors, Lemma S4 improves on Yan and Wainwright (2024, Equation B.4) by replacing the second-order term $\frac{\sigma^2(N+T_{1,p})}{\gamma_{\min}^2 \rho_T} \|\mathbf{U}^\top \mathbf{x}\|_2$

with $\frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} \|\mathbf{U}^\top \mathbf{x}\|_2$, while leaving the first-order term unchanged. This is an independent contribution of interest, and ensures that the SNR requirement improves with K .

3.2 Discussion of the Assumptions

We now discuss the four assumptions in Theorem 1. Assumption (A1) requires $\mathbf{U}_{1j}^\top \mathbf{U}_{1j}$ and $\mathbf{V}_{1j}^\top \mathbf{V}_{1j}$ to be uniformly well-conditioned across $j \in [K]$, with eigenvalues proportional to $\frac{N_{1j}}{N}$ and $\frac{T_{1j}}{T}$. Propositions S16–S17 in the Supplement show that $c_\ell > 0$ is necessary for (3) to be identifiable. This condition also implies $r \leq \min(N_{1j}, T_{1j})$ for all $j \in [K]$, and hence $r \leq \min(N, T, N_{1,p}, T_{1,p})$, which is the minimal dimensional requirement for the rank- r SVDs in Algorithm 1. Assumption (A2) consists of mild dimension-regularity conditions, which are introduced to simplify the statement of the final result and make it more transparent, while (A4) is a standard incoherence condition adapted to the directions $\mathbf{x}$ and $\mathbf{y}$ of interest.

Finally, (A3) is a SNR condition analogous to that in Agterberg (2026) for estimating $\mathbf{U}$ in a shared-subspace model with complete observations under the loss $\|\sin \Theta(\widehat{\mathbf{U}}, \mathbf{U})\|_F^2$. By analogy, their Theorem 1 suggests that estimating (3) would be information-theoretically impossible without (A3). Furthermore, compared to Yan and Wainwright (2024, Assumption 4.3), (A3) becomes progressively less stringent as the number of layers K increases, reflecting the benefit of pooling information across layers. This improvement continues until the requirement saturates at a local term which cannot be further reduced by pooling, thereby highlighting that a sufficiently strong slice-specific signal is still needed to learn $\mathcal{C}_{\bullet, \bullet, k}$.

3.3 Local minimax lower bound

We next complement the upper bound in Theorem 1 by establishing a local minimax lower bound in neighborhoods of fixed tensors $\mathcal{M}_0$ that satisfy suitable conditions. For $\mathcal{M} \in \mathcal{F}(c_\ell, c_u)$, we write $\mu_{\mathbf{x}\mathbf{y}}^{(k)}(\mathcal{M}) := \mathbf{x}^\top \mathcal{M}_{\bullet, \bullet, k}^{(d)} \mathbf{y}$ and $Z_\Omega := \{ \mathcal{M}_{itj} + \mathcal{E}_{itj} : \Omega_{i,t,j} = 1, (i, t, j) \in$

$[N] \times [T] \times [K]$ for the observed entries, and use $\mathbb{P}_{\mathcal{M}}$ and $\mathbb{E}_{\mathcal{M}}$ for probability and expectation under the law of Z_{Ω} . We also suppose for simplicity that $N_{1j} = N_1$ and $T_{1j} = T_1$ for all $j \in [K]$. Under this assumption, $\mathbf{U}_{0,1j_1} = \mathbf{U}_{0,1j_2}$ and $\mathbf{U}_{0,2j_1} = \mathbf{U}_{0,2j_2}$ for all $j_1, j_2 \in [K]$. We denote these common matrices by $\mathbf{U}_{0,1}$ and $\mathbf{U}_{0,2}$, respectively, and adopt the analogous convention for $\mathbf{V}_{0,1}$ and $\mathbf{V}_{0,2}$.

Theorem 2. Fix $k \in [K]$ and unit vectors $\mathbf{x} \in \mathcal{S}^{N_{2k}-1}$, $\mathbf{y} \in \mathcal{S}^{T_{2k}-1}$, and set $N_{1j} = N_1$ and $T_{1j} = T_1$ for all $j \in [K]$. Let $\mathcal{M}_0 = \mathcal{C}_0 \times_1 \mathbf{U}_0 \times_2 \mathbf{V}_0 \times_3 \mathbf{I}_K \in \mathcal{F}(c_{\ell}, c_u)$, and suppose that (A1) holds with margin $0 < \delta_{A1} < (c_u - c_{\ell})/2$, in the sense that

$$\begin{aligned} (c_{\ell} + \delta_{A1}) \frac{N_1}{N} \mathbf{I}_r &\preceq \mathbf{U}_{0,1}^{\top} \mathbf{U}_{0,1} \preceq (c_u - \delta_{A1}) \frac{N_1}{N} \mathbf{I}_r, \\ (c_{\ell} + \delta_{A1}) \frac{T_1}{T} \mathbf{I}_r &\preceq \mathbf{V}_{0,1}^{\top} \mathbf{V}_{0,1} \preceq (c_u - \delta_{A1}) \frac{T_1}{T} \mathbf{I}_r. \end{aligned}$$

Assume that the singular values of the k -th core matrix are separated from the boundary of the admissible interval, in the sense that $\delta_{\gamma} := \min\{\sigma_{\min}((\mathcal{C}_0)_{\bullet, \bullet, k}) - \gamma_{\min}, \gamma_{\max} - \sigma_{\max}((\mathcal{C}_0)_{\bullet, \bullet, k})\} > 0$. Also suppose that $c_u \max(\frac{N_1}{N}, \frac{T_1}{T}) \leq 1 - \delta_{\text{sep}}$ with $\delta_{\text{sep}} > 0$, and that $\delta_{\text{sep}}^{-1} \max(\|\mathbf{U}_{0,2}^{\top} \mathbf{x}\|_2^2, \|\mathbf{V}_{0,2}^{\top} \mathbf{y}\|_2^2) \leq 1/2$. For $\varsigma > 0$ define $\mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma) := \{\mathcal{M} \in \mathcal{F}(c_{\ell}, c_u) : \|\mathcal{M} - \mathcal{M}_0\|_F \leq \varsigma\}$. There exists a constant $c \equiv c(c_u, \gamma_{\min}, \gamma_{\max}) > 0$ such that

$$\begin{aligned} &\inf_{\phi} \sup_{\mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma)} \mathbb{E}_{\mathcal{M}} \left[\left\{ \phi(Z_{\Omega}) - \mu_{\mathbf{x}\mathbf{y}}^{(k)}(\mathcal{M}) \right\}^2 \right] \\ &\geq c \min \left(\frac{\sigma^2 N}{K N_1}, \varepsilon_V^2 \right) \|\mathbf{U}_{0,2}^{\top} \mathbf{x}\|_2^2 + c \min \left(\frac{\sigma^2 T}{K T_1}, \varepsilon_U^2 \right) \|\mathbf{V}_{0,2}^{\top} \mathbf{y}\|_2^2 \\ &\quad + c \min \left\{ \sigma^2 \min \left(\frac{N}{N_{1k}}, \frac{T}{T_{1k}} \right), \varsigma^2, \delta_{\gamma}^2 \right\} \|\mathbf{U}_{0,2k}^{\top} \mathbf{x}\|_2^2 \|\mathbf{V}_{0,2k}^{\top} \mathbf{y}\|_2^2, \end{aligned}$$

where $\varepsilon_U := \min(8^{-1/2}, \sqrt{T_1/T}, \varsigma \gamma_{\max}^{-1} K^{-1/2}, \sqrt{\delta_{A1}/c_{\ell}})$, $\varepsilon_V := \min(8^{-1/2}, \sqrt{N_1/N}, \varsigma \gamma_{\max}^{-1} K^{-1/2}, \sqrt{\delta_{A1}/c_{\ell}})$, and the infimum is over all Borel-measurable functions ϕ of Z_{Ω} .

This result shows the necessity of the elbow behavior in the rate as a function of K . In particular, when σ^2 is sufficiently small, the right-hand side of the preceding display

matches the upper bound in Theorem 1 up to constants, rank factors, and logarithmic factors. Moreover, although the results above are local and stated around a fixed $\mathcal{M}_0$, they also imply global minimax lower bounds by taking ς sufficiently large so that it contains the entire parameter space.

Finally, we observe that the assumption $\delta_{\text{sep}}^{-1} \max(\|\mathbf{U}_{0,2}^\top \mathbf{x}\|_2^2, \|\mathbf{V}_{0,2}^\top \mathbf{y}\|_2^2) \leq 1/2$ is mild in the regime relevant for comparison with Theorem 1. Indeed, under (A4) we have $\|\mathbf{U}_{0,2}^\top \mathbf{x}\|_2^2 \asymp \frac{r}{N}$ and $\|\mathbf{V}_{0,2}^\top \mathbf{y}\|_2^2 \asymp \frac{r}{T}$, hence, when r and δ_{sep} are of constant order, the condition holds for sufficiently large N and T .

4 Bilinear form estimation under staggered adoption

4.1 Proposed methodology for staggered missingness

We now extend the four-block setting to general staggered-adoption designs. Our methodology reduces the staggered missingness problem to simpler four-block missingness patterns by constructing pooled upper and left matrices in the same spirit as before. The main additional challenge is to accommodate the more complex missingness structure induced by staggered adoption. In particular, for a fixed layer $j \in [K]$, staggered adoption means that missingness is irreversible, i.e. for each unit $i \in [N]$, there is an adoption time A_{ij} such that $\Omega_{i,t,j} = \mathbb{1}\{t < A_{ij}\}$. For completeness, we set $A_{ij} = \infty$ for never-adopters.

It is useful to note that each layer has its own natural row ordering under which the corresponding missingness mask is a staircase; these orderings need not agree across layers. Since our target will be a bilinear form in layer k , we use the adoption-time ordering of the target layer as the common row ordering for all slices. In other words, we permute the rows so that $A_{1k} \geq \dots \geq A_{Nk}$. This entails no loss of generality, since applying a common row permutation to all layers preserves the Tucker2 structure. Under this convention, the mask $\Omega_{\bullet, \bullet, k}$ admits an equivalent staircase characterization: there exists

an integer $o_k \geq 2$ and ordered non-empty contiguous partitions $[N] = R_{1k} \cup \dots \cup R_{o_k,k}$ and $[T] = C_{1k} \cup \dots \cup C_{o_k,k}$, with $|R_{ak}| = N_{ak}$ and $|C_{bk}| = T_{bk}$, such that $\Omega_{i,t,k} = \mathbb{1}\{(i, t) \in R_{ak} \times C_{bk} \text{ for some } a, b \text{ with } a + b \leq o_k + 1\}$. This block representation will be useful in what follows, as it allows us to describe the observed and missing regions of the target layer in terms of the staircase partitions $\{R_{ak}\}_{a=1}^{o_k}$ and $\{C_{bk}\}_{b=1}^{o_k}$.

As is apparent from the staircase representation above, the assumption we make on Ω is that the missingness pattern for the target slice $\Omega_{\bullet,\bullet,k}$ contains fully observed rows, corresponding to never-adopting units, as well as an initial time period during which no unit in that slice has adopted. This is the basic requirement that allows us to reuse the methodology developed for the four-block design. An example of a staggered-adoption pattern covered by our framework is illustrated in Fig. 4.

Having specified the structure of the missingness masks, we now introduce the signal and noise model. As in the previous sections, we assume that the signal tensor $\mathcal{M}$ admits a Tucker2 decomposition of rank (r, r, K) , so that $\mathcal{M} = \mathcal{C} \times_1 \mathbf{U} \times_2 \mathbf{V} \times_3 \mathbf{I}_K$, where $\mathbf{U} \in \mathbb{R}^{N \times r}$ and $\mathbf{V} \in \mathbb{R}^{T \times r}$ have orthonormal columns. The noise tensor $\mathcal{E}$ has independent Gaussian entries with mean zero and variance σ^2 , and for each $j \in [K]$ we observe $\mathcal{Y}_{\bullet,\bullet,j} = P_{\Omega_{\bullet,\bullet,j}}(\mathcal{M}_{\bullet,\bullet,j} + \mathcal{E}_{\bullet,\bullet,j})$. Our goal here is to estimate bilinear forms over all missing entries in layer $k \in [K]$. For simplicity, we first focus on a specific missing block, since this is the key step needed for the general case. In this regard, choose indices (a, b) such that $a + b > o_k + 1$, so that the block $R_{ak} \times C_{bk}$ is unobserved. For unit vectors $\mathbf{x} \in \mathcal{S}^{N_{ak}-1}$ and $\mathbf{y} \in \mathcal{S}^{T_{bk}-1}$, our target estimand is the bilinear form

$$\mu_{\mathbf{x}\mathbf{y}}^{(k,a,b)} := \mathbf{x}^\top \mathcal{M}_{\bullet,\bullet,k}^{(a,b)} \mathbf{y}, \quad (5)$$

where $\mathcal{M}_{\bullet,\bullet,k}^{(a,b)} := \mathbf{U}_{ak} \mathcal{C}_{\bullet,\bullet,k} \mathbf{V}_{bk}^\top \in \mathbb{R}^{N_{ak} \times T_{bk}}$, with $\mathbf{U}_{ak} := \mathbf{U}_{R_{ak},\bullet} \in \mathbb{R}^{N_{ak} \times r}$, $\mathbf{V}_{bk} := \mathbf{V}_{C_{bk},\bullet} \in \mathbb{R}^{T_{bk} \times r}$ denoting the restrictions of the Tucker2 factors to specified row and column blocks.

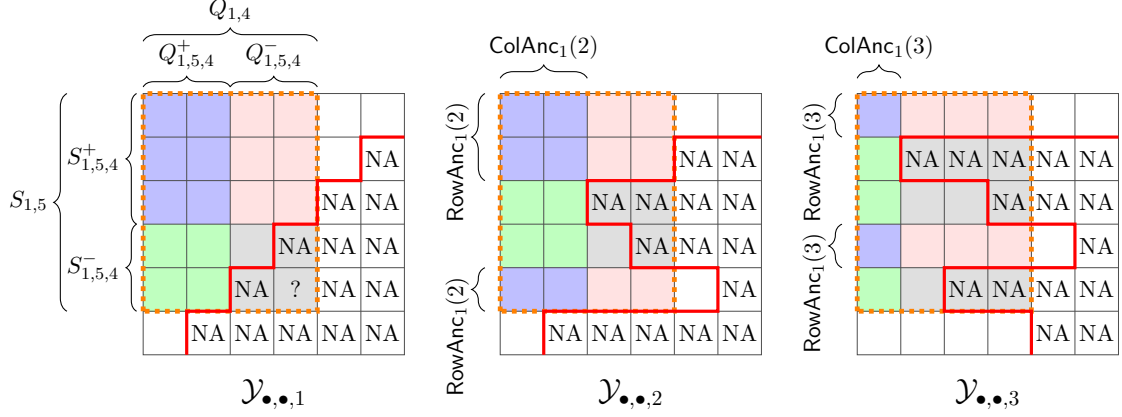


Figure 4: Layer-specific staggered adoption for $K=3$, target layer $k=1$, and target block $(a,b)=(5,4)$. Panel 1 orders the target layer by its own rows, yielding a staircase missingness pattern; the orange dotted rectangle identifies the target rows $S_{1,5}$ and columns $Q_{1,4}$ used to form auxiliary matrices. Panels 2–3 show non-target layers under the same target-layer ordering, with general staggered missingness. These layers provide fully observed anchor rows $\text{RowAnc}_1(j)$ and anchor columns $\text{ColAnc}_1(j)$: blue/pink blocks form the upper pooled matrix, blue/green blocks form the left pooled matrix, and gray blocks are discarded.

The following definitions are needed to present our algorithm. We set

$$S_{k,a,b}^+ := \bigcup_{a'=1}^{o_k+1-b} R_{a'k}, \quad S_{k,a,b}^- := \bigcup_{a'=o_k+2-b}^a R_{a'k}, \quad Q_{k,a,b}^+ := \bigcup_{b'=1}^{o_k+1-a} C_{b'k}, \quad Q_{k,a,b}^- := \bigcup_{b'=o_k+2-a}^b C_{b'k}.$$

Since $a+b > o_k+1$, the lower limits o_k+2-b and o_k+2-a are at most a and b , respectively, so the sets $S_{k,a,b}^-$ and $Q_{k,a,b}^-$ are non-empty and contain R_{ak} and C_{bk} . We also write $S_{k,a} := S_{k,a,b}^+ \cup S_{k,a,b}^-$, $Q_{k,b} := Q_{k,a,b}^+ \cup Q_{k,a,b}^-$, and observe that $S_{k,a}$ and $Q_{k,a,b}^+$ depend only on a , whereas $Q_{k,b}$ and $S_{k,a,b}^+$ depend only on b .

The four index sets in the display define a four-block structure in which the target missing block (a,b) is contained in $S_{k,a,b}^- \times Q_{k,a,b}^-$. Fig. 4 illustrates this construction, with the target block marked by a question mark. Some entries of $S_{k,a,b}^- \times Q_{k,a,b}^-$ may be observed under the original staggered pattern, but we discard them to obtain a literal four-block construction. Furthermore, to leverage the shared subspaces $\mathbf{U}$ and $\mathbf{V}$ in the Tucker2 model, we define anchor sets that identify auxiliary fully observed rows and columns. For $j \neq k$, we write $\text{ColAnc}_k(j) := \{t \in Q_{k,b} : \Omega_{i,t,j} = 1 \text{ for all } i \in S_{k,a}\}$ and $\text{RowAnc}_k(j) := \{i \in S_{k,a} : \Omega_{i,t,j} = 1 \text{ for all } t \in Q_{k,b}\}$. These correspond to periods in $Q_{k,b}$ that are observed for all units in $S_{k,a}$, and to units in $S_{k,a}$ that are observed throughout

all periods in $Q_{k,b}$, respectively. For completeness, we also set $\text{RowAnc}_k(k) := S_{k,a,b}^+$ and $\text{ColAnc}_k(k) := Q_{k,a,b}^+$. These sets identify the auxiliary submatrices used to construct the upper and left pooled matrices, as outlined in the following algorithm.

Algorithm 2 BILINEAR TENSOR STAGGERED for the estimation of $\mu_{\mathbf{x}\mathbf{y}}^{(k,a,b)} = \mathbf{x}^\top \mathcal{M}_{\bullet,\bullet,k}^{(a,b)} \mathbf{y}$ for a fixed missing block (a,b) in slice k of a tensor with staggered adoption missingness

Require: index $k \in [K]$, missing block (a,b) with $a+b > o_k+1$, rank r , unit vectors $\mathbf{x} \in \mathcal{S}^{N_{ak}-1}$ and $\mathbf{y} \in \mathcal{S}^{T_{bk}-1}$, data $\mathcal{Y}$, parameter $\tau > 0$.

- 1: Permute rows so that $\Omega_{\bullet,\bullet,k}$ is in staircase form.
 - 2: Compute $S_{k,a,b}^+, S_{k,a}, Q_{k,a,b}^+, Q_{k,b}$, and $\text{RowAnc}_k(j), \text{ColAnc}_k(j)$ for all $j \in [K]$.
 - 3: Form pooled left matrix $\mathbf{Y}_{\text{left}}^p \leftarrow (\mathcal{Y}_{S_{k,a}, \text{ColAnc}_k(1),1} \cdots \mathcal{Y}_{S_{k,a}, \text{ColAnc}_k(K),K}) \in \mathbb{R}^{|S_{k,a}| \times \sum_{j=1}^K |\text{ColAnc}_k(j)|}$.
 - 4: Compute rank- r truncated singular value decomposition $(\hat{\mathbf{U}}_{\text{left}}, \hat{\mathbf{\Sigma}}_{\text{left}}, \hat{\mathbf{V}}_{\text{left}}) \leftarrow \text{SVD}_r(\mathbf{Y}_{\text{left}}^p)$.
 - 5: Set $\hat{\mathbf{U}}_{+k} \leftarrow (\hat{\mathbf{U}}_{\text{left}})_{S_{k,a,b}^+, \bullet}$ and $\hat{\mathbf{U}}_{ak} \leftarrow (\hat{\mathbf{U}}_{\text{left}})_{R_{ak}, \bullet}$.
 - 6: Compute $\hat{\mathbf{H}}_k \leftarrow \hat{\mathbf{U}}_{+k}^\top \hat{\mathbf{U}}_{+k} \in \mathbb{R}^{r \times r}$, take the eigendecomposition $\hat{\mathbf{H}}_k = \mathbf{Q} \text{diag}(\lambda_1, \dots, \lambda_r) \mathbf{Q}^\top$, and set
$$\hat{\mathbf{H}}_{k,\tau}^{\text{inv}} \leftarrow \mathbf{Q} \text{diag} \left(\left\{ \frac{1}{\max[\lambda_i, \tau]} \right\}_{i=1}^r \right) \mathbf{Q}^\top.$$
 - 7: Compute $\hat{\boldsymbol{\alpha}}_x^{(k,a,b)} \leftarrow \hat{\mathbf{U}}_{ak}^\top \mathbf{x} \in \mathbb{R}^r$.
 - 8: Form pooled upper matrix $\mathbf{Y}_{\text{up}}^p \leftarrow (\mathcal{Y}_{\text{RowAnc}_k(1), Q_{k,b},1} ; \cdots ; \mathcal{Y}_{\text{RowAnc}_k(K), Q_{k,b},K}) \in \mathbb{R}^{\sum_{j=1}^K |\text{RowAnc}_k(j)| \times |Q_{k,b}|}$.
 - 9: Compute rank- r truncated singular value decomposition $(\hat{\mathbf{U}}_{\text{up}}, \hat{\mathbf{\Sigma}}_{\text{up}}, \hat{\mathbf{V}}_{\text{up}}) \leftarrow \text{SVD}_r(\mathbf{Y}_{\text{up}}^p)$.
 - 10: Let $s_k \leftarrow \sum_{j=1}^{k-1} |\text{RowAnc}_k(j)|$, and extract $\hat{\mathbf{U}}_{\text{up}}^{(k)} \leftarrow (\hat{\mathbf{U}}_{\text{up}})_{\{s_k+1, \dots, s_k+|S_{k,a,b}^+|\}, \bullet}$, $\hat{\mathbf{V}}_{bk} \leftarrow (\hat{\mathbf{V}}_{\text{up}})_{C_{bk}, \bullet}$.
 - 11: Compute $\mathbf{T}_y \leftarrow \hat{\mathbf{V}}_{bk}^\top \mathbf{y} \in \mathbb{R}^r$, $\mathbf{W}_y \leftarrow \hat{\mathbf{\Sigma}}_{\text{up}} \mathbf{T}_y \in \mathbb{R}^r$, and $\mathbf{X}_y \leftarrow \hat{\mathbf{U}}_{\text{up}}^{(k)} \mathbf{W}_y \in \mathbb{R}^{|S_{k,a,b}^+|}$.
 - 12: Compute $\hat{\boldsymbol{\beta}}_y^{(k,a,b)} \leftarrow \hat{\mathbf{H}}_{k,\tau}^{\text{inv}} \hat{\mathbf{U}}_{+k}^\top \mathbf{X}_y \in \mathbb{R}^r$.
 - 13: **return** $\hat{\mu}_{\mathbf{x}\mathbf{y}}^{(k,a,b)} \leftarrow \langle \hat{\boldsymbol{\alpha}}_x^{(k,a,b)}, \hat{\boldsymbol{\beta}}_y^{(k,a,b)} \rangle$.
-

Algorithm 2 extends Algorithm 1 to the staggered-adoption setting, and reduces to it when the missingness pattern has four-block form. For a fixed missing block (a,b) in layer k , the algorithm restricts attention to $\mathcal{Y}_{S_{k,a}, Q_{k,b}, \bullet}$, and uses the observations lying in $\bigcup_{j=1}^K (S_{k,a} \times \text{ColAnc}_k(j) \times \{j\})$ and $\bigcup_{j=1}^K (\text{RowAnc}_k(j) \times Q_{k,b} \times \{j\})$ to construct an auxiliary four-block problem, discarding all remaining entries. The left pooled matrix is formed from the anchor-column blocks $\mathcal{Y}_{S_{k,a}, \text{ColAnc}_k(j), j}$, while the upper pooled matrix is formed from the anchor-row blocks $\mathcal{Y}_{\text{RowAnc}_k(j), Q_{k,b}, j}$. This construction is illustrated in Fig. 4.

Our procedure generalizes Yan and Wainwright (2024, Algorithm 2) to the tensor setting and targets the bilinear form directly, rather than reconstructing the entire missing block. Related denoising techniques, using anchor sets and combined with PCA, were employed in Liu et al. (2026) for a different statistical problem, where the goal is to recover the global

left subspace from matrix data with blockwise missingness, with error measured with $\|\cdot\|_F$.

The auxiliary four-block construction also imposes a basic dimensional feasibility condition. In applications, the rank must satisfy $r \leq \min(\sum_{j \in [K]} |\text{RowAnc}_k(j)|, \sum_{j \in [K]} |\text{ColAnc}_k(j)|, |S_{k,a}|, |Q_{k,b}|)$, so that the two rank- r truncated SVDs are well defined. This is only a minimal requirement for running the procedure. Even when this condition holds, additional assumptions are needed to guarantee that the resulting estimator is accurate; the theoretical analysis of Algorithm 2 is the object of the next section.

Finally, when aggregate quantities over multiple missing blocks of the same slice are required, such as those introduced in Section 5.2, the most direct strategy is to apply Algorithm 2 separately to each missing block and then aggregate the resulting estimates. This blockwise implementation recomputes two rank- r SVDs for every missing block, leading to the quadratic-cost procedure described in Algorithm S1 in Appendix S2.2. We also propose a reduced-anchor variant that reuses computations by caching the pooled left SVD once for each active row block a , and the pooled upper SVD once for each active column block b . This yields the linear-SVD-cost procedure in Algorithm S2, at the price of some loss in statistical efficiency. Appendix S2.2 provides an extensive discussion of this computational–statistical tradeoff, and Fig. 6 in Section 5.1 compares the two procedures in simulation.

4.2 Theoretical analysis

Algorithm 2 inherits the desirable properties of Algorithm 1 under analogous assumptions, with particular care needed in adapting Assumption (A1) to the auxiliary four-block reduction. To state these assumptions and the resulting corollary, we suppress the dependence on (k, a, b) for simplicity, and set $\mathbf{n} := |S|$ and $\mathbf{t} := |Q|$. For the target layer, define $\mathbf{n}_{1k} := |S^+|$ and $\mathbf{t}_{1k} := |Q^+|$. For each $j \neq k$, define $\mathbf{n}_{1j} := |\text{RowAnc}_k(j)|$ and $\mathbf{t}_{1j} := |\text{ColAnc}_k(j)|$, and set $\mathbf{n}_{2j} := \mathbf{n} - \mathbf{n}_{1j}$ and $\mathbf{t}_{2j} := \mathbf{t} - \mathbf{t}_{1j}$. Finally, let $\mathbf{n}_{1,p} := \sum_{j=1}^K \mathbf{n}_{1j}$, $\mathbf{t}_{1,p} := \sum_{j=1}^K \mathbf{t}_{1j}$, $\rho_{\mathbf{n}} := \mathbf{n}_{1,p}/\mathbf{n}$,

$\rho_{\mathbf{t}} := \mathbf{t}_{1,p}/\mathbf{t}$, $p_{\mathbf{n}} := \max(\mathbf{n}_{1,p}, \mathbf{t})$, $p_{\mathbf{t}} := \max(\mathbf{n}, \mathbf{t}_{1,p})$, $\zeta_{\mathbf{n}} := \log(\mathbf{n}_{1,p} + \mathbf{t})$, $\zeta_{\mathbf{t}} := \log(\mathbf{n} + \mathbf{t}_{1,p})$,
 $\tilde{\gamma}_{\min} := c_{\ell} \gamma_{\min} \sqrt{\mathbf{n}\mathbf{t}/NT}$ and $\tilde{\gamma}_{\max} := c_u \gamma_{\max} \sqrt{\mathbf{n}\mathbf{t}/NT}$. We assume the following.

Assumption A5. *There exist constants $0 < c_{\ell} \leq c_u$ such that*

$$c_{\ell} \frac{\mathbf{n}_{1k}}{N} \mathbf{I}_r \preceq \mathbf{U}_{S^+}^{\top} \mathbf{U}_{S^+} \preceq c_u \frac{\mathbf{n}_{1k}}{N} \mathbf{I}_r, \quad c_{\ell} \frac{\mathbf{n}}{N} \mathbf{I}_r \preceq \mathbf{U}_S^{\top} \mathbf{U}_S \preceq c_u \frac{\mathbf{n}}{N} \mathbf{I}_r,$$

and, for every $j \neq k$,

$$c_{\ell} \frac{\mathbf{n}_{1j}}{N} \mathbf{I}_r \preceq \mathbf{U}_{\text{RowAnc}_k(j)}^{\top} \mathbf{U}_{\text{RowAnc}_k(j)} \preceq c_u \frac{\mathbf{n}_{1j}}{N} \mathbf{I}_r.$$

We require that the column factors satisfy the analogous conditions with $\mathbf{V}, Q, Q^+, \text{ColAnc}_k(j), \mathbf{t}_{1j}$ and T in place of $\mathbf{U}, S, S^+, \text{RowAnc}_k(j), \mathbf{n}_{1j}$ and N , respectively.

Assumption A6. *We have $r + \max(\zeta_{\mathbf{n}}, \zeta_{\mathbf{t}}) \leq c_{\dim} \min(\mathbf{n} - r, \mathbf{t} - r, \mathbf{n}_{1k}, \mathbf{t}_{1k})$ and $c_{\text{blk}} \min(\zeta_{\mathbf{n}}, \zeta_{\mathbf{t}}) \leq r$. Also, the noise level satisfies*

$$\tilde{\theta} := \frac{\sigma}{\tilde{\gamma}_{\min}} \max \left\{ \sqrt{\mathbf{n}}, \sqrt{\mathbf{t}}, \sqrt{\frac{\mathbf{n}}{\rho_{\mathbf{t}}}}, \sqrt{\frac{\mathbf{n}\mathbf{t}}{\mathbf{n}_{1k}}} \right\} \leq c_0.$$

Finally, we write $\tilde{\mathbf{U}} := \mathbf{U}_S(\mathbf{U}_S^{\top} \mathbf{U}_S)^{-1/2}$ and $\tilde{\mathbf{V}} := \mathbf{V}_Q(\mathbf{V}_Q^{\top} \mathbf{V}_Q)^{-1/2}$, and assume that $\tilde{\nu}_x := \sqrt{\mathbf{n}/r} \|\tilde{\mathbf{U}}_{R_{ak}}^{\top} \mathbf{x}\|_2$ and $\tilde{\nu}_y := \sqrt{\mathbf{t}/r} \|\tilde{\mathbf{V}}_{C_{bk}}^{\top} \mathbf{y}\|_2$ are of constant order.

Assumption (A6) is the adaptation of (A2), (A3) and (A4) for staggered designs. Moreover, the first and third conditions in (A5) are the direct analogues of (A1), and require the observed row and column blocks retained in the auxiliary four-block problem to contain all r latent directions in a well-conditioned way. The middle condition is the restricted analogue of the orthonormality condition $\mathbf{U}^{\top} \mathbf{U} = \mathbf{V}^{\top} \mathbf{V} = \mathbf{I}_r$, and requires that the Gram matrices associated to $\mathbf{U}_S$ and $\mathbf{V}_Q$ remain full rank and well conditioned.

Corollary 3. *Consider a tensor $\mathcal{M}$ satisfying $\mathcal{M} = \mathcal{C} \times_1 \mathbf{U} \times_2 \mathbf{V} \times_3 \mathbf{I}_K$, where $\mathbf{U} \in \mathbb{R}^{N \times r}$ and $\mathbf{V} \in \mathbb{R}^{T \times r}$ have orthonormal columns, and the core tensor is such that $0 < \gamma_{\min} \leq$*

$\sigma_{\min}(\mathcal{C}_{\bullet,\bullet,j}) \leq \sigma_{\max}(\mathcal{C}_{\bullet,\bullet,j}) \leq \gamma_{\max} < \infty$ for all $j \in [K]$. Choose $k \in [K]$, indices (a, b) such that $a + b > o_k + 1$, and unit vectors $\mathbf{x} \in \mathcal{S}^{N_{ak}-1}$ and $\mathbf{y} \in \mathcal{S}^{T_{bk}-1}$. Let $\mu_{\mathbf{xy}}^{(k,a,b)}$ be as in (5), and define $\hat{\mu}_{\mathbf{xy}}^{(k,a,b)}$ to be the output of Algorithm 2 run with $0 < \tau \leq \frac{c_\ell \mathbf{n}_{1k}}{2c_u \mathbf{n}}$. Fix also absolute constants $0 < c_\ell \leq c_u < \infty$, and assume (A5), (A6) with $\tilde{v}_x \neq 0, \tilde{v}_y \neq 0$. Let

$$\tilde{\Upsilon}_{xy} := \frac{\sigma^2(r + \zeta_{\mathbf{n}})}{\rho_{\mathbf{n}}} \|\tilde{\mathbf{U}}_{R_{ak}}^\top \mathbf{x}\|_2^2 + \frac{\sigma^2(r + \zeta_{\mathbf{t}})}{\rho_{\mathbf{t}}} \|\tilde{\mathbf{V}}_{C_{bk}}^\top \mathbf{y}\|_2^2 + \frac{\sigma^2 \mathbf{n}}{\mathbf{n}_{1k}} \|\tilde{\mathbf{U}}_{R_{ak}}^\top \mathbf{x}\|_2^2 \|\tilde{\mathbf{V}}_{C_{bk}}^\top \mathbf{y}\|_2^2,$$

and further assume that

$$\frac{\tilde{\gamma}_{\max}^2 \mathbf{n}_{1k}}{\tau \mathbf{n}} (p_{\mathbf{n}}^{-10} + p_{\mathbf{t}}^{-10}) + \frac{\sigma^2}{\tau} (\mathbf{n}_{1k} + \mathbf{t}) (p_{\mathbf{n}}^{-5} + p_{\mathbf{t}}^{-5}) \leq c_0 \tilde{\Upsilon}_{xy}. \quad (6)$$

There exists a constant $c_1 = c_1(c_\ell, c_u, c_0, c_{\dim}, c_{\text{blk}}, \kappa, \tilde{v}_x, \tilde{v}_y) < \infty$ such that $\mathbb{E}_{\mathcal{M}}[\{\hat{\mu}_{xy}^{(k,a,b)} - \mu_{xy}^{(k,a,b)}\}^2] \leq c_1 \tilde{\Upsilon}_{xy}$.

This result follows directly from Theorem 1, with the original dimensions and block sizes replaced by their auxiliary counterparts. To see why, observe that $\mathbf{U}_S^\top \mathbf{U}_S$ and $\mathbf{V}_Q^\top \mathbf{V}_Q$ are invertible under (A5), hence $\tilde{\mathbf{U}} := \mathbf{U}_S(\mathbf{U}_S^\top \mathbf{U}_S)^{-1/2}$, $\tilde{\mathbf{V}} := \mathbf{V}_Q(\mathbf{V}_Q^\top \mathbf{V}_Q)^{-1/2}$, and $\tilde{\mathcal{C}}_{\bullet,\bullet,j} := (\mathbf{U}_S^\top \mathbf{U}_S)^{1/2} \mathcal{C}_{\bullet,\bullet,j} (\mathbf{V}_Q^\top \mathbf{V}_Q)^{1/2}$ give an orthonormal Tucker2 representation of the auxiliary signal $\mathcal{M}_{S,Q,j} = \tilde{\mathbf{U}} \tilde{\mathcal{C}}_{\bullet,\bullet,j} \tilde{\mathbf{V}}^\top$. The auxiliary core tensor satisfies $0 < \tilde{\gamma}_{\min} \leq \sigma_{\min}(\tilde{\mathcal{C}}_{\bullet,\bullet,j}) \leq \sigma_{\max}(\tilde{\mathcal{C}}_{\bullet,\bullet,j}) \leq \tilde{\gamma}_{\max}$, while $\tilde{\mathbf{U}}$ and $\tilde{\mathbf{V}}$ satisfy (A1) with c_ℓ/c_u and c_u/c_ℓ in place of c_ℓ and c_u , respectively; see the proof of Corollary 3 for the precise details. This, together with (A6), implies that the pooled upper and left matrices obey the same conditions as in the four-block setting, with $N, T, N_{1k}, T_{1k}, \rho_N, \rho_T, p_N, p_T, \zeta_N, \zeta_T, \gamma_{\min}, \gamma_{\max}, c_\ell, c_u$ replaced by $\mathbf{n}, \mathbf{t}, \mathbf{n}_{1k}, \mathbf{t}_{1k}, \rho_{\mathbf{n}}, \rho_{\mathbf{t}}, p_{\mathbf{n}}, p_{\mathbf{t}}, \zeta_{\mathbf{n}}, \zeta_{\mathbf{t}}, \tilde{\gamma}_{\min}, \tilde{\gamma}_{\max}, c_\ell/c_u, c_u/c_\ell$. This is precisely what is needed to apply Theorem 1, even though the auxiliary observation pattern need not consist of four contiguous blocks, and with these substitutions the stated result follows.

5 Numerical results

5.1 Synthetic data

The code and datasets for reproducing the simulations are available on [GitHub](#). In this subsection we empirically validate our theoretical claims on synthetic data. We verify that pooling improves performance for moderate values of K , while saturation occurs for large K , thereby confirming the phase transitions predicted by Theorem 1. We include robustness checks with respect to rank misspecification, SNR levels, vector inputs, and mask dimensions. For staggered adoption designs, we show that pooling across layers improves accuracy and demonstrate how to lower computational costs when computing an average counterfactual component over multiple blocks.

In Fig. 5(a), we fix $N = 100$, $T = 80$, $r = 6$ and vary $K \in \{1, 2, 5, 10, 20, 50, 200\}$. The matrices $\mathbf{U}$ and $\mathbf{V}$ are generated by drawing Gaussian random matrices and orthonormalizing their columns. For each slice $j \in [K]$, we generate $\mathcal{C}_{\bullet,\bullet,j} = \mathbf{O}_j \text{diag}(\sigma_1, \dots, \sigma_r) \tilde{\mathbf{O}}_j^\top$, where $\mathbf{O}_j, \tilde{\mathbf{O}}_j \in \mathbb{R}^{r \times r}$ are independent random orthonormal matrices, and the singular values are fixed at $(\sigma_1, \dots, \sigma_r) = (2, 1.72, 1.44, 1.16, 0.88, 0.60)$. The observations are generated according to (2), with noise variance σ^2 chosen so that $\text{SNR}^{-1} := \sigma^2 N / \sigma_r^2$ and $\text{SNR} = 1$, and with block sizes $N_{1j} = 70$ and $T_{1j} = 60$ for all $j \in [K]$. We fix $k = 1$ and consider unit vectors $\mathbf{x} \in \mathbb{R}^{N_{21}}$ and $\mathbf{y} \in \mathbb{R}^{T_{21}}$ independently drawn from standard Gaussian distributions and then normalized to have unit norm. We compare five procedures for estimating (3):

1. ESTIMATED POOLED: Algorithm 1 with $\tau = 0.01$;
2. ESTIMATED NO-POOL: Algorithm 1 with $\tau = 0.01$, applied to the target slice $\mathcal{Y}_{\bullet,\bullet,k}$;
3. ORACLE POOLED: returns $\mathbf{x}^\top \mathbf{U}_{2k} \mathcal{C}_{\bullet,\bullet,k} (\mathbf{W}_{\text{up}}^\top \mathbf{W}_{\text{up}})^{-1} \mathbf{W}_{\text{up}}^\top (\mathbf{E}_{\text{up}}^{\text{p}})_{\bullet,\{T_{1k}+1,\dots,T\}} \mathbf{y} + \mathbf{x}^\top (\mathbf{E}_{\text{left}}^{\text{p}})_{\{N_{1k}+1,\dots,N\},\bullet} \mathbf{W}_{\text{left}} (\mathbf{W}_{\text{left}}^\top \mathbf{W}_{\text{left}})^{-1} \mathcal{C}_{\bullet,\bullet,k} \mathbf{V}_{2k}^\top \mathbf{y} + \mathbf{x}^\top \mathcal{M}_{\bullet,\bullet,k}^{(d)} \mathbf{y}$;
4. ORACLE NO-POOL: layer-specific counterpart of ORACLE POOLED, which returns $\mathbf{x}^\top \mathbf{U}_{2k} (\mathbf{U}_{1k}^\top \mathbf{U}_{1k})^{-1} \mathbf{U}_{1k}^\top \mathcal{E}_{\bullet,\bullet,k}^{(b)} \mathbf{y} + \mathbf{x}^\top \mathcal{E}_{\bullet,\bullet,k}^{(c)} \mathbf{V}_{1k} (\mathbf{V}_{1k}^\top \mathbf{V}_{1k})^{-1} \mathbf{V}_{1k}^\top \mathbf{y} + \mathbf{x}^\top \mathcal{M}_{\bullet,\bullet,k}^{(d)} \mathbf{y}$;

$$5. \text{ ORACLE LOCAL: } \mathbf{x}^\top \mathbf{U}_{2k} (\mathbf{U}_{1k}^\top \mathbf{U}_{1k})^{-1} \mathbf{U}_{1k}^\top (\mathbf{E}_{\text{up}}^{\text{p}})_{\{s_k+1, \dots, s_k+N_{1k}\}, \bullet} \mathbf{V} \mathbf{V}_{2k}^\top \mathbf{y} + \mathbf{x}^\top \mathcal{M}_{\bullet, \bullet, k}^{(d)} \mathbf{y}.$$

The oracle quantities correspond to the Gaussian terms in the expansion of $\hat{\mu}_{xy}^{(k)} - \mu_{xy}^{(k)}$ given in Lemma S12 in the Supplement, and capture the leading contribution to the stochastic error of our procedure. For each value of K and for each of these estimators, we run 500 replications and report the average squared error. The results show that the NO-POOL estimators do not decay with K , as expected. By contrast, the two POOLED estimators exhibit an approximate $1/K$ decay and appear to approach the line corresponding to ORACLE LOCAL. This line is much lower than the others because it is related to both $\|\mathbf{U}_{2k}^\top \mathbf{x}\|_2^2$ and $\|\mathbf{V}_{2k}^\top \mathbf{y}\|_2^2$, yielding an additional factor of roughly $r/T = 6/80 = 0.075$. To better understand the relationship among these latter three methods, in Fig. 5(b) we repeat the same simulation study for $K \in \{50, 150, 300, 500, 1000\}$. The results show that ORACLE POOLED continues to decay with K , whereas Algorithm 1 saturates at approximately the level of ORACLE LOCAL. This is in accordance with Theorem 1.

Fig. 5(c) uses the same setup as Fig. 5(a), with a rank-6 signal tensor, but draws the mask dimensions randomly over $N_{1j} \in \{30, \dots, 70\}$ and $T_{1j} \in \{30, \dots, 60\}$. The query vectors are chosen to be aligned with the leading eigenspaces of $\mathbf{U}_{2k}$ and $\mathbf{V}_{2k}$, thereby increasing $\|\mathbf{U}_{2k}^\top \mathbf{x}\|_2$ and $\|\mathbf{V}_{2k}^\top \mathbf{y}\|_2$ and deliberately violating the incoherence condition in (A4). In addition, all estimators are run with misspecified rank $r + 5 = 11$. The results show that ESTIMATED POOLED still exhibits a similar phase transition as in Fig. 5(a), despite the misspecification and incoherence violation. By contrast, ESTIMATED NO-POOL is more sensitive to both sources of misspecification and has larger error than ORACLE NO-POOL.

Finally, Fig. 5(d) uses the same setup as Fig. 5(b), but varies $\text{SNR} \in \{1, 10^{-2}, 10^{-4}\}$ to examine sensitivity under increasingly noisy regimes. The results indicate that the proposed methodology remains reasonably stable even at lower signal levels, with ESTIMATED POOLED appearing close to ORACLE LOCAL across the considered SNR values.

We also empirically evaluate Algorithm 2 in a synthetic staggered-adoption design. We

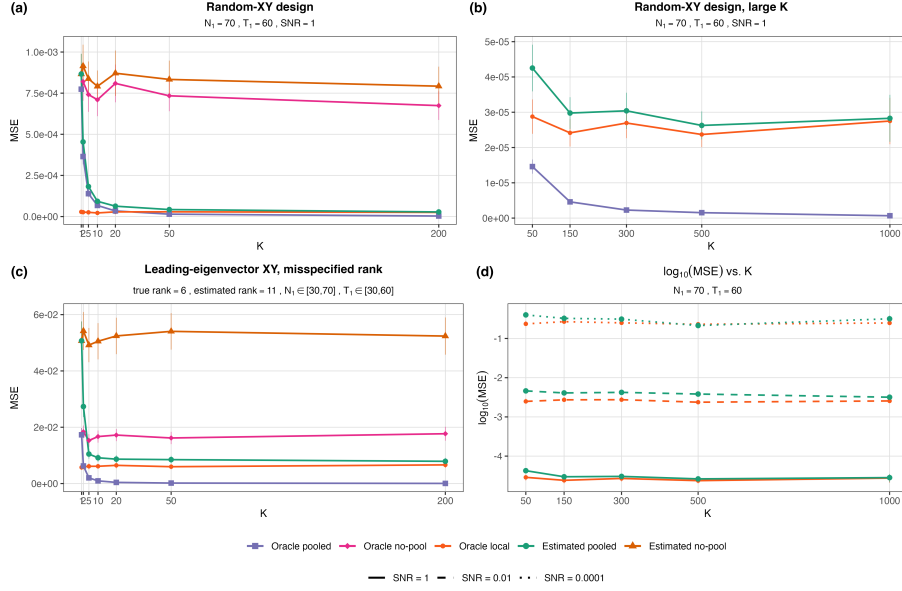


Figure 5: (a) Mean squared error of the five estimators as a function of $K \in \{1, 2, 5, 10, 20, 50, 200\}$, with $N = 100, T = 80, r = 6, \text{SNR} = 1$, and block sizes $N_{1j} = 70, T_{1j} = 60$ for all $j \in [K]$. Results are averaged over 500 replications. (b) Mean squared error of the pooled and local estimators for $K \in \{50, 150, 300, 500, 1000\}$ under the same simulation setting. (c) Analogue of (a) with violation of incoherence, rank-misspecification and mask sizes $N_{1j} \in \{30, \dots, 70\}$ and $T_{1j} \in \{30, \dots, 60\}$. (d) Sensitivity analysis with different SNR values. Error bars show ± 1.96 standard errors.

generate a rank-5 Tucker2 signal tensor with $N = 150, T = 200$, and $K = 10$, where

$\mathbf{U} \in \mathbb{R}^{N \times r}$ and $\mathbf{V} \in \mathbb{R}^{T \times r}$ are orthonormal and generated as before, and the entries of the core matrices $\mathcal{C}_{\bullet, \bullet, j} \in \mathbb{R}^{r \times r}$ are independent standard normal variables. We observe $\mathcal{Y}_{\bullet, \bullet, j} = P_{\Omega_{\bullet, \bullet, j}}(\mathcal{M}_{\bullet, \bullet, j} + \mathcal{E}_{\bullet, \bullet, j})$, where the entries of $\mathcal{E}$ are independent $\mathcal{N}(0, 0.03^2)$. The missingness masks $\Omega_{\bullet, \bullet, j}$ are generated by an irreversible adoption process in which, for each unit i and layer j , the adoption time A_{ij} is sampled independently from a layer-specific grid of adoption times, with probability 0.20 of never adopting. We then set $\Omega_{i, t, j} = \mathbb{1}\{t < A_{ij}\}$. We take $k = 1$ as the target layer and use the adoption ordering in this slice to reorder the units across all layers. Under this ordering, $\Omega_{\bullet, \bullet, 1}$ is a staircase and satisfies $\Omega_{i, t, 1} = \mathbb{1}\{(i, t) \in R_{a1} \times C_{b1} \text{ for some } a, b \text{ with } a + b \leq 5\}$. We then target all six missing blocks and, for each of them, we generate 100 independent query pairs $\mathbf{x} \in \mathcal{S}^{N_{a1}-1}$ and $\mathbf{y} \in \mathcal{S}^{T_{b1}-1}$ by drawing independent standard normal vectors and normalizing them to have unit norm. For each query we estimate (5) using both the pooled estimator outlined in Algorithm 2, and its matrix-only counterpart, which runs the procedure on $\mathcal{Y}_{\bullet, \bullet, k}$ only and does not borrow information from the other layers. In particular, this algorithm

uses $\mathcal{Y}_{S_{k,a}, Q_{k,a,b}^+, k}$ and $\mathcal{Y}_{S_{k,a,b}^+, Q_{k,b}, k}$ in place of $\mathbf{Y}_{\text{left}}^p$ and $\mathbf{Y}_{\text{up}}^p$, respectively, and is essentially equivalent to a plug-in estimator based on the matrix completion estimator proposed in Yan and Wainwright (2024). Both procedures are run with $\tau \in \{10^{-0.5+0.025j} : j = 0, \dots, 20\}$.

Fig. 6(a) reports the mean absolute estimation error over the 100 random bilinear queries for each selected block and each value of τ . The pooled estimator has substantially lower estimation error than the matrix-only estimator. This is expected, since the pooled method exploits the shared row and column subspaces across layers, whereas the matrix-only method uses only the two anchor blocks available in $\mathcal{Y}_{\bullet, \bullet, k}$. In general, the fact that certain blocks have larger errors than others can be attributed to differences in the size of the missing blocks: smaller missing blocks may have a lower signal-to-noise ratio, which can in turn reduce estimation accuracy. Furthermore, we see that both estimators are stable across values of τ , particularly when $\tau \ll 1$. This agrees with our theory: under Assumption (A1) with $c_\ell > 0$, choosing τ small enough ensures that clipping is inactive with high probability, while stabilizing the inverse on the complementary event. Thus, τ acts primarily as a safeguard rather than a tuning parameter, and careful tuning appears unnecessary.

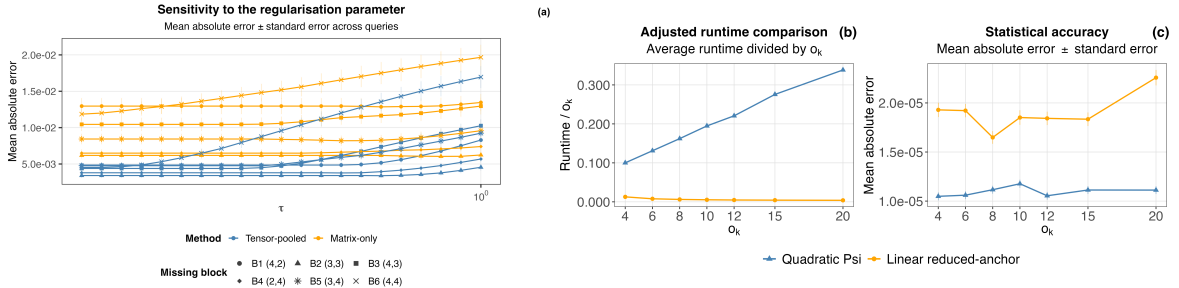


Figure 6: (a) Mean absolute estimation error over 100 random bilinear queries for each target block as a function of τ for Algorithm 2 and its matrix counterpart. Runtime (b) and accuracy (c) comparison for Algorithms S1 and S2 when estimating the aggregate AVG functional in a synthetic staggered-adoption design. Here $o_k \in \{4, 6, 8, 10, 12, 15, 20\}$ denotes the number of row/time blocks induced by the staircase-adoption pattern in the target slice.

Finally, we compare Algorithm S1 and Algorithm S2 in terms of statistical efficiency and runtime when estimating the aggregate AVG functional defined as a weighted average of the bilinear forms in (5) across all missing blocks in the target slice; this quantity and the two procedures are presented in detail in Appendix S2.2. We generate a rank-5 Tucker2 signal tensor with $N = 150$, $T = 200$, and $K = 5$, using random orthonormal matrices $\mathbf{U} \in \mathbb{R}^{N \times r}$

and $\mathbf{V} \in \mathbb{R}^{T \times r}$ and independent standard normal entries in each core matrix $\mathcal{C}_{\bullet, \bullet, j}$. The error tensor $\mathcal{E}$ has independent $\mathcal{N}(0, 0.01^2)$ entries. The target layer is $k = 1$ and has staircase missingness with $o_k \in \{4, 6, 8, 10, 12, 15, 20\}$; all non-target layers are fully observed. For each value of o_k , we run both procedures with $\tau = 10^{-3}$ over 500 replications and report the average runtime divided by o_k and the mean absolute deviation from the true value.

The results in Figs. 6(b, c) are consistent with the discussion in Appendix S2.2. The runtime divided by o_k appears approximately linear for Algorithm S1, reflecting its quadratic dependence on the number of target blocks, while it is nearly constant for Algorithm S2, reflecting the fact that the dominant SVD cost is linear in o_k . Also, Algorithm S1 has better statistical accuracy, as expected, since Algorithm S2 uses reduced anchor sets to improve runtime and therefore sacrifices some statistical efficiency.

5.2 Real-data application: Castle Doctrine data

In this and the next section, we consider real-data applications involving two signal tensors, $\mathcal{M}(0)$ and $\mathcal{M}(1)$, corresponding to the untreated and treated responses. Following the potential-outcomes framework (Rubin, 1974), the entries of the fully observed data tensor $\mathcal{Y}$ satisfy $\mathcal{Y}_{itj} = \Omega_{itj} \mathcal{Y}_{itj}(0) + (1 - \Omega_{itj}) \mathcal{Y}_{itj}(1)$, where $\mathcal{Y}(0)$ denotes the untreated potential outcome, which is observed on $\{(i, t, j) : \Omega_{itj} = 1\}$ and missing on the complementary set, and $\mathcal{Y}(1)$ denotes the treated potential outcome, which is observed only over $\{(i, t, j) : \Omega_{itj} = 0\}$. Here, we focus on estimating bilinear forms of $\mathcal{M}(0)$ and $\mathcal{M}(1)$ over the treated region. The latter problem is straightforward because $\mathcal{Y}(1)$ is observed on this region, so simple plug-in estimators can be used. By contrast, $\mathcal{Y}(0)$ is unobserved, hence bilinear functionals of $\mathcal{M}(0)$ require different approaches such as those introduced in the previous sections.

The empirical study uses the Castle Doctrine policy data from the PolicyEval repository, available on GitHub, a standard staggered-adoption benchmark in the difference-in-differences literature (e.g. Cheng and Hoekstra, 2013). After processing as described

in Appendix S2.3, we obtain tensors $\mathcal{Y}(0), \mathcal{Y}(1) \in \mathbb{R}^{50 \times 11 \times 4}$ for policy-off and policy-on potential outcomes. Here, $\mathcal{Y}(0)$ follows a staggered-adoption missingness pattern, while $\mathcal{Y}(1)$ is observed on the complementary policy-on region; see Fig. S3. Rows are U.S. states, columns are years 2000–2010, and slices are four logged crime rates. Importantly, because the tensor slices represent different outcomes rather than policy regimes, the staggered-adoption missingness pattern is shared across all layers of $\mathcal{Y}(0)$.

To assess the accuracy of our method, we conduct a masking experiment in which we artificially remove entries observed in the original data and use their known values to evaluate the resulting estimates. Starting from the original staggered observation pattern, we progressively introduce additional missingness in the target layer $k = 1$, corresponding to motor vehicle theft, while preserving the staggered structure by masking the last currently observed entry of selected rows. At each step, we estimate the average of the masked entries using the pooled estimator of Algorithm 2 and its matrix counterpart, as well as the nuclear-norm estimator of Choi and Yuan (2026) and its debiased counterpart; we will refer to the latter two as CY. All rank-dependent procedures are run with $r \in \{1, 2, 3\}$, with $\tau = 10^{-2}$ for the tensor and matrix methods. We repeat the procedure independently 10 times, using a different masking path for each replicate, all converging to a terminal mask that retains 9 fully observed rows and 4 fully observed columns.

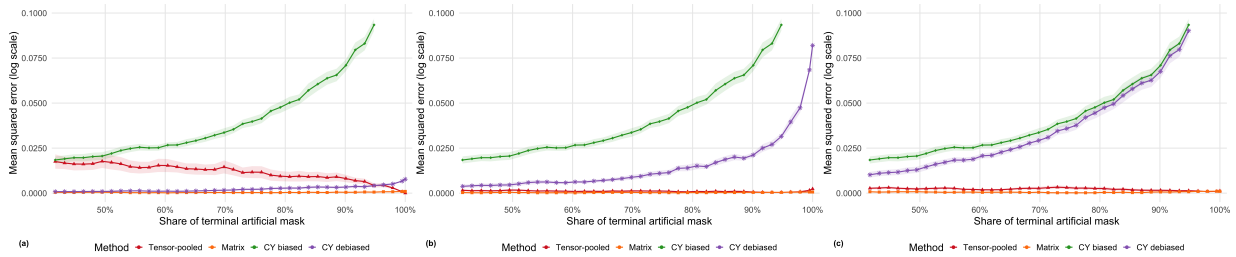


Figure 7: Castle masking experiment with additional missingness in the target layer $k = 1$. The panels correspond to ranks $r = 1, 2, 3$ used for the four methods, with $\tau = 0.01$ for the tensor and matrix estimators. The terminal mask retains 9 fully observed rows and 4 fully observed columns. The curves report the average squared error over 10 replicates, with one standard error. The percentage on the x-axis represents the percentage of the terminal mask reached.

Fig. 7 shows the squared errors averaged over 10 repetitions. In this setting, the two CY procedures perform worse than the others. One possible explanation is that these

procedures accommodate more general missingness patterns, whereas the tensor and matrix estimators can be favored by structured staggered designs that are close to a four-block.

To better compare these methods, in Fig. 8 we repeat the experiment under a more severe masking scheme, in which the terminal mask retains only 3 fully observed rows and 4 fully observed columns. For $r = 1$ and $r = 2$, the matrix estimator performs better when additional masking is limited. As masking becomes more severe, however, its error increases relative to that of the pooled estimator, which achieves lower error in the high-missingness regime. This is consistent with the target layer initially containing enough information to estimate its low-rank structure on its own, while, as more information is removed, the auxiliary layers become increasingly useful; similar qualitative behavior arises in the synthetic masking experiment of Appendix S2.1. For $r = 3$, the pooled estimator performs uniformly better than its matrix-only counterpart. Its performance is also comparable to that of the matrix-only estimators for $r = 1$ and $r = 2$, and may even be slightly better when the fraction of missing entries is large.

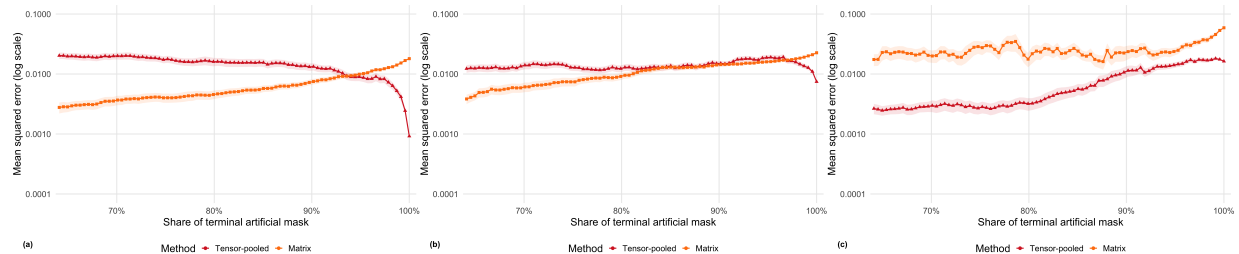


Figure 8: Castle masking experiment with additional missingness in the target layer $k = 1$. The panels correspond to ranks $r = 1, 2, 3$ used for the two methods; we again use $\tau = 0.01$. The terminal mask retains 3 fully observed rows and 4 fully observed columns. The curves report the average squared error over 30 replicates, with one standard error. The percentage on the x -axis represents the percentage of the terminal mask reached.

Furthermore, to examine patterns in motor vehicle theft, we consider several linear aggregate quantities over the policy-on region. We order the rows of the first layer so that the adoption pattern has staircase form, with $\Omega_{i,t,1} = \mathbb{1}\{(i,t) \in R_{a_1} \times C_{b_1} \text{ for } a,b \text{ satisfying } a + b \leq o_1 + 1\}$ for some integer $o_1 \geq 2$, and let $\mathcal{D}_1 := \{(a,b) : a + b > o_1 + 1\}$ denote the policy-on target blocks. For $c \in \{0,1\}$ and $h \in \{\text{AVG}, \text{ROWHET}, \text{LOCAL-}i_0, \text{TREND}\}$, we consider weighted aggregates of the

block-level bilinear forms $\Psi_c^{(h)}(1) := \{W_h(1)\}^{-1} \sum_{(a,b) \in \mathcal{D}_1} c_{ab}^{(h)} x_{a,h}^\top \mathcal{M}_{\bullet,\bullet,1}^{(a,b)}(c) y_{b,h}$. Formal definitions of the query directions, weights, and normalizations are given in Appendix S2.2. Informally, these are chosen so that $\Psi_c^{(\text{AVG})}(1)$ is the average potential outcome over all target entries, $\Psi_c^{(\text{ROWHET})}(1)$ is the corresponding signed contrast across rows, $\Psi_c^{(\text{LOCAL-}i_0)}(1)$ is the average potential outcome for a specified row i_0 over its target periods, and $\Psi_c^{(\text{TREND})}(1)$ captures the average slope of the row-averaged trajectory across blocks. We define the corresponding policy contrast as $\Delta^{(h)}(1) := \Psi_1^{(h)}(1) - \Psi_0^{(h)}(1)$. For the LOCAL- i_0 functional, we take i_0 to correspond to Florida, Montana, and Texas, while for ROWHET we set $\eta_i = +1$ for states that voted Republican in the 2000 presidential election and $\eta_i = -1$ for states that voted Democratic, so that this estimand measures a contrast between these two groups. We estimate $\Psi_0^{(h)}(1)$ using Algorithm S1, its matrix analogue, and the two CY plug-in estimators. Our procedure uses $r = 3$ and $\tau = 10^{-2}$, the matrix-only estimator uses $r = 1$ and $\tau = 10^{-2}$, while the debiased CY method uses $r = 1$; these choices are motivated by the preceding masking experiment. Because $\mathcal{Y}(1)$ is fully observed over $\mathcal{D}_1$, we estimate $\Psi_1^{(h)}(1)$ with a plug-in estimator. Finally, we estimate the policy contrast $\Delta^{(h)}(1)$ by subtracting the corresponding estimators.

Table 2: Estimates of $\Delta^{(h)}(1)$ using the pooled estimator of Algorithm S1, its matrix counterpart, and two plug-in approaches based on the nuclear-norm estimator of Choi and Yuan (2026) and its debiased counterpart.

	AVG	LOCAL-FLORIDA	LOCAL-MONTANA	LOCAL-TEXAS	ROWHET	TREND
$\hat{\Delta}^{(h)}$	0.1080	-0.2440	0.1430	0.1160	0.0992	-0.0184
$\hat{\Delta}_{\text{mat}}^{(h)}$	0.0746	-0.1070	0.2170	0.1210	0.0764	-0.0323
$\hat{\Delta}_{\text{CY,nuc}}^{(h)}$	0.1900	0.0239	0.3020	0.2370	0.1790	-0.0380
$\hat{\Delta}_{\text{CY,deb}}^{(h)}$	0.0924	-0.0832	0.2260	0.1380	0.0922	-0.0336

The estimates for $\Delta^{(h)}(1)$ in Table 2 are generally positive for the AVG, LOCAL, and ROWHET functionals, although their magnitudes vary across methods and states. In particular, the LOCAL-MONTANA and LOCAL-TEXAS estimates are positive across all four estimators, while the LOCAL-FLORIDA estimates are more heterogeneous. In contrast, the TREND estimates are consistently negative but small, ranging from -0.0380 to -0.0184 .

Taken together, the results point to an increase in motor vehicle theft following adoption that appears to diminish over time; this is broadly consistent with [Cheng and Hoekstra \(2013, Section 4.1\)](#). The results also underscore the importance of considering functionals beyond simple averages, particularly when treatment responses may vary substantially across states.

5.3 Real-data application: COVID-19 data

We construct a panel by merging policy-response indicators from the [Oxford COVID-19 Government Response Tracker](#) with epidemiological outcomes from the [Our World in Data COVID-19 dataset](#). We focus on stay-at-home requirements and contact tracing over the period from March 5, 2020, to April 5, 2020, and retain 18 countries, mostly European, where adoption of both policies was irreversible. This yields a tensor $\mathcal{Y} \in \mathbb{R}^{18 \times 32 \times 2}$, where the modes correspond to countries, days, and policy–outcome pairs. The first layer is associated with the 28-day delayed outcome `new_deaths_smoothed_per_million` and the stay-at-home policy, while the second is associated with the 28-day delayed outcome `new_cases_smoothed_per_million` and the contact-tracing policy. The 28-day delay allows time for policy effects to appear in reported outcomes. The pairing is also natural: stay-at-home requirements may affect downstream mortality, while contact tracing more directly targets contagion and reported cases. The outcome tensor is plotted in [Fig. 1](#). In each panel, blue entries indicate periods in which the policy is inactive, while red entries indicate periods in which it is active. From the perspective of the untreated potential outcomes, the blue entries are observed and the red entries are missing, with the resulting missingness presumably MNAR because policy adoption is more likely where disease burden is higher.

We first conduct a masking experiment similar to that in [Fig. 7](#), where the terminal mask retains 2 rows and 3 columns. The target layer is $k = 1$, corresponding to new deaths. [Fig. 9](#) shows that, at ranks 1 and 2, the two CY methods perform much better than in

the Castle data application, likely due to the less structured staggered missingness in this application. Since the dataset contains only two fully observed rows, the matrix and CY methods cannot accommodate higher ranks, whereas Algorithm 2 can. The plot on the right shows results for ranks $r = 3$ and $r = 4$; for these ranks, (A5) fails for some target blocks, so the corresponding results are not covered by Corollary 3. We nevertheless report them to examine the empirical behavior of the estimator beyond the assumptions required by our theory. Our method, particularly for $r = 4$, is substantially more stable as the percentage of missingness increases and achieves smaller errors than all competing methods, with the only exception being the matrix method with $r = 1$ between 60% and 90% missingness.

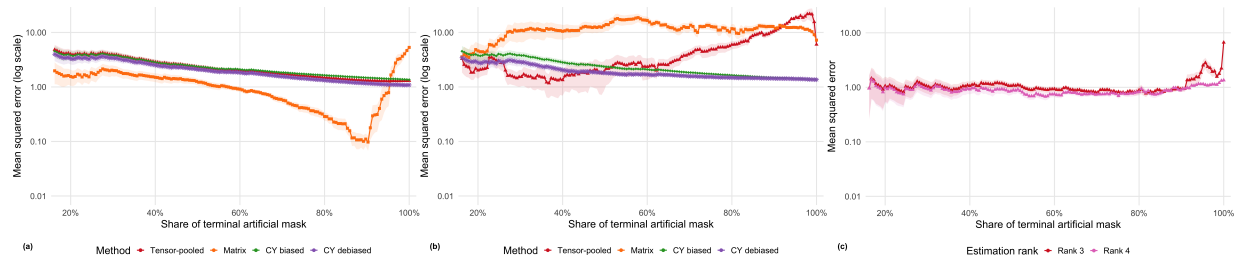


Figure 9: COVID-19 masking experiment with additional missingness in the target layer $k = 1$. Results are shown for ranks $r = 1, 2$ (left and center) and $r = 3, 4$ (right), with $\tau = 0.0001$. The terminal mask retains 2 fully observed rows and 3 fully observed columns. Curves show average squared error $\pm$ one standard error over 10 replicates (left and center) and 30 replicates (right). The x-axis gives the percentage of the terminal mask reached.

Furthermore, to explore the relationship between stay-at-home policies and new deaths, Table 3 reports estimates of $\Delta^{(h)}(1)$ for $h \in \{\text{AVG}, \text{TREND}\}$ across the feasible ranks for each method. Our pooled procedure is run with $r \in \{1, 2, 3, 4\}$, while the remaining methods can accommodate only $r \in \{1, 2\}$ in this application. As in the masking experiment, we also utilize $r = 3, 4$, which remain feasible for the SVD computations and can be competitive despite falling outside our current theory.

Across methods and ranks, the estimated trend contrast is generally close to zero, although there is some sensitivity to the choice of rank. The average contrast shows greater variation: our pooled estimator yields a negative estimate for $r \in \{2, 3, 4\}$, whereas the comparison methods always yield positive estimates. As a robustness check, since the cases and deaths layers may differ in scale and thereby affect estimation of the pooled subspaces,

we also run our estimator after rescaling the cases layer to match the root mean square of the deaths layer over observed policy-off entries; the resulting estimates are similar to the original ones. Under the maintained low-rank counterfactual model and the assumed interpretation of policy timing, these negative estimates from our procedure are consistent with a reduction in new deaths following stay-at-home requirements. The discrepancy in the estimated average contrast across methods may reflect the limited information in the target layer, where only Norway and Iceland are fully observed. By borrowing information on the latent unit and time factors from the second layer, our method may be better able to handle this setting, illustrating the potential benefit of exploiting information across layers.

r	$\hat{\Delta}^{(\text{AVG})}$	$\hat{\Delta}_{\text{STD}}^{(\text{AVG})}$	$\hat{\Delta}_{\text{mat}}^{(\text{AVG})}$	$\hat{\Delta}_{\text{CY,nuc}}^{(\text{AVG})}$	$\hat{\Delta}_{\text{CY,deb}}^{(\text{AVG})}$
1	1.525	1.302	0.428	1.485	1.338
2	-0.831	-0.519	0.121	1.485	1.291
3	-1.833	-1.993	—	—	—
4	-1.196	-0.750	—	—	—

r	$\hat{\Delta}^{(\text{TREND})}$	$\hat{\Delta}_{\text{STD}}^{(\text{TREND})}$	$\hat{\Delta}_{\text{mat}}^{(\text{TREND})}$	$\hat{\Delta}_{\text{CY,nuc}}^{(\text{TREND})}$	$\hat{\Delta}_{\text{CY,deb}}^{(\text{TREND})}$
1	-0.029	-0.020	-0.011	-0.038	-0.030
2	0.154	0.028	-0.049	-0.038	-0.031
3	0.004	0.051	—	—	—
4	-0.040	-0.038	—	—	—

Table 3: Estimates of $\Delta^{(h)}(1)$ for $h \in \{\text{AVG}, \text{TREND}\}$ using the pooled estimator of Algorithm S1, its matrix counterpart, and the nuclear-norm and debiased estimators of Choi and Yuan (2026), across the feasible ranks for each method. We also report results from Algorithm S1 after standardizing the cases layer relative to the deaths layer.

6 Disclosure statement

The authors report there are no competing interests to declare. The research of the second author was supported by European Research Council Starting Grant 101163546.

OpenAI GPT-5.6 Sol was used for language editing and improvements to clarity and presentation. All scientific content, arguments, results, and conclusions were developed and verified by the authors. The authors reviewed all AI-assisted modifications and take full responsibility for the accuracy and integrity of the manuscript.

SUPPLEMENTARY MATERIAL

Proofs and technical details are provided in the appendices. Code and dataset to replicate the paper’s figures and tables are included in the accompanying ZIP folder.

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Supplementary Material for “Estimation of Linear Functionals in Multilayer Panels under Staggered Adoption”

These Appendices are organized as follows. Appendix S1 contains the proofs of all results stated in the main text. Appendix S2 provides additional simulation studies and introduces some aggregate estimands of interest, together with a detailed discussion of procedures for estimating them. Appendix S3 discusses the computational cost of Algorithm 1. Appendix S4 develops key matrix-denoising results and the technical tools required to analyze Algorithm 1 in the four-block setting. Appendix S5 contains additional results on identifiability and tensor structure that are only briefly mentioned in the main text. Finally, Appendix S6 collects standard probabilistic results and tail bounds, together with background material on the Stiefel manifold and Haar measure.

S1 Proofs

S1.1 Proofs for Section 3.1

Proof of Theorem 1. The proof of this result relies on the matrix denoising theory developed in Appendix S4. Start by writing $\hat{\mu}_{xy}^{(k)} - \mu_{xy}^{(k)} = Z_{xy}^{(1)} + Z_{xy}^{(2)} + Z_{xy}^{(3)} + Z_{xy}^{(4)} + \Delta_{xy} =: Z_{xy} + \Delta_{xy}$, where

$$\begin{aligned} Z_{xy}^{(1)} &:= x^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{J}_k} y, \\ Z_{xy}^{(2)} &:= x^\top U_{2k} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{J}_k} y, \\ Z_{xy}^{(3)} &:= x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y, \\ Z_{xy}^{(4)} &:= x^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \mathcal{C}_{\bullet, \bullet, k} V_{2k}^\top y. \end{aligned}$$

Under the assumptions of Theorem 1, Lemma S13 gives $\mathbb{E}[Z_{xy}^2] \leq c_1 \Upsilon_{xy}$ for a sufficiently large constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\dim}, c_{\text{blk}}, \kappa, \nu_x, \nu_y) > 0$. Moreover, the same result ensures that there exists an event $\mathcal{G}_1$ with $\mathbb{P}(\mathcal{G}_1) \geq 1 - \mathcal{O}(p_N^{-10} + p_T^{-10})$ such that $\Delta_{xy}^2 \leq c_1 \Upsilon_{xy}$ under $\mathcal{G}_1$.

Possibly enlarging the constant c_1 and allowing it to change from line to line, and writing $\hat{\mu} \equiv \hat{\mu}_{xy}^{(k)}$, $\mu \equiv \mu_{xy}^{(k)}$ to simplify the notation, we then decompose the mean squared error as

$$\begin{aligned} \mathbb{E} \left[\left\{ \hat{\mu} - \mu \right\}^2 \right] &= \mathbb{E} \left[\left\{ \hat{\mu} - \mu \right\}^2 \mathbb{1}_{\mathcal{G}_1} \right] + \mathbb{E} \left[\left\{ \hat{\mu} - \mu \right\}^2 \mathbb{1}_{\mathcal{G}_1^c} \right] \\ &= \mathbb{E} \left[\left\{ Z_{xy} + \Delta_{xy} \right\}^2 \mathbb{1}_{\mathcal{G}_1} \right] + \mathbb{E} \left[\left\{ \hat{\mu} - \mu \right\}^2 \mathbb{1}_{\mathcal{G}_1^c} \right] \\ &\leq 2\mathbb{E} \left[Z_{xy}^2 \right] + 2\mathbb{E} \left[\Delta_{xy}^2 \mathbb{1}_{\mathcal{G}_1} \right] + 2\mathbb{E} \left[\hat{\mu}^2 \mathbb{1}_{\mathcal{G}_1^c} \right] + 2\mu^2 \mathbb{P}(\mathcal{G}_1^c) \\ &\leq 2\mathbb{E}[Z_{xy}^2] + 2c_1 \Upsilon_{xy} + 2\mathbb{E} \left[\hat{\mu}^2 \mathbb{1}_{\mathcal{G}_1^c} \right] + 2\mu^2 \mathbb{P}(\mathcal{G}_1^c) \\ &\leq c_1 \Upsilon_{xy} + 2\mathbb{E} \left[\hat{\mu}^2 \mathbb{1}_{\mathcal{G}_1^c} \right] + c_1 \mu^2 (p_N^{-10} + p_T^{-10}), \end{aligned}$$

where the previous decomposition and the bound on the second moment of Z_{xy} from Lemma S13 are used to control the contribution on the good event $\mathcal{G}_1$, while the probability bound for $\mathcal{G}_1^c$ is used in the last inequality. We next derive separate bounds for each of the remaining two terms. For the third term, we have $|\mu| \leq \|\mathcal{C}_{\bullet, \bullet, k}\|_{\text{op}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 \leq \gamma_{\max}$, which yields

$$\mu^2 (p_N^{-10} + p_T^{-10}) \leq \gamma_{\max}^2 (p_N^{-10} + p_T^{-10}) \leq c_1 \tau^{-1} \gamma_{\max}^2 (p_N^{-10} + p_T^{-10}) \frac{N_{1k}}{N}.$$

The last inequality follows from $\tau \leq c_\ell N_{1k} / (2N)$. For the second term, we start by noticing that $\left\| \hat{H}_{k, \tau}^{\text{inv}} \hat{U}_{1k}^\top \right\|_{\text{op}}^2 = \left\| \hat{H}_{k, \tau}^{\text{inv}} \hat{H}_k \hat{H}_{k, \tau}^{\text{inv}} \right\|_{\text{op}} = \max_{i \in [r]} \lambda_i / (\lambda_i \vee \tau)^2 \leq \tau^{-1}$. This allows showing

$$\left| \hat{\mu} \right| = \left| x^\top \hat{U}_{2k} \hat{H}_{k, \tau}^{\text{inv}} \hat{U}_{1k}^\top \hat{U}_{\text{up}}^{(k)} \hat{\Sigma}_{\text{up}} \hat{V}_{2k}^\top y \right| \leq \left\| \hat{U}_{2k}^\top x \right\|_2 \left\| \hat{H}_{k, \tau}^{\text{inv}} \hat{U}_{1k}^\top \right\|_{\text{op}} \left\| \hat{U}_{\text{up}}^{(k)} \hat{\Sigma}_{\text{up}} \right\|_{\text{op}} \left\| \hat{V}_{2k}^\top y \right\|_2$$

$$\begin{aligned}
&\leq \tau^{-1/2} \|(Y_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} \hat{V}_{\text{up}}\|_{\text{op}} \leq \tau^{-1/2} \|(Y_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet}\|_{\text{op}} = \tau^{-1/2} \|Y_{\text{up}}^{(k)}\|_{\text{op}} \\
&\leq \tau^{-1/2} \|M_{\text{up}}^{(k)}\|_{\text{op}} + \tau^{-1/2} \|E_{\text{up}}^{(k)}\|_{\text{op}} = \tau^{-1/2} \|U_{1k} \mathcal{C}_{\bullet, \bullet, k} V^{\top}\|_{\text{op}} + \tau^{-1/2} \|E_{\text{up}}^{(k)}\|_{\text{op}} \\
&\leq \tau^{-1/2} c_u^{1/2} \gamma_{\max} \sqrt{N_{1k}/N} + \tau^{-1/2} \|E_{\text{up}}^{(k)}\|_{\text{op}},
\end{aligned}$$

where the last inequality follows from (A1). This, together with an application of the Cauchy–Schwarz inequality and Lemma S20 with $p = 4$, gives

$$\begin{aligned}
\mathbb{E} [\hat{\mu}^2 \mathbb{1}_{\mathcal{G}_1^{\mathfrak{C}}}] &\leq 2\tau^{-1} c_u \gamma_{\max}^2 \frac{N_{1k}}{N} \mathbb{P}(\mathcal{G}_1^{\mathfrak{C}}) + 2\tau^{-1} \mathbb{E} \left[\|E_{\text{up}}^{(k)}\|_{\text{op}}^2 \mathbb{1}_{\mathcal{G}_1^{\mathfrak{C}}} \right] \\
&\leq c_1 \tau^{-1} \gamma_{\max}^2 \frac{N_{1k}}{N} (p_N^{-10} + p_T^{-10}) + c_1 \tau^{-1} (p_N^{-5} + p_T^{-5}) \mathbb{E}^{1/2} \left[\|E_{\text{up}}^{(k)}\|_{\text{op}}^4 \right] \\
&\leq c_1 \left\{ \tau^{-1} \gamma_{\max}^2 \frac{N_{1k}}{N} (p_N^{-10} + p_T^{-10}) + \tau^{-1} (p_N^{-5} + p_T^{-5}) \sigma^2 (N_{1k} + T) \right\}.
\end{aligned}$$

Combining the previous bounds and using $\tau^{-1} \gamma_{\max}^2 (p_N^{-10} + p_T^{-10}) \frac{N_{1k}}{N} + \tau^{-1} (p_N^{-5} + p_T^{-5}) (\sigma^2 N_{1k} + \sigma^2 T) \leq c_0 \Upsilon_{xy}$ concludes the proof. $\square$

S1.2 Proofs for Section 3.3

The proof of Theorem 2 follows by combining the next two results. The first addresses the large- K regime.

Theorem S1. Fix $k \in [K]$ and unit vectors $x \in \mathcal{S}^{N_{2k}-1}, y \in \mathcal{S}^{T_{2k}-1}$. Let $\mathcal{M}_0 = \mathcal{C}_0 \times_1 U_0 \times_2 V_0 \times_3 I_K \in \mathcal{F}(c_\ell, c_u)$. Assume that the k -th core matrix is separated from the boundary of the admissible singular-value interval, in the sense that $\delta_\gamma := \min\{\sigma_{\min}((\mathcal{C}_0)_{\bullet, \bullet, k}) - \gamma_{\min}, \gamma_{\max} - \sigma_{\max}((\mathcal{C}_0)_{\bullet, \bullet, k})\} > 0$. For $\varsigma > 0$ define $\mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma) := \{\mathcal{M} \in \mathcal{F}(c_\ell, c_u) : \|\mathcal{M} - \mathcal{M}_0\|_F \leq \varsigma\}$. There exists $c \equiv c(c_u) > 0$ such that

$$\begin{aligned}
&\inf_{\phi} \sup_{\mathcal{M} \in \mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma)} \mathbb{E}_{\mathcal{M}} \left[\{\phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M})\}^2 \right] \\
&\geq c \min \left\{ \sigma^2 \min \left(\frac{N}{N_{1k}}, \frac{T}{T_{1k}} \right), \varsigma^2, \delta_\gamma^2 \right\} \|U_{0,2k}^{\top} x\|_2^2 \|V_{0,2k}^{\top} y\|_2^2,
\end{aligned}$$

where the infimum is over all Borel-measurable functions ϕ of the observed entries Z_Ω .

Proof of Theorem S1. We prove the result by reducing the problem to a one-dimensional parametric submodel. To this end, we assume that $U_{0,2k}^\top x \neq 0$ and $V_{0,2k}^\top y \neq 0$; if either of these conditions fails, the desired lower bound is trivially satisfied. Let

$$F_k := (U_{0,2k}^\top x)(V_{0,2k}^\top y)^\top, \quad G_k := \frac{F_k}{\|F_k\|_F^2}.$$

Write $C_{0,j} := (\mathcal{C}_0)_{\bullet,\bullet,j}$ to simplify the notation. For $\theta \in \mathbb{R}$, define a perturbed core tensor $\mathcal{C}(\theta)$ by keeping all slices except the k -th one fixed, and setting $C_k(\theta) := C_{0,k} + \theta G_k$. We then set $\mathcal{M}(\theta) := \mathcal{C}(\theta) \times_1 U_0 \times_2 V_0 \times_3 I_K$. We next verify the range of θ for which the path remains in the local parameter space. Since U_0 and V_0 have orthonormal columns, we have

$$\|C_k(\theta) - C_{0,k}\|_{\text{op}} \leq \|C_k(\theta) - C_{0,k}\|_F = \frac{|\theta|}{\|F_k\|_F},$$

$$\|\mathcal{M}(\theta) - \mathcal{M}_0\|_F = \|U_0\{C_k(\theta) - C_{0,k}\}V_0^\top\|_F = \frac{|\theta|}{\|F_k\|_F}.$$

We thus deduce that $\{\mathcal{M}(\theta) : |\theta| \leq h\}$ is contained in $\{\mathcal{M} \in \mathcal{F}(c_\ell, c_u) : \|\mathcal{M} - \mathcal{M}_0\|_F \leq \varsigma\}$ whenever $h \leq \|F_k\|_F \min(\varsigma, \delta_\gamma)$. In particular, the singular values of $C_k(\theta)$ remain in $[\gamma_{\min}, \gamma_{\max}]$ by the operator-norm bound above and Weyl's inequality (Lemma S21). Assumption (A1) continues to hold along the path because U_0 and V_0 are fixed.

We next compute the induced change in the target functional. Since the bottom-right block of the k -th slice is $U_{0,2k}C_k(\theta)V_{0,2k}^\top$, we have

$$\mu(\theta) := \mu_{xy}^{(k)}(\mathcal{M}(\theta)) = x^\top U_{0,2k}C_k(\theta)V_{0,2k}^\top y = \mu_{xy}^{(k)}(\mathcal{M}_0) + \theta x^\top U_{0,2k}G_kV_{0,2k}^\top y = \mu_{xy}^{(k)}(\mathcal{M}_0) + \theta.$$

Hence estimating $\mu_{xy}^{(k)}(\mathcal{M}(\theta))$ along this path is equivalent to estimating the scalar parame-

ter θ , up to the known additive constant $\mu_{xy}^{(k)}(\mathcal{M}_0)$. It is also useful to compute the Fisher information for θ . Using the Gaussianity of the error and the four-block structure of $\Omega_{\bullet, \bullet, k}$, the Fisher information is constant and equal to

$$\begin{aligned}
\mathfrak{I}_\theta &= \sigma^{-2} \left\| P_{\Omega_{\bullet, \bullet, k}}(U_0 G_k V_0^\top) \right\|_F^2 \\
&= \sigma^{-2} \left\{ \|U_{0,1k} G_k V_{0,1k}^\top\|_F^2 + \|U_{0,1k} G_k V_{0,2k}^\top\|_F^2 + \|U_{0,2k} G_k V_{0,1k}^\top\|_F^2 \right\} \\
&= \sigma^{-2} \left\{ \|U_{0,1k} G_k\|_F^2 + \|G_k V_{0,1k}^\top\|_F^2 - \|U_{0,1k} G_k V_{0,1k}^\top\|_F^2 \right\} \\
&= \sigma^{-2} \|F_k\|_F^{-4} \left\{ \|U_{0,1k} F_k\|_F^2 + \|F_k V_{0,1k}^\top\|_F^2 - \|U_{0,1k} F_k V_{0,1k}^\top\|_F^2 \right\} \\
&\leq \sigma^{-2} \|F_k\|_F^{-4} \left\{ \|U_{0,1k} F_k\|_F^2 + \|F_k V_{0,1k}^\top\|_F^2 \right\} \\
&\leq \sigma^{-2} c_u \|F_k\|_F^{-2} \left\{ \frac{N_{1k}}{N} + \frac{T_{1k}}{T} \right\},
\end{aligned}$$

where the last inequality follows from Assumption (A1). Note that only the k -th slice contributes to $\mathfrak{I}_\theta$, since the submodel is fixed along all other slices.

We now apply the van Trees inequality (Gill and Levit, 1995, Equation 3) on the interval $[-h, h]$. Let p be an arbitrary absolutely continuous prior density on $[-h, h]$ satisfying $p(-h) = p(h) = 0$. For any estimator $\phi(Z_\Omega)$, define $\hat{\theta} := \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}_0)$. Since $\mu(\theta) = \mu_{xy}^{(k)}(\mathcal{M}_0) + \theta$, the risk for estimating the functional is exactly the risk for estimating θ along the submodel. Writing $J(p) := \int_{-h}^h \frac{\{p'(\theta)\}^2}{p(\theta)} d\theta$ and choosing $h = \|F_k\|_F \min(\varsigma, \delta_\gamma)$, the van Trees inequality yields

$$\begin{aligned}
\sup_{|\theta| \leq h} \mathbb{E}_\theta \left[\{\phi(Z_\Omega) - \mu(\theta)\}^2 \right] &\geq \int_{-h}^h \mathbb{E}_\theta[(\hat{\theta} - \theta)^2] p(\theta) d\theta \\
&\geq \frac{1}{\int_{-h}^h \mathfrak{I}_\theta p(\theta) d\theta + \inf_{p: p(\pm h)=0} J(p)} = \frac{1}{\int_{-h}^h \mathfrak{I}_\theta p(\theta) d\theta + \pi^2/h^2} \\
&\geq \left\{ \sigma^{-2} c_u \|F_k\|_F^{-2} \left(\frac{N_{1k}}{N} + \frac{T_{1k}}{T} \right) + \pi^2 \|F_k\|_F^{-2} \min^{-2}(\varsigma, \delta_\gamma) \right\}^{-1} \\
&= \|F_k\|_F^2 \left\{ \sigma^{-2} c_u \left(\frac{N_{1k}}{N} + \frac{T_{1k}}{T} \right) + \pi^2 \min^{-2}(\varsigma, \delta_\gamma) \right\}^{-1}
\end{aligned}$$

$$\begin{aligned}
&\geq \|F_k\|_F^2 \left\{ 2c_u \sigma^{-2} \max\left(\frac{N_{1k}}{N}, \frac{T_{1k}}{T}\right) + \pi^2 \min^{-2}(\varsigma, \delta_\gamma) \right\}^{-1} \\
&\geq \frac{1}{2} \min \left\{ \frac{\sigma^2}{2c_u} \min\left(\frac{N}{N_{1k}}, \frac{T}{T_{1k}}\right), \pi^{-2} \min^2(\varsigma, \delta_\gamma) \right\} \|F_k\|_F^2 \\
&= \min \left\{ \frac{\sigma^2}{4c_u} \min\left(\frac{N}{N_{1k}}, \frac{T}{T_{1k}}\right), \frac{\varsigma^2}{2\pi^2}, \frac{\delta_\gamma^2}{2\pi^2} \right\} \|F_k\|_F^2 \\
&\geq \min \left\{ \frac{1}{4c_u}, \frac{1}{2\pi^2} \right\} \min \left\{ \sigma^2 \min\left(\frac{N}{N_{1k}}, \frac{T}{T_{1k}}\right), \varsigma^2, \delta_\gamma^2 \right\} \|F_k\|_F^2 \\
&= \min \left\{ \frac{1}{4c_u}, \frac{1}{2\pi^2} \right\} \min \left\{ \sigma^2 \min\left(\frac{N}{N_{1k}}, \frac{T}{T_{1k}}\right), \varsigma^2, \delta_\gamma^2 \right\} \|U_{0,2k}^\top x\|_2^2 \|V_{0,2k}^\top y\|_2^2,
\end{aligned}$$

where the first equality follows from the fact that $J(p)$ is minimized by $p(\theta) = h^{-1} \cos^2(\pi \theta / 2h)$, while the successive inequality follows from our previous bound on $\mathfrak{J}_\theta$. Since the path is contained in the local parameter space, this lower bound also applies to the local minimax risk over $\mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma)$. This concludes the proof. $\square$

For the second result, which concerns the small- K regime, we suppose for simplicity that $N_{1j} = N_1$ and $T_{1j} = T_1$ for all $j \in [K]$. Under this assumption, $U_{0,1j_1} = U_{0,1j_2}$ and $U_{0,2j_1} = U_{0,2j_2}$ for all $j_1, j_2 \in [K]$. As in the main body, we denote these common matrices by $U_{0,1}$ and $U_{0,2}$, respectively, and adopt the analogous convention for $V_{0,1}$ and $V_{0,2}$. We define the projections $P_{V_{0,2}} := V_{0,2} (V_{0,2}^\top V_{0,2})^{-1} V_{0,2}^\top$, $P_{U_{0,2}} := U_{0,2} (U_{0,2}^\top U_{0,2})^{-1} U_{0,2}^\top$, $P_{V_{0,2}}^\perp := I_{T_2} - P_{V_{0,2}}$, and $P_{U_{0,2}}^\perp := I_{N_2} - P_{U_{0,2}}$. The inverses are well defined when (A1) holds with $c_u \max(\frac{N_1}{N}, \frac{T_1}{T}) < 1$. We also introduce $\omega_V := \|P_{V_{0,2}}^\perp y\|_2$, $\omega_U := \|P_{U_{0,2}}^\perp x\|_2$, which measure the components of y and x orthogonal to the column spaces of $V_{0,2}$ and $U_{0,2}$.

Theorem S2. Fix $k \in [K]$ and unit vectors $x \in \mathcal{S}^{N_{2k}-1}, y \in \mathcal{S}^{T_{2k}-1}$. Set $N_{1j} = N_1$ and $T_{1j} = T_1$ for all $j \in [K]$. Let $\mathcal{M}_0 = \mathcal{C}_0 \times_1 U_0 \times_2 V_0 \times_3 I_K \in \mathcal{F}(c_\ell, c_u)$, and suppose that (A1) holds with margin $0 < \delta_{A1} < (c_u - c_\ell)/2$, in the sense that

$$(c_\ell + \delta_{A1}) \frac{N_1}{N} I_r \preceq U_{0,1}^\top U_{0,1} \preceq (c_u - \delta_{A1}) \frac{N_1}{N} I_r,$$

$$(c_\ell + \delta_{A1}) \frac{T_1}{T} I_r \preceq V_{0,1}^\top V_{0,1} \preceq (c_u - \delta_{A1}) \frac{T_1}{T} I_r.$$

Also assume that $c_u \max(\frac{N_1}{N}, \frac{T_1}{T}) < 1$. There exists $c \equiv c(c_u, \gamma_{\min}, \gamma_{\max}) > 0$ such that

$$\begin{aligned} \inf_{\phi} \sup_{\mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma)} \mathbb{E}_{\mathcal{M}} \left[\left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}) \right\}^2 \right] &\geq c \omega_V^2 \min \left(\frac{\sigma^2 N}{K N_1}, \varepsilon_V^2 \right) \|U_{0,2}^\top x\|_2^2 \\ &\quad + c \omega_U^2 \min \left(\frac{\sigma^2 T}{K T_1}, \varepsilon_U^2 \right) \|V_{0,2}^\top y\|_2^2, \end{aligned}$$

where $\varepsilon_U := \min(\omega_U/2, \sqrt{T_1/T}, \varsigma \gamma_{\max}^{-1} K^{-1/2}, \sqrt{\delta_{A1}/c_\ell})$, $\varepsilon_V := \min(\omega_V/2, \sqrt{N_1/N}, \varsigma \gamma_{\max}^{-1} K^{-1/2}, \sqrt{\delta_{A1}/c_\ell})$, and the infimum is over all Borel-measurable functions ϕ of Z_Ω .

Proof of Theorem S2. We prove the first term in the lower bound by reducing the problem to a one-dimensional parametric submodel where only V_0 is perturbed. The second follows by the symmetric argument in which U_0 is perturbed instead of V_0 . To this end, we will assume that $U_{0,2}^\top x \neq 0$ and $\omega_V > 0$; if either of these conditions is not met, the desired lower bound is trivially satisfied. Let $C_{0,j} := (\mathcal{C}_0)_{\bullet, \bullet, j}$ satisfying $\gamma_{\min} \leq \sigma_r(C_{0,j}) \leq \sigma_1(C_{0,j}) \leq \gamma_{\max}$ for all $j \in [K]$. Let $G_V := C_{0,k}^\top U_{0,2}^\top x y^\top P_{V_{0,2}}^\perp \neq 0$ since $U_{0,2}^\top x \neq 0$, $\sigma_r(C_{0,k}) \geq \gamma_{\min} > 0$, and $\omega_V > 0$. Since G_V is rank one, write $G_V = dab^\top$, where

$$d := \|G_V\|_F = \omega_V \|C_{0,k}^\top U_{0,2}^\top x\|_2, \quad a := \frac{C_{0,k}^\top U_{0,2}^\top x}{\|C_{0,k}^\top U_{0,2}^\top x\|_2}, \quad b := \frac{P_{V_{0,2}}^\perp y}{\omega_V}.$$

Then $\|a\|_2 = \|b\|_2 = 1$ and $G_V b = da$. By definition of b we also have $b \in \text{Im}(P_{V_{0,2}}^\perp)$, hence $V_{0,2}^\top b = 0$ and $P_{V_{0,2}}^\perp b = b$. Setting $w := (\mathbf{0}_{T_1}^\top; b^\top)^\top \in \mathbb{R}^T$, we obtain a vector supported only on the last T_2 entries that is orthogonal to V_0 , meaning that $V_0^\top w = 0$.

We now introduce a one-dimensional submodel by perturbing V_0 only. For $\theta \in \mathbb{R}$, define $V(\theta) := (V_0 + \theta w a^\top)(I_r + \theta^2 a a^\top)^{-1/2} = (V_0 + \theta w a^\top)(I_r + [\{1 + \theta^2\}^{-1/2} - 1] a a^\top)$. We have

$$V(\theta)^\top V(\theta) = I_r \text{ and}$$

$$V'(\theta) = (1 + \theta^2)^{-3/2} (w - \theta V_0 a) a^\top.$$

We leave U_0 and $\mathcal{C}_0$ untouched, and set $\mathcal{M}(\theta) := \mathcal{C}_0 \times_1 U_0 \times_2 V(\theta) \times_3 I_K$. We next verify the range of θ for which the path $\{\mathcal{M}(\theta) : |\theta| \leq h\}$ remains in the local parameter space. As $\mathcal{C}_0$ and U_0 are fixed, the only conditions to check are the Frobenius-norm bound, and assumption (A1) for $V(\theta)$. As for the former, since $U_0^\top U_0 = I_r$, $V_0^\top w = 0$, $w^\top w = 1$, $a^\top a = 1$, we have

$$\begin{aligned} \|\mathcal{M}(\theta) - \mathcal{M}_0\|_F^2 &= \sum_{j=1}^K \left\| U_0 C_{0,j} \{V(\theta) - V_0\}^\top \right\|_F^2 \leq \gamma_{\max}^2 \sum_{j=1}^K \|V(\theta) - V_0\|_F^2 \\ &= \gamma_{\max}^2 K \|V(\theta) - V_0\|_F^2 = 2 \gamma_{\max}^2 K \left\{ 1 - (1 + \theta^2)^{-1/2} \right\} \leq \gamma_{\max}^2 K \theta^2 \leq \gamma_{\max}^2 K h^2, \end{aligned}$$

where in the penultimate inequality we used the standard bound $2 \{1 - (1 + x)^{-1/2}\} \leq x$ for $x \geq 0$. This is upper bounded by ς^2 for $h \leq \varsigma \gamma_{\max}^{-1} K^{-1/2}$. As for (A1), start by observing that $(I_r + \theta^2 a a^\top)^{-1/2}$ is symmetric and has eigenvalues bounded between $(1 + \theta^2)^{-1/2}$ and 1. As a result, for $V_1(\theta) := V(\theta)_{[T_1], \bullet} = V_{0,1} (I_r + \theta^2 a a^\top)^{-1/2}$ we have

$$\begin{aligned} V_1(\theta)^\top V_1(\theta) &= \{I_r + \theta^2 a a^\top\}^{-1/2} V_{0,1}^\top V_{0,1} \{I_r + \theta^2 a a^\top\}^{-1/2} \\ &\preceq (c_u - \delta_{A1}) \frac{T_1}{T} \{I_r + \theta^2 a a^\top\}^{-1} \preceq (c_u - \delta_{A1}) \frac{T_1}{T} I_r \preceq c_u \frac{T_1}{T} I_r. \end{aligned}$$

Similarly,

$$V_1(\theta)^\top V_1(\theta) \succeq (c_\ell + \delta_{A1}) \frac{T_1}{T} \{I_r + \theta^2 a a^\top\}^{-1} \succeq \frac{c_\ell + \delta_{A1}}{1 + \theta^2} \frac{T_1}{T} I_r,$$

which is lower bounded by $c_\ell T_1 T^{-1} I_r$ whenever $h^2 \leq \delta_{A1} c_\ell^{-1}$. We thus deduce that the path $\{\mathcal{M}(\theta) : |\theta| \leq h\}$ is contained in $\{\mathcal{M} \in \mathcal{F}(c_\ell, c_u) : \|\mathcal{M} - \mathcal{M}_0\|_F \leq \varsigma\}$ whenever $h \leq$

$\min(\varsigma \gamma_{\max}^{-1} K^{-1/2}, \delta_{A1}^{1/2} c_\ell^{-1/2})$. In order to simplify the computations below while remaining in the Frobenius neighborhood of $\mathcal{M}_0$, we will set $h = \varepsilon_V$.

We next compute the induced change in $\mu(\theta) := \mu_{xy}^{(k)}(\mathcal{M}(\theta)) = x^\top U_{0,2} C_{0,k} V_2(\theta)^\top y$. Using $G_V = C_{0,k}^\top U_{0,2}^\top x y^\top P_{V_{0,2}}^\perp = dab^\top$, $P_{V_{0,2}}^\perp b = b$, $(x^\top U_{0,2} C_{0,k} a)(b^\top y) = d$, and the expression for $V'(\theta)$ derived above, we have

$$\begin{aligned}
\mu'(\theta) &= x^\top U_{0,2} C_{0,k} V_2'(\theta)^\top y \\
&= (1 + \theta^2)^{-3/2} x^\top U_{0,2} C_{0,k} a (b - \theta V_{0,2} a)^\top y \\
&= d(1 + \theta^2)^{-3/2} \left\{ 1 - \theta d^{-1} a^\top (C_{0,k}^\top U_{0,2}^\top x y^\top V_{0,2}) a \right\} \\
&\geq d(1 + \theta^2)^{-3/2} \left\{ 1 - |\theta| d^{-1} |a^\top (C_{0,k}^\top U_{0,2}^\top x y^\top V_{0,2}) a| \right\} \\
&\geq d(1 + h^2)^{-3/2} (1 - h d^{-1} \|C_{0,k}^\top U_{0,2}^\top x y^\top V_{0,2}\|_{\text{op}}) \\
&\geq d(1 + h^2)^{-3/2} (1 - h \omega_V^{-1}) \geq 2^{-5/2} d \\
&= 2^{-5/2} \omega_V \|C_{0,k}^\top U_{0,2}^\top x\|_2 \geq 2^{-5/2} \gamma_{\min} \omega_V \|U_{0,2}^\top x\|_2 > 0,
\end{aligned}$$

where in the penultimate inequality we used $h \leq \min(\sqrt{N_1/N}, \omega_V/2) \leq \min(1, \omega_V/2)$.

Coming now to the Fisher information for this model, it is useful to compute $\|V_2'(\theta)\|_F^2 \leq \|V'(\theta)\|_F^2 = (1 + \theta^2)^{-2} \leq 1$. Similarly, we can show that $\|V_1'(\theta)\|_F^2 = \theta^2(1 + \theta^2)^{-3} \|V_{0,1} a\|_2^2 \leq \theta^2(c_u - \delta_{A1}) \frac{T_1}{T} (1 + \theta^2)^{-3}$. We thus have

$$\begin{aligned}
\mathfrak{I}_\theta &= \sigma^{-2} \sum_{j=1}^K \|P_{\Omega_{\bullet, \bullet, j}}(U_0 C_{0,j} [V'(\theta)]^\top)\|_F^2 \\
&= \sigma^{-2} \sum_{j=1}^K \left\{ \|V_1'(\theta) C_{0,j}^\top U_0^\top\|_F^2 + \|V_2'(\theta) C_{0,j}^\top U_{0,1}^\top\|_F^2 \right\} \\
&\leq \gamma_{\max}^2 \sigma^{-2} K \left\{ \|V_1'(\theta)\|_F^2 + (c_u - \delta_{A1}) \frac{N_1}{N} \|V_2'(\theta)\|_F^2 \right\} \\
&\leq \gamma_{\max}^2 \sigma^{-2} K \left\{ \|V_1'(\theta)\|_F^2 + (c_u - \delta_{A1}) \frac{N_1}{N} \right\} \\
&\leq \gamma_{\max}^2 \sigma^{-2} K (c_u - \delta_{A1}) \left\{ \theta^2(1 + \theta^2)^{-3} \frac{T_1}{T} + \frac{N_1}{N} \right\}
\end{aligned}$$

$$\leq \gamma_{\max}^2 \sigma^{-2} K (c_u - \delta_{A1}) \left\{ h^2 + \frac{N_1}{N} \right\} \leq 2 \gamma_{\max}^2 \sigma^{-2} (c_u - \delta_{A1}) \frac{K N_1}{N},$$

where in the last inequality we used $h^2 \leq N_1/N$.

We now apply the van Trees inequality (Gill and Levit, 1995, Equation 4) on the interval $[-h, h]$ with $h = \varepsilon_V$. Let p be an arbitrary absolutely continuous prior density on $[-h, h]$ satisfying $p(-h) = p(h) = 0$. Writing $J(p) := \int_{-h}^h \frac{\{p'(\theta)\}^2}{p(\theta)} d\theta$, and setting $c_V := 2^{-6} \gamma_{\min}^2 \min \{2^{-1} \gamma_{\max}^{-2} c_u^{-1}, \pi^{-2}\}$, the van Trees inequality yields

$$\begin{aligned} \sup_{|\theta| \leq h} \mathbb{E}_\theta \left[\{\phi(Z_\Omega) - \mu(\theta)\}^2 \right] &\geq \int_{-h}^h \mathbb{E}_\theta \left[\{\hat{\mu} - \mu(\theta)\}^2 \right] p(\theta) d\theta \\ &\geq \frac{\left\{ \int_{-h}^h \mu'(\theta) p(\theta) d\theta \right\}^2}{\int_{-h}^h \mathfrak{I}_\theta p(\theta) d\theta + \inf_{p: p(\pm h)=0} J(p)} \\ &= \frac{\left\{ \int_{-h}^h \mu'(\theta) p(\theta) d\theta \right\}^2}{\int_{-h}^h \mathfrak{I}_\theta p(\theta) d\theta + \pi^2/h^2} \\ &\geq \frac{2^{-5} \gamma_{\min}^2 \omega_V^2 \|U_{0,2}^\top x\|_2^2}{2 \gamma_{\max}^2 \sigma^{-2} (c_u - \delta_{A1}) K N_1 / N + \pi^2/h^2} \\ &= \frac{2^{-5} \gamma_{\min}^2 \omega_V^2 \|U_{0,2}^\top x\|_2^2}{2 \gamma_{\max}^2 (c_u - \delta_{A1}) \sigma^{-2} K N_1 / N + \pi^2 h^{-2}} \\ &\geq \frac{2^{-5} \gamma_{\min}^2 \omega_V^2 \|U_{0,2}^\top x\|_2^2}{2 \max (2 \gamma_{\max}^2 (c_u - \delta_{A1}) \sigma^{-2} K N_1 / N, \pi^2 h^{-2})} \\ &= 2^{-6} \gamma_{\min}^2 \omega_V^2 \|U_{0,2}^\top x\|_2^2 \min \left(\frac{\sigma^2 N}{2 \gamma_{\max}^2 (c_u - \delta_{A1}) K N_1}, \frac{h^2}{\pi^2} \right) \\ &\geq 2^{-6} \gamma_{\min}^2 \min \left\{ \frac{1}{2 \gamma_{\max}^2 (c_u - \delta_{A1})}, \frac{1}{\pi^2} \right\} \omega_V^2 \min \left(\frac{\sigma^2 N}{K N_1}, h^2 \right) \|U_{0,2}^\top x\|_2^2 \\ &\geq c_V \omega_V^2 \min \left(\frac{\sigma^2 N}{K N_1}, \varepsilon_V^2 \right) \|U_{0,2}^\top x\|_2^2, \end{aligned}$$

where the first equality follows from the fact that $J(p)$ is minimized by $p(\theta) = h^{-1} \cos^2(\pi \theta / 2h)$, while the successive inequalities follow from our previous bounds on $\mu'(\theta)$ and $\mathfrak{I}_\theta$. Since the path is contained in the local parameter space, this lower bound also applies to the local minimax risk over $\mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma)$.

Interchanging the roles of U_0 and V_0 and applying the same argument to $P_{U_0,2}^\perp x y^\top V_{0,2} C_{0,k}^\top$, yields the second term in the lower bound. In particular, we will assume that $V_{0,2}^\top y \neq 0$ and $\omega_U > 0$; if either of these conditions is not met, the desired lower bound is trivially satisfied. We then define $G_U := P_{U_0,2}^\perp x y^\top V_{0,2} C_{0,k}^\top$. Since $y^\top V_{0,2} C_{0,k}^\top \neq 0$ and $\omega_U = \|P_{U_0,2}^\perp x\|_2 > 0$, this matrix is rank one. Writing $G_U = d_U a_U b_U^\top$, where

$$d_U = \omega_U \|C_{0,k} V_{0,2}^\top y\|_2, \quad a_U = \frac{P_{U_0,2}^\perp x}{\omega_U}, \quad b_U = \frac{C_{0,k} V_{0,2}^\top y}{\|C_{0,k} V_{0,2}^\top y\|_2},$$

and perturbing U_0 along $(\mathbf{0}_{N_1}^\top; a_U^\top)^\top$ gives an analogous one-dimensional path $U(\theta)$ with V_0 and C_0 fixed. The same calculations, with N_1/N and T_1/T interchanged, yield

$$\inf_{\phi} \sup_{\mathcal{M} \in \mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma)} \mathbb{E}_{\mathcal{M}} [\{\phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M})\}^2] \geq c_V \omega_U^2 \min\left(\frac{\sigma^2 T}{K T_1}, \varepsilon_U^2\right) \|V_{0,2}^\top y\|_2^2.$$

Combining this with the lower bound obtained from the V_0 -perturbation, and using that the maximum of two lower bounds is at least their average, gives

$$\begin{aligned} \inf_{\phi} \sup_{\mathcal{M} \in \mathcal{F}_{\text{loc}}(\mathcal{M}_0, \varsigma)} \mathbb{E}_{\mathcal{M}} [\{\phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M})\}^2] &\geq \frac{c_V}{2} \omega_V^2 \min\left(\frac{\sigma^2 N}{K N_1}, \varepsilon_V^2\right) \|U_{0,2}^\top x\|_2^2 \\ &\quad + \frac{c_V}{2} \omega_U^2 \min\left(\frac{\sigma^2 T}{K T_1}, \varepsilon_U^2\right) \|V_{0,2}^\top y\|_2^2. \end{aligned}$$

Setting $c = c_V/2$ yields the desired bound. $\square$

For any $\mathcal{M}_0$ covered by both sets of assumptions, the combined results of Theorems [S1](#) and [S2](#) immediately imply the lower bound given in Theorem [2](#) by taking the maximum of the respective right-hand sides.

In particular, the assumptions $c_u \max(\frac{N_1}{N}, \frac{T_1}{T}) \leq 1 - \delta_{\text{sep}}$ and $\delta_{\text{sep}}^{-1} \max(\|U_{0,2}^\top x\|_2^2, \|V_{0,2}^\top y\|_2^2) \leq 1/2$ allow simplifying the statement to avoid the presence of ω_U^2 and ω_V^2 . To see why,

start by noticing that $(1 - c_u \frac{N_1}{N})(1 - \omega_U^2) \leq \sigma_{\min}^2(U_{0,2})(1 - \omega_U^2) \leq \|U_{0,2}^\top x\|_2^2 \leq \sigma_{\max}^2(U_{0,2})(1 - \omega_U^2) \leq 1 - \omega_U^2$. An analogous statement holds for ω_V and $\|V_{0,2}^\top y\|_2$. Under $c_u \max(\frac{N_1}{N}, \frac{T_1}{T}) \leq 1 - \delta_{\text{sep}}$, we thus get $\|U_{0,2}^\top x\|_2^2 \geq \delta_{\text{sep}}(1 - \omega_U^2)$ and $\|V_{0,2}^\top y\|_2^2 \geq \delta_{\text{sep}}(1 - \omega_V^2)$. Finally, $\delta_{\text{sep}}^{-1} \max(\|U_{0,2}^\top x\|_2^2, \|V_{0,2}^\top y\|_2^2) \leq 1/2$ implies $\min(\omega_U^2, \omega_V^2) \geq 1/2$, allowing us to remove the dependence on ω_U and ω_V both from the prefactor multiplying the constant c and from the quantities $\varepsilon_U, \varepsilon_V$ appearing inside the minima.

S1.3 Proofs for Section 4

Proof of Corollary 3. We verify that the auxiliary problem obtained by restricting to $S \times Q$ satisfies the hypotheses of Theorem 1. Although, for $j \neq k$, the missingness masks $\Omega_{S,Q,j}$ are not necessarily in four-block form, this is not an essential requirement. What is needed in order to apply Theorem 1, and in particular Lemma S12 in Appendix S4, is the relevant subblock conditioning assumption for the rows and columns corresponding to fully observed rows and columns, respectively. These are exactly the submatrices that enter the definitions of the upper and left pooled matrices, and they are what enable improved estimation of the shared subspaces.

We start by verifying the analogue of (A1). Let $G_U := U_S^\top U_S$ and $G_V := V_Q^\top V_Q$. By Assumption (A5) we have

$$c_\ell \frac{\mathbf{n}}{N} I_r \preceq G_U \preceq c_u \frac{\mathbf{n}}{N} I_r, \quad c_\ell \frac{\mathbf{t}}{T} I_r \preceq G_V \preceq c_u \frac{\mathbf{t}}{T} I_r,$$

hence G_U and G_V are nonsingular. Letting $\tilde{U} := U_S G_U^{-1/2}$, $\tilde{V} := V_Q G_V^{-1/2}$ and $\tilde{\mathcal{C}}_{\bullet,\bullet,j} := G_U^{1/2} \mathcal{C}_{\bullet,\bullet,j} G_V^{1/2}$, we have that $\tilde{U}^\top \tilde{U} = I_r$, $\tilde{V}^\top \tilde{V} = I_r$ and $\mathcal{M}_{S,Q,j} = U_S \mathcal{C}_{\bullet,\bullet,j} V_Q^\top = \tilde{U} \tilde{\mathcal{C}}_{\bullet,\bullet,j} \tilde{V}^\top$. This shows that the restriction of the signal to $S \times Q$ admits an orthonormal Tucker2 representation with row and column dimensions $\mathbf{n}$ and $\mathbf{t}$, respectively.

We next check that the restricted Gram matrices are well conditioned. For the target layer, we have $\tilde{U}_{S^+}^\top \tilde{U}_{S^+} = G_U^{-1/2} U_{S^+}^\top U_{S^+} G_U^{-1/2}$, hence, using (A5) and the preceding bounds on G_U gives

$$\frac{c_\ell}{c_u} \frac{\mathbf{n}_{1k}}{\mathbf{n}} I_r \preceq \tilde{U}_{S^+}^\top \tilde{U}_{S^+} \preceq \frac{c_u}{c_\ell} \frac{\mathbf{n}_{1k}}{\mathbf{n}} I_r.$$

Similarly, for each $j \neq k$ we have $\tilde{U}_{\text{RowAnc}_k(j)}^\top \tilde{U}_{\text{RowAnc}_k(j)} = G_U^{-1/2} U_{\text{RowAnc}_k(j)}^\top U_{\text{RowAnc}_k(j)} G_U^{-1/2}$, thus

$$\frac{c_\ell}{c_u} \frac{\mathbf{n}_{1j}}{\mathbf{n}} I_r \preceq \tilde{U}_{\text{RowAnc}_k(j)}^\top \tilde{U}_{\text{RowAnc}_k(j)} \preceq \frac{c_u}{c_\ell} \frac{\mathbf{n}_{1j}}{\mathbf{n}} I_r.$$

Since the same argument applies to the column factors, we can conclude that the auxiliary sub-block conditioning assumption holds with $c_\ell/c_u, c_u/c_\ell$ in place of c_ℓ, c_u , respectively.

It remains to identify the signal strengths in the auxiliary model. Since $\tilde{\mathcal{C}}_{\bullet, \bullet, j} = G_U^{1/2} \mathcal{C}_{\bullet, \bullet, j} G_V^{1/2}$, standard bounds on the singular values of a matrix product yield

$$\tilde{\gamma}_{\min} := c_\ell \gamma_{\min} \sqrt{\frac{\mathbf{n}\mathbf{t}}{NT}} \leq \sigma_{\min}(\tilde{C}_j) \leq \sigma_{\max}(\tilde{C}_j) \leq c_u \gamma_{\max} \sqrt{\frac{\mathbf{n}\mathbf{t}}{NT}} =: \tilde{\gamma}_{\max},$$

thereby showing that the effective lower and upper signals are $\tilde{\gamma}_{\min}$ and $\tilde{\gamma}_{\max}$.

The noise distribution is unchanged by restriction since, on the observed auxiliary coordinates, the errors are still independent centered Gaussian variables with variance σ^2 . Moreover, Assumption (A6) is the analogue of (A2), (A3) and (A4) after replacing $N, T, N_{1k}, T_{1k}, \rho_N, \rho_T, p_N, p_T, \zeta_N, \zeta_T, \gamma_{\min}, \gamma_{\max}$ by $\mathbf{n}, \mathbf{t}, \mathbf{n}_{1k}, \mathbf{t}_{1k}, \rho_{\mathbf{n}}, \rho_{\mathbf{t}}, p_{\mathbf{n}}, p_{\mathbf{t}}, \zeta_{\mathbf{n}}, \zeta_{\mathbf{t}}, \tilde{\gamma}_{\min}, \tilde{\gamma}_{\max}$. Also, (6) is the analogue of (4), again with the same substitution of auxiliary dimensions and signal strengths. We can thus conclude that all hypotheses of Theorem 1 hold for the auxiliary problem, hence applying this result gives $\mathbb{E}_{\mathcal{M}}[\{\hat{\mu}_{xy}^{(k,a,b)} - \mu_{xy}^{(k,a,b)}\}^2] \leq c_1 \tilde{\Upsilon}_{xy}$. This concludes the proof. $\square$

S2 Additional details on the numerical results

S2.1 Additional simulation studies

In this section we complement the empirical results on the masking experiments in Sections 5.2-5.3 by running similar masking experiments on synthetic data. In particular, this gives us an idea of what to expect in the favorable setting where the signal tensor follows a Tucker2 model and its true rank is known.

In this regard, we generate a tensor $\mathcal{Y} \in \mathbb{R}^{50 \times 11 \times 4}$ according to $\mathcal{Y}_{\bullet,\bullet,j} = \mathcal{M}_{\bullet,\bullet,j} + \mathcal{E}_{\bullet,\bullet,j}$, where $\mathcal{M}_{\bullet,\bullet,j} = U\mathcal{C}_{\bullet,\bullet,j}V^\top$ for all $j \in [4]$, so that the signal $\mathcal{M}$ has Tucker2 rank $(2, 2, 4)$. The unit factor $U \in \mathbb{R}^{50 \times 2}$ is obtained by generating a matrix with independent standard Gaussian entries and orthonormalizing its columns. The two columns of $V \in \mathbb{R}^{11 \times 2}$ are obtained by orthonormalizing a constant and a centered linear function evaluated over an equally spaced grid on $[0, 1]$. For each layer j , the entries of the core matrix $\mathcal{C}_{\bullet,\bullet,j} \in \mathbb{R}^{2 \times 2}$ are independently drawn from a standard Gaussian distribution, after which the core is normalized to have unit Frobenius norm. The noise tensor has entries $\mathcal{E}_{itj} \sim \mathcal{N}(0, 1)$.

We impose the same staggered-adoption pattern on all four layers. Of the 50 units, 10 are never treated, while the remaining 40 units are equally divided across adoption times $4, \dots, 11$, with five units adopting at each time. As usual, an entry is observed before the corresponding adoption time and missing thereafter. We take $k = 2$ as the target layer. Starting from this observation pattern, we generate additional missingness while preserving the staggered structure. More precisely, at each step, we randomly select an eligible row and mask its last currently observed entry, so that the observed entries in every row always form an initial segment of the time periods. The terminal mask is chosen to retain 6 fully observed rows and 4 fully observed columns, with all remaining rows observed only over the first 4 periods. For each of these intermediate masks, our target is the average of the

artificially masked entries in $\mathcal{Y}_{\bullet,\bullet,2}$, whose values before masking are treated as ground truth. We estimate this quantity using Algorithm 2, aggregating the block-specific estimates with weights proportional to the number of artificially masked entries in each block. We compare it with its matrix-only counterpart, which is essentially equivalent to a plug-in approach based on the estimator of Yan and Wainwright (2024), as well as with two plug-in approaches based on the nuclear-norm estimator of Choi and Yuan (2026) and its debiased counterpart.

All rank-dependent procedures are run with $r \in \{1, 2, 3\}$, corresponding to under-specification, correct specification, and over-specification of the true rank $r = 2$. For Algorithm 2 and its matrix counterpart, we set $\tau = 0.01$. Fig. S1 reports the average squared error over 30 replicates, together with its standard error. Each replicate begins after 80 additional entries have been masked and then adds one masked entry at a time until reaching the common terminal configuration. We repeat this procedure independently 30 times, using a different masking path for each replicate.

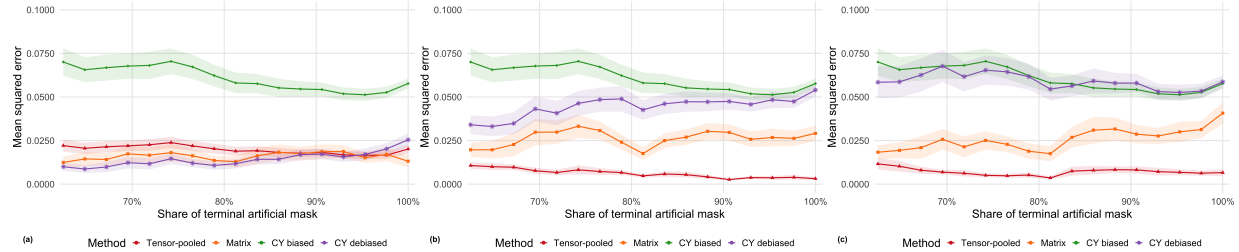


Figure S1: Synthetic masking experiment with additional missingness in the target layer $k = 2$. The panels correspond to ranks $r = 1, 2, 3$ used to run the four methods. The terminal mask retains 6 fully observed rows and 4 fully observed columns. The curves report the average squared error over 30 replicates, with one standard error. The percentage on the x -axis represents the percentage of the terminal mask reached.

The results show that the two CY procedures generally perform worse than the tensor and matrix estimators. A possible explanation is that the method of Choi and Yuan (2026) accommodates more general missingness patterns, whereas Algorithm 2 and its matrix counterpart are tailored to the staggered design and therefore perform particularly well when the missingness is close to a four-block structure. This should not be interpreted as a general ordering between the methods, as the COVID-19 application provides an example

with less structured missingness where the CY procedures are highly competitive.

Furthermore, to better understand the relationship between the tensor and matrix estimators, we run a similar experiment in which the terminal mask is chosen to retain 3 fully observed rows and 3 fully observed columns. The results are shown in Fig. S2.

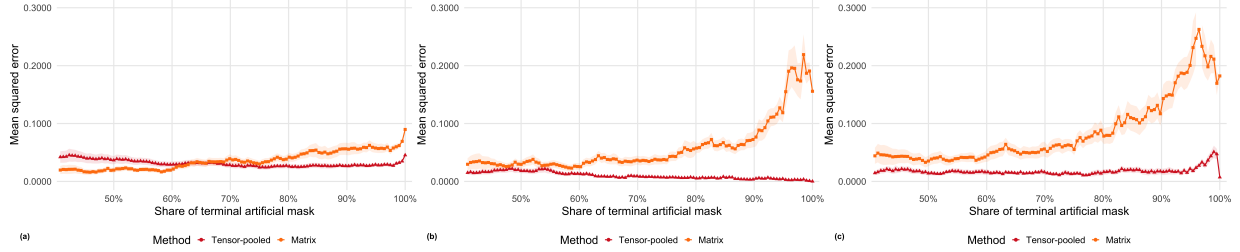


Figure S2: Synthetic masking experiment with additional missingness in the target layer $k = 2$. The panels correspond to ranks $r = 1, 2, 3$ used to run the two methods. The terminal mask retains 3 fully observed rows and 3 fully observed columns. The curves report the average squared error over 30 replicates, with one standard error. The percentage on the x -axis represents the percentage of the terminal mask reached.

In the first panel, where the procedures are run with $r = 1$, the matrix method performs better when the number of artificially masked entries is small. This may be because the target layer initially contains enough information on its own, while Algorithm 1 must estimate a common one-dimensional subspace across all layers. As the masking becomes more severe, however, the information in the target layer becomes scarcer and the matrix estimator deteriorates, while the pooled estimator can exploit additional information from the auxiliary layers.

The second and third panels show this advantage more clearly. When the rank is correctly specified at $r = 2$, Algorithm 2 achieves smaller error. A similar pattern holds for $r = 3$, showing that the gain from pooling persists under rank over-specification.

S2.2 Target estimands used in Sections 5.2 and 5.3

We provide more details on the aggregate quantities used in the real-data experiments. In these applications, we work with two signal tensors, $\mathcal{M}(0)$ and $\mathcal{M}(1)$, corresponding to the untreated and treated responses, respectively. The staggered-adoption mask Ω is

constructed from the treatment variable, with $\Omega_{itj} = 1$ if entry $(i, t, j) \in [N] \times [T] \times [K]$ lies in the untreated region, and $\Omega_{itj} = 0$ otherwise. The fully observed tensor $\mathcal{Y}$ satisfies $\mathcal{Y}_{itj} = \Omega_{itj} \mathcal{Y}_{itj}(0) + (1 - \Omega_{itj}) \mathcal{Y}_{itj}(1)$, where $\mathcal{Y}(0)$ denotes the untreated potential outcome, which is observed on $\{(i, t, j) : \Omega_{itj} = 1\}$ and missing on the complementary set, and $\mathcal{Y}(1)$ denotes the treated potential outcome, which is observed only over $\{(i, t, j) : \Omega_{itj} = 0\}$.

We now introduce the four functionals used in the empirical applications: AVG, ROWHET, LOCAL- i_0 , and TREND. Fix a target slice k with staircase adoption, and let $\mathcal{D}_k = \{(a, b) : a + b > o_k + 1\}$ for some integer $o_k \geq 2$ be the collection of policy-on target blocks. For all $(a, b) \in \mathcal{D}_k$ and $c \in \{0, 1\}$, we also denote by $\mathcal{M}_{\bullet, \bullet, k}^{(a, b)}(c)$ the restriction of $\mathcal{M}(c)$ to rows R_{ak} and columns C_{bk} . For $i \in R_{ak}$ and $t \in C_{bk}$, define the local-index maps $\ell_a(i) := \text{pos}_{R_{ak}}(i)$ and $\ell_b(t) := \text{pos}_{C_{bk}}(t)$, so that $\ell_a(i)$ is the position of row i within R_{ak} , and $\ell_b(t)$ is the position of column t within C_{bk} . For the ROWHET functional, choose a sign vector $\eta = (\eta_1, \dots, \eta_N)^\top \in \{\pm 1\}^N$. For the LOCAL- i_0 functional, choose a row-block index a_0 such that $\{b : (a_0, b) \in \mathcal{D}_k\} \neq \emptyset$, and then fix a row index $i_0 \in R_{a_0 k}$. We also define $\mathcal{D}_k^{\text{tr}} := \{(a, b) \in \mathcal{D}_k : T_{bk} \geq 2\}$, and assume $\mathcal{D}_k^{\text{tr}} \neq \emptyset$ whenever the TREND functional is considered. For $h \in \{\text{AVG}, \text{ROWHET}, \text{LOCAL-}i_0, \text{TREND}\}$, we write

$$\mathcal{D}_{k, h} := \begin{cases} \mathcal{D}_k, & h \in \{\text{AVG}, \text{ROWHET}\}, \\ \{(a_0, b) : (a_0, b) \in \mathcal{D}_k\}, & h = \text{LOCAL-}i_0, \\ \mathcal{D}_k^{\text{tr}}, & h = \text{TREND} \end{cases}$$

for the active block set, and, for fixed $(a, b) \in \mathcal{D}_{k,h}$, we consider the query directions

$$\begin{aligned}
\text{AVG} : \quad & x_{a,h} = N_{ak}^{-1/2} \mathbf{1}_{N_{ak}}, \quad y_{b,h} = T_{bk}^{-1/2} \mathbf{1}_{T_{bk}}, \\
\text{ROWHET} : \quad & x_{a,h} = N_{ak}^{-1/2} \eta_{R_{ak}}, \quad y_{b,h} = T_{bk}^{-1/2} \mathbf{1}_{T_{bk}}, \\
\text{LOCAL-}i_0 : \quad & x_{a,h} = \mathbf{e}_{\ell_a(i_0)}, \quad y_{b,h} = T_{bk}^{-1/2} \mathbf{1}_{T_{bk}}, \\
\text{TREND} : \quad & x_{a,h} = N_{ak}^{-1/2} \mathbf{1}_{N_{ak}}, \quad y_{b,h} = \frac{z_b - \bar{z}_b \mathbf{1}_{T_{bk}}}{\|z_b - \bar{z}_b \mathbf{1}_{T_{bk}}\|_2},
\end{aligned}$$

with $z_b = (1, \dots, T_{bk})^\top$ and $\bar{z}_b = (T_{bk} + 1)/2$.

Based on this, for $h \in \{\text{AVG}, \text{ROWHET}, \text{LOCAL-}i_0, \text{TREND}\}$ and $(a, b) \in \mathcal{D}_{k,h}$, we define the block-level bilinear forms

$$\mu_h^{(k,a,b)}(c) := x_{a,h}^\top \mathcal{M}_{\bullet, \bullet, k}^{(a,b)}(c) y_{b,h}.$$

As for the interpretation of these functionals, AVG averages all entries in the block, ROWHET averages a signed row contrast over the block's columns, LOCAL- i_0 averages only row i_0 over the block's columns, while TREND averages over rows and contrasts later columns with earlier columns. In particular, this latter bilinear form also has a simple slope interpretation. For $t \in C_{bk}$ and $c \in \{0, 1\}$, define the row-averaged trajectory $\bar{m}_t^{(a,b)}(c) := N_{ak}^{-1} \sum_{i \in R_{ak}} \mathcal{M}_{i,t,k}^{(a,b)}(c)$. If this trajectory is linear in local time, i.e. $\bar{m}_t^{(a,b)}(c) = \alpha_{a,b}^c + \beta_{a,b}^c \ell_b(t)$, then

$$x_{a,\text{TREND}}^\top \mathcal{M}_{\bullet, \bullet, k}^{(a,b)}(c) y_{b,\text{TREND}} = \sqrt{N_{ak}} \beta_{a,b}^c \left\{ \frac{T_{bk}(T_{bk}^2 - 1)}{12} \right\}^{1/2},$$

thus showing that TREND recovers the slope of the row-averaged trajectory up to a known normalization.

We then aggregate these block-level summaries over all missing blocks in the target slice by

$$\Psi_c^{(h)}(k) := \{W_h(k)\}^{-1} \sum_{(a,b) \in \mathcal{D}_{k,h}} c_{ab}^{(h)} \mu_h^{(k,a,b)}(c),$$

where the choices of weights and normalizing constants, together with the simplified form of each estimand, are given below:

h	$c_{ab}^{(h)}$	$W_h(k)$	$\Psi_c^{(h)}(k)$
AVG	$\sqrt{N_{ak}T_{bk}}$	$\sum_{(a,b) \in \mathcal{D}_k} N_{ak}T_{bk}$	$\frac{\sum_{(a,b) \in \mathcal{D}_k} \sum_{i \in R_{ak}} \sum_{t \in C_{bk}} \mathcal{M}_{i,t,k}^{(a,b)}(c)}{\sum_{(a,b) \in \mathcal{D}_k} N_{ak}T_{bk}}$
ROWHET	$\sqrt{N_{ak}T_{bk}}$	$\sum_{(a,b) \in \mathcal{D}_k} N_{ak}T_{bk}$	$\frac{\sum_{(a,b) \in \mathcal{D}_k} \sum_{i \in R_{ak}} \sum_{t \in C_{bk}} \eta_i \mathcal{M}_{i,t,k}^{(a,b)}(c)}{\sum_{(a,b) \in \mathcal{D}_k} N_{ak}T_{bk}}$
LOCAL- i_0	$\sqrt{T_{bk}}$	$\sum_{b: (a_0,b) \in \mathcal{D}_k} T_{bk}$	$\frac{\sum_{b: (a_0,b) \in \mathcal{D}_k} \sum_{t \in C_{bk}} \mathcal{M}_{i_0,t,k}^{(a_0,b)}(c)}{\sum_{b: (a_0,b) \in \mathcal{D}_k} T_{bk}}$
TREND	$\frac{1}{\sqrt{N_{ak}T_{bk}(T_{bk}^2 - 1)/12}}$	$ \mathcal{D}_k^{\text{tr}} $	$\frac{1}{ \mathcal{D}_k^{\text{tr}} } \sum_{(a,b) \in \mathcal{D}_k^{\text{tr}}} \beta_{a,b}^c.$

(S1)

The final expression for $\Psi_c^{(\text{Trend})}(k)$ uses the linearity condition on the row-averaged trajectory and the assumption $\mathcal{D}_k^{\text{tr}} \neq \emptyset$. These four quantities have the following interpretations: $\Psi_c^{(\text{Avg})}(k)$ is the average potential outcome over the missing entries in slice k , $\Psi_c^{(\text{RowHET})}(k)$ is the corresponding signed row contrast, $\Psi_c^{(\text{Local-}i_0)}(k)$ is the average potential outcome for row i_0 over the missing target blocks containing that row, and $\Psi_c^{(\text{Trend})}(k)$ is the average within-block slope of the row-averaged trajectory.

Based on these potential-outcome summaries, we also define the aggregate policy contrast for functional h by $\Delta^{(h)}(k) := \Psi_1^{(h)}(k) - \Psi_0^{(h)}(k)$.

Coming now to the estimation of these aggregate quantities, it is useful to recall that $\mathcal{Y}_{\bullet,\bullet,k}(1)$ is observed over $\mathcal{D}_k$, since these are exactly the policy-on target blocks in which $\mathcal{Y}_{\bullet,\bullet,k}(0)$ has missing entries. As a result, functionals with $c = 1$ are easier to target and can be estimated by simple plug-in estimators. On the other hand, quantities such as $\mu_h^{(k,a,b)}(0)$ require an alternative approach, and can be estimated using Algorithm 2. This immediately leads to a naive estimator of $\Psi_0^{(h)}(k)$ that applies Algorithm 2 separately to each missing

target block, and then aggregates the resulting block-level estimates using the weights $c_{ab}^{(h)}$ and normalizing constants $W_h(k)$ defined in (S1).

Algorithm S1 QUADRATICSTAGGEREDAGGREGATE for estimating $\Psi_0^{(h)}(k)$

Require: target slice $k \in [K]$, functional $h \in \{\text{AVG}, \text{ROWHET}, \text{LOCAL-}i_0, \text{TREND}\}$, rank r , data $\mathcal{Y}$, staircase partitions $\{R_{ak}\}_{a=1}^{o_k}$ and $\{C_{bk}\}_{b=1}^{o_k}$, parameter $\tau > 0$, and, when needed, sign vector η and row index i_0 .

- 1: Initialize $S_h \leftarrow 0$.
- 2: **for** $(a, b) \in \mathcal{D}_{k,h}$ **do**
- 3: Construct $x_{a,h}$ and $y_{b,h}$ according to the definitions above.
- 4: Run Algorithm 2 with inputs $(k, a, b, r, x_{a,h}, y_{b,h}, \mathcal{Y}, \tau)$, and denote its output by $\hat{\mu}_h^{(k,a,b)}(0)$.
- 5: Update

$$S_h \leftarrow S_h + c_{ab}^{(h)} \hat{\mu}_h^{(k,a,b)}(0).$$
- 6: **end for**
- 7: **return** $\hat{\Psi}_0^{(h)}(k) \leftarrow \{W_h(k)\}^{-1} S_h$.

The blockwise plug-in estimator in Algorithm S1 applies Algorithm 2 separately to every target block $(a, b) \in \mathcal{D}_{k,h} \subseteq \mathcal{D}_k$, hence it recomputes two rank- r singular value decompositions for each missing block. In the AVG and ROWHET cases, we have $|\mathcal{D}_{k,h}| = |\mathcal{D}_k| = o_k(o_k - 1)/2$, so the cost is quadratic in o_k .

This cost can be reduced by trading some statistical efficiency for computational savings through a reduced-anchor construction. In particular, for fixed a , we keep the target-slice column anchor $\text{ColAnc}_{k,a,b}(k) = Q_{k,a,b}^+$ but replace $\text{ColAnc}_{k,a,b}(j)$ by $\text{ColAnc}_{k,a,b}(j) \cap Q_{k,1}$ for each $j \neq k$. Because $Q_{k,a,b}^+$ depends only on a , and because $Q_{k,1} \subseteq Q_{k,b}$ for all b , the resulting pooled left matrix depends only on a . Similarly, for fixed b , we keep the target-slice row anchor $\text{RowAnc}_{k,a,b}(k) = S_{k,a,b}^+$ but replace $\text{RowAnc}_{k,a,b}(j)$ by $\text{RowAnc}_{k,a,b}(j) \cap S_{k,1}$ for each $j \neq k$. Because $S_{k,a,b}^+$ depends only on b , and because $S_{k,1} \subseteq S_{k,a}$ for all a , the resulting pooled upper matrix depends only on b .

With this reduced-anchor construction, the SVDs of the pooled left and upper matrices can be cached and reused, as illustrated in the following algorithm. We will use the shorthand $Q_{k,a,b}^+ \equiv Q_{k,a}^+$ and $S_{k,a,b}^+ \equiv S_{k,b}^+$ to emphasize that these sets depend only on a and b , respectively.

Algorithm S2 avoids recomputing two spectral decompositions for each missing block by

Algorithm S2 LINEARSTAGGEREDAGGREGATE for estimating $\Psi_0^{(h)}(k)$

Require: target slice $k \in [K]$, functional $h \in \{\text{AVG}, \text{ROWHET}, \text{LOCAL-}i_0, \text{TREND}\}$, rank r , data $\mathcal{Y}$, staircase partitions $\{R_{ak}\}_{a=1}^{o_k}$ and $\{C_{bk}\}_{b=1}^{o_k}$, parameter $\tau > 0$, and, when needed, sign vector η and row index i_0 .

- 1: Let $\mathcal{A}_h := \{a : \exists b \text{ such that } (a, b) \in \mathcal{D}_{k,h}\}$ and $\mathcal{B}_h := \{b : \exists a \text{ such that } (a, b) \in \mathcal{D}_{k,h}\}$.
- 2: **for** $a \in \mathcal{A}_h$ **do**
- 3: Set $\text{ColAnc}_{k,a}^{\text{red}}(k) := Q_{k,a}^+$ and, for $j \neq k$, $\text{ColAnc}_{k,a}^{\text{red}}(j) := \{t \in Q_{k,1} : \Omega_{i,t,j} = 1 \text{ for all } i \in S_{k,a}\}$.
- 4: Form reduced pooled left matrix $Y_{\text{left},a}^{\text{red}} \leftarrow (\mathcal{Y}_{S_{k,a}, \text{ColAnc}_{k,a}^{\text{red}}(1),1} \cdots \mathcal{Y}_{S_{k,a}, \text{ColAnc}_{k,a}^{\text{red}}(K),K})$.
- 5: Compute rank- r truncated singular value decomposition $(\hat{U}_{\text{left},a}^{\text{red}}, \hat{\Sigma}_{\text{left},a}^{\text{red}}, \hat{V}_{\text{left},a}^{\text{red}}) \leftarrow \text{SVD}_r(Y_{\text{left},a}^{\text{red}})$.
- 6: Construct $x_{a,h}$, and cache $\hat{\alpha}_{\text{red}}^{(k,a,h)} \leftarrow (\hat{U}_{\text{left},a}^{\text{red}})_{R_{ak},\bullet}^\top x_{a,h} \in \mathbb{R}^r$.
- 7: **end for**
- 8: **for** $b \in \mathcal{B}_h$ **do**
- 9: Set $\text{RowAnc}_{k,b}^{\text{red}}(k) := S_{k,b}^+$ and, for $j \neq k$, $\text{RowAnc}_{k,b}^{\text{red}}(j) := \{i \in S_{k,1} : \Omega_{i,t,j} = 1 \text{ for all } t \in Q_{k,b}\}$.
- 10: Form reduced pooled upper matrix $Y_{\text{up},b}^{\text{red}} \leftarrow (\mathcal{Y}_{\text{RowAnc}_{k,b}^{\text{red}}(1),Q_{k,b,1}} ; \cdots ; \mathcal{Y}_{\text{RowAnc}_{k,b}^{\text{red}}(K),Q_{k,b,K}})$.
- 11: Compute rank- r truncated singular value decomposition $(\hat{U}_{\text{up},b}^{\text{red}}, \hat{\Sigma}_{\text{up},b}^{\text{red}}, \hat{V}_{\text{up},b}^{\text{red}}) \leftarrow \text{SVD}_r(Y_{\text{up},b}^{\text{red}})$.
- 12: Set $s_{k,b} := \sum_{j=1}^{k-1} |\text{RowAnc}_{k,b}^{\text{red}}(j)|$, $\hat{U}_{\text{up},b}^{(k)} \leftarrow (\hat{U}_{\text{up},b}^{\text{red}})_{\{s_{k,b}+1, \dots, s_{k,b}+|S_{k,b}^+|\},\bullet}$, $\hat{V}_{bk} \leftarrow (\hat{V}_{\text{up},b}^{\text{red}})_{C_{bk},\bullet}$.
- 13: Construct $y_{b,h}$, and compute $T_{b,h} \leftarrow \hat{V}_{bk}^\top y_{b,h}$, $W_{b,h} \leftarrow \hat{\Sigma}_{\text{up},b}^{\text{red}} T_{b,h}$, $X_{b,h} \leftarrow \hat{U}_{\text{up},b}^{(k)} W_{b,h}$.
- 14: **end for**
- 15: Set $S_h \leftarrow 0$.
- 16: **for** $(a, b) \in \mathcal{D}_{k,h}$ **do**
- 17: Set $\hat{U}_{+k}^{(a,b)} \leftarrow (\hat{U}_{\text{left},a}^{\text{red}})_{S_{k,b}^+,\bullet}$.
- 18: Compute $\hat{H}_{a,b} \leftarrow (\hat{U}_{+k}^{(a,b)})^\top \hat{U}_{+k}^{(a,b)}$ and $\hat{H}_{a,b} = Q_{a,b} \text{diag}(\lambda_{a,b,1}, \dots, \lambda_{a,b,r}) Q_{a,b}^\top$. Then set

$$\hat{H}_{a,b,\tau}^{\text{inv}} \leftarrow Q_{a,b} \text{diag} \left(\left\{ \frac{1}{\max[\lambda_{a,b,i}, \tau]} \right\}_{i=1}^r \right) Q_{a,b}^\top.$$
- 19: Compute $\hat{\beta}_{\text{red}}^{(k,a,b,h)} \leftarrow \hat{H}_{a,b,\tau}^{\text{inv}} (\hat{U}_{+k}^{(a,b)})^\top X_{b,h} \in \mathbb{R}^r$.
- 20: Update $S_h \leftarrow S_h + c_{ab}^{(h)} \langle \hat{\alpha}_{\text{red}}^{(k,a,h)}, \hat{\beta}_{\text{red}}^{(k,a,b,h)} \rangle$.
- 21: **end for**
- 22: **return** $\hat{\Psi}_{0,\text{lin}}^{(h)}(k) \leftarrow \{W_h(k)\}^{-1} S_h$.

exploiting the reduced-anchor construction. Instead, it computes at most $o_k - 1$ left decompositions and at most $o_k - 1$ upper decompositions, thus making the dominant spectral cost linear in o_k rather than quadratic. This reduction is obtained at the expense of statistical efficiency. While the blockwise auxiliary construction uses all anchor rows and columns available for each target block, the reduced-anchor construction uses a smaller common set of anchors, restricting the non-target-slice column anchors to $Q_{k,1}$ and the non-target-slice row anchors to $S_{k,1}$. The resulting pooled matrices may therefore contain less information, which can weaken the conditioning of the auxiliary Gram matrices and make the estimated shared row and column subspaces less accurate. Nevertheless, when $Q_{k,1}$ and $S_{k,1}$ are sufficiently large and well conditioned, Algorithm S2 provides a computationally

cheaper alternative to Algorithm S1. A numerical comparison between Algorithm S1 and Algorithm S2 is illustrated in Figs. 6(b, c).

S2.3 Castle Doctrine data

We provide more details on the data-processing setup of Section 5.2. We use the Castle Doctrine data from the PolicyEval repository, [available on GitHub](#). The data contain state identifiers, calendar years, a Castle Doctrine treatment variable, and several state-level public-safety and socioeconomic variables. We use four logged crime-rate outcomes corresponding to the log motor-theft rate, log robbery rate, log aggravated-assault rate, and log murder rate.

We organize the observed outcomes into a fully observed tensor $\mathcal{Y} \in \mathbb{R}^{50 \times 11 \times 4}$, whose modes correspond to U.S. states, calendar years, and crime outcomes. Thus, each entry $\mathcal{Y}_{itj}$ records outcome j for state i in year t . The four outcome slices are `l_motor`, `l_robbery`, `l_assault`, and `l_homicide`, respectively.

We construct the staggered-adoption mask Ω from the Castle Doctrine treatment variable. For all $j \in [K]$, we set $\Omega_{itj} = 1$ if entry (i, t, j) lies in the untreated region and zero otherwise. Because treatment status is common across crime outcomes, the same treatment pattern applies to each outcome slice; equivalently, $\Omega_{\bullet, \bullet, 1} = \Omega_{\bullet, \bullet, 2} = \Omega_{\bullet, \bullet, 3} = \Omega_{\bullet, \bullet, 4}$. Rows are ordered with never-adopting states at the top. Among adopting states, rows are arranged from later to earlier adopters, so that the treatment boundary moves smoothly across the panel. The resulting observation patterns are shown in Fig. S3. Blue cells indicate untreated observations, red cells indicate treated observations, and darker shades correspond to larger logged crime-rate values. In the potential-outcomes notation $\mathcal{Y}_{itj} = \Omega_{itj} \mathcal{Y}_{itj}(0) + (1 - \Omega_{itj}) \mathcal{Y}_{itj}(1)$, the blue cells are therefore the observed entries of $\mathcal{Y}(0)$, while the corresponding treated entries are treated as missing. Conversely, the red cells are the

observed entries of $\mathcal{Y}(1)$, with the corresponding untreated entries treated as missing.

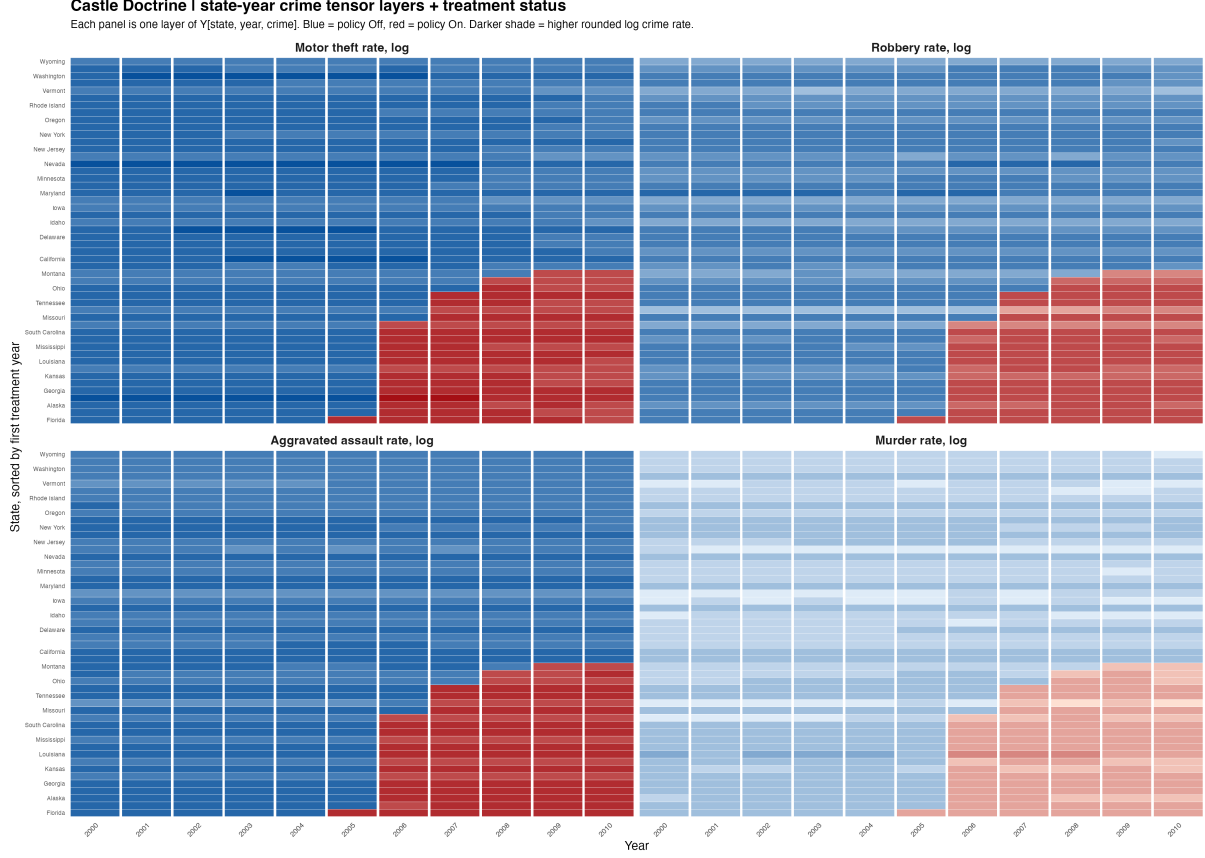


Figure S3: Castle Doctrine state-year-crime tensor used in the real-data simulation. Panels show logged crime rates for motor theft, robbery, aggravated assault, and murder. Rows are U.S. states, with never-adopters first and adopters ordered from later to earlier adoption years; columns are calendar years from 2000 to 2010. Blue cells are untreated, red cells are treated, and darker shades indicate larger logged crime-rate values.

S3 Computational cost of Algorithm 1

We further elaborate on the computational complexity of Algorithm 1. The dominant cost is given by the rank- r truncated singular value decompositions of $Y_{\text{left}}^p \in \mathbb{R}^{N \times T_{1,p}}$ and $Y_{\text{up}}^p \in \mathbb{R}^{N_{1,p} \times T}$, which are of the order $\mathcal{O}(NT_{1,p}r + N_{1,p}Tr)$. After these decompositions are computed, the bilinear form is targeted directly. Indeed, Steps 9 and 10 compute $T_y = \hat{V}_{2k}^\top y$, $W_y = \hat{\Sigma}_{\text{up}} T_y$, $X_y = \hat{U}_{\text{up}}^{(k)} W_y$, and $\hat{\beta}_y^{(k)}$ without ever materializing the full matrix. This avoids an $\mathcal{O}(N_{1k}T_{2k}r)$ block-construction cost, and computes the required action on y in $\mathcal{O}((T_{2k} + N_{1k})r)$ time; including the computation of $\hat{\alpha}_x^{(k)} = \hat{U}_{2k}^\top x$ in Step 5, the total

cost per query is $\mathcal{O}((N_{2k} + T_{2k} + N_{1k})r)$. Furthermore, for the clipped inverse in Step 4, forming $\hat{H}_k = \hat{U}_{1k}^\top \hat{U}_{1k}$ costs $\mathcal{O}(N_{1k}r^2)$, while its eigendecomposition and computing $\hat{H}_{k,\tau}^{\text{inv}}$ cost $\mathcal{O}(r^3)$. Taken together, the runtime analysis of these steps also indicate that our algorithm is well suited to caching. In particular, for fixed k , once the pooled singular value decompositions and the slice-specific regression factorization have been computed, each additional query costs only $\mathcal{O}((N_{2k} + T_{2k} + N_{1k})r)$. When k varies, the pooled singular value decompositions can still be reused, and the only additional slice-specific computation is the $\mathcal{O}(N_{1k}r^2)$ regression factorization, giving a total cost of $\mathcal{O}(N_{1k}r^2 + (N_{2k} + T_{2k} + N_{1k})r)$.

S4 Matrix denoising in the pooled four-block setting

This appendix collects matrix denoising results specialized to the tensor four-block framework introduced in Section 2. Throughout, we allow the constant c_1 to vary from line to line, while still depending only on $c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa$. We use the notation introduced in Table 1. We also denote by $\mathbb{O}(d) := \{Q \in \mathbb{R}^{d \times d} : Q^\top Q = QQ^\top = I_d\}$ the set of orthogonal matrices of dimension d , and write $\mathcal{I}_k := \{N_{1k} + 1, \dots, N\}$, $\mathcal{J}_k := \{T_{1k} + 1, \dots, T\}$ and $\mathcal{I}_k^{\text{up}} := \{s_k + 1, \dots, s_k + N_{1k}\}$, where we recall $s_k = \sum_{j=1}^{k-1} N_{1j}$.

The main results of this section are Lemmas S12 and S13. Their proofs rely on the intermediate results presented below, which provide first-order expansions of some relevant quantities appearing in the definition of $\hat{\mu}_{xy}^{(k)}$ in terms of the matrices $W_{\text{left}}, W_{\text{up}}, E_{\text{left}}^{\text{p}}, E_{\text{up}}^{\text{p}}$. The first step in applying these results to $Y_{\text{left}}^{\text{p}}$ and Y_{up}^{p} is to characterize the spectrum of the corresponding signal matrices $M_{\text{left}}^{\text{p}}$ and M_{up}^{p} .

Lemma S3. *Grant Assumption (A1) with fixed constants c_ℓ, c_u satisfying $0 < c_\ell \leq c_u < \infty$,*

and suppose $0 < \gamma_{\min} \leq \sigma_{\min}(\mathcal{C}_{\bullet,\bullet,j}) \leq \sigma_{\max}(\mathcal{C}_{\bullet,\bullet,j}) \leq \gamma_{\max} < \infty$ for all $j \in [K]$. We have

$$\begin{cases} c_\ell^{1/2} \gamma_{\min} \rho_T^{1/2} \leq \sigma_r(M_{\text{left}}^p) \leq \sigma_1(M_{\text{left}}^p) \leq c_u^{1/2} \gamma_{\max} \rho_T^{1/2}, \\ c_\ell^{1/2} \gamma_{\min} \rho_N^{1/2} \leq \sigma_r(M_{\text{up}}^p) \leq \sigma_1(M_{\text{up}}^p) \leq c_u^{1/2} \gamma_{\max} \rho_N^{1/2}. \end{cases}$$

Proof. Assumption (A1) implies

$$c_\ell \sum_{j=1}^K \frac{T_{1j}}{T} \mathcal{C}_{\bullet,\bullet,j} \mathcal{C}_{\bullet,\bullet,j}^\top \preceq W_{\text{left}}^\top W_{\text{left}} = \sum_{k=1}^K \mathcal{C}_{\bullet,\bullet,k} V_{1k}^\top V_{1k} \mathcal{C}_{\bullet,\bullet,k}^\top \preceq c_u \sum_{j=1}^K \frac{T_{1j}}{T} \mathcal{C}_{\bullet,\bullet,j} \mathcal{C}_{\bullet,\bullet,j}^\top,$$

$$c_\ell \sum_{j=1}^K \frac{N_{1j}}{N} \mathcal{C}_{\bullet,\bullet,j}^\top \mathcal{C}_{\bullet,\bullet,j} \preceq W_{\text{up}}^\top W_{\text{up}} = \sum_{k=1}^K \mathcal{C}_{\bullet,\bullet,k}^\top U_{1k}^\top U_{1k} \mathcal{C}_{\bullet,\bullet,k} \preceq c_u \sum_{j=1}^K \frac{N_{1j}}{N} \mathcal{C}_{\bullet,\bullet,j}^\top \mathcal{C}_{\bullet,\bullet,j}.$$

Since $M_{\text{left}}^p = U W_{\text{left}}^\top$, $M_{\text{up}}^p = W_{\text{up}} V^\top$ and U, V have orthonormal columns, we also have

$\sigma_j(M_{\text{left}}^p) = \sigma_j(W_{\text{left}})$ and $\sigma_j(M_{\text{up}}^p) = \sigma_j(W_{\text{up}})$ for all $j \in [r]$. We thus get

$$\sqrt{c_\ell \lambda_r \left(\sum_{j=1}^K \frac{T_{1j}}{T} \mathcal{C}_{\bullet,\bullet,j} \mathcal{C}_{\bullet,\bullet,j}^\top \right)} \leq \sigma_r(M_{\text{left}}^p) \leq \sigma_1(M_{\text{left}}^p) \leq \sqrt{c_u \lambda_1 \left(\sum_{j=1}^K \frac{T_{1j}}{T} \mathcal{C}_{\bullet,\bullet,j} \mathcal{C}_{\bullet,\bullet,j}^\top \right)},$$

$$\sqrt{c_\ell \lambda_r \left(\sum_{j=1}^K \frac{N_{1j}}{N} \mathcal{C}_{\bullet,\bullet,j}^\top \mathcal{C}_{\bullet,\bullet,j} \right)} \leq \sigma_r(M_{\text{up}}^p) \leq \sigma_1(M_{\text{up}}^p) \leq \sqrt{c_u \lambda_1 \left(\sum_{j=1}^K \frac{N_{1j}}{N} \mathcal{C}_{\bullet,\bullet,j}^\top \mathcal{C}_{\bullet,\bullet,j} \right)}.$$

Combining this with $0 < \gamma_{\min} \leq \sigma_{\min}(\mathcal{C}_{\bullet,\bullet,j}) \leq \sigma_{\max}(\mathcal{C}_{\bullet,\bullet,j}) \leq \gamma_{\max} < \infty$ concludes the proof. $\square$

The following lemma provides an upper bound on the estimation error of $\hat{U}_{\text{left}}$ relative to U measured by the operator norm of the projected error $\Pi_N^\top(\hat{U}_{\text{left}} H_U - U)$. The proof relies on tools from Haar compression and properties of the Stiefel manifold, as outlined in Appendix S6.

Lemma S4. *Grant Assumptions (A1) with fixed constants $0 < c_\ell \leq c_u$, (A2) and (A3). Suppose further that $0 < \gamma_{\min} \leq \sigma_{\min}(\mathcal{C}_{\bullet, \bullet, j}) \leq \sigma_{\max}(\mathcal{C}_{\bullet, \bullet, j}) \leq \gamma_{\max} < \infty$ for all $j \in [K]$, and let $\kappa := \gamma_{\max}/\gamma_{\min}$. Write $Y_{\text{left}}^{\text{p}} = M_{\text{left}}^{\text{p}} + E_{\text{left}}^{\text{p}}$, with $M_{\text{left}}^{\text{p}} = UW_{\text{left}}^{\top}$, and set $\Lambda := W_{\text{left}}^{\top} W_{\text{left}}$. Let $(\hat{U}_{\text{left}}, \hat{\Sigma}_{\text{left}}, \hat{V}_{\text{left}}) := \text{SVD}_r(Y_{\text{left}}^{\text{p}})$ and $H_U := \text{sgn}(\hat{U}_{\text{left}}^{\top} U)$, and define the centered empirical eigenvalue matrix $\hat{\Lambda}_c := H_U^{\top} (\hat{\Sigma}_{\text{left}}^2 - \sigma^2 T_{1,p} I_r) H_U$. Also fix $1 \leq p \leq N$ and $\Pi_N \in \mathbb{R}^{N \times p}$ with $\Pi_N^{\top} \Pi_N = I_p$. There exists a constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$ such that, with probability at least $1 - \mathcal{O}(p_T^{-10})$, the following statements hold:*

(i) *The centered empirical eigenvalue matrix is well-conditioned, i.e.*

$$\lambda_r(\hat{\Lambda}_c) \geq \frac{3}{4} \lambda_r(\Lambda), \quad \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \leq \frac{4}{3} \lambda_r(\Lambda)^{-1}. \quad (\text{S2})$$

(ii) *We have*

$$\|\Pi_N^{\top} (\hat{U}_{\text{left}} H_U - U)\|_{\text{op}} \leq c_1 \frac{\sigma \sqrt{p+r+\zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + c_1 \frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} \|\Pi_N^{\top} U\|_{\text{op}}. \quad (\text{S3})$$

Proof. For readability, only in this proof we write $Y = Y_{\text{left}}^{\text{p}}$, $M = M_{\text{left}}^{\text{p}} = UW_{\text{left}}^{\top}$, and $E = E_{\text{left}}^{\text{p}}$. Define $\hat{S} := YY^{\top} - \sigma^2 T_{1,p} I_N$, $S_0 := MM^{\top} = U\Lambda U^{\top}$, and $\Xi := \hat{S} - S_0$. Let $U_{\perp} \in \mathbb{R}^{N \times (N-r)}$ be such that $[U \ U_{\perp}] \in \mathcal{O}(N)$.

Since $YY^{\top}$ and $\hat{S}$ differ by a scalar multiple of the identity, they have the same eigenvectors. Hence $\hat{U}_{\text{left}}$ is also the top- r eigenspace of $\hat{S}$, and

$$\hat{S} \hat{U}_{\text{left}} H_U = \hat{U}_{\text{left}} H_U \hat{\Lambda}_c. \quad (\text{S4})$$

Set $G_1 := U^{\top} E \in \mathbb{R}^{r \times T_{1,p}}$ and $G_2 := U_{\perp}^{\top} E \in \mathbb{R}^{(N-r) \times T_{1,p}}$. Since E has independent $\mathcal{N}(0, \sigma^2)$ entries, rotational invariance ensures that G_1 and G_2 are independent Gaussian matrices with i.i.d. $\mathcal{N}(0, \sigma^2)$ entries. Expanding Ξ gives $\Xi = UW_{\text{left}}^{\top} E^{\top} + EW_{\text{left}} U^{\top} + (EE^{\top} - \sigma^2 T_{1,p} I_N)$,

and therefore

$$\begin{aligned}
U^\top \Xi U &= W_{\text{left}}^\top G_1^\top + G_1 W_{\text{left}} + (G_1 G_1^\top - \sigma^2 T_{1,p} I_r), \\
U_\perp^\top \Xi U &= G_2 W_{\text{left}} + G_2 G_1^\top = G_2 K, \quad K := W_{\text{left}} + G_1^\top, \\
U_\perp^\top \Xi U_\perp &= G_2 G_2^\top - \sigma^2 T_{1,p} I_{N-r}.
\end{aligned} \tag{S5}$$

Write the aligned empirical eigenspace as $\hat{U}_{\text{left}} H_U = UC + U_\perp S$, where $C := U^\top \hat{U}_{\text{left}} H_U$ and $S := U_\perp^\top \hat{U}_{\text{left}} H_U$. Since H_U is the Procrustes alignment, C is symmetric and satisfies $C \succeq 0$, $C^\top C + S^\top S = I_r$. We can thus write

$$\Pi_N^\top (\hat{U}_{\text{left}} H_U - U) = \Pi_N^\top U_\perp S + \Pi_N^\top U (C - I_r), \tag{S6}$$

which shows that it is enough to control the two terms on the right-hand side.

More precisely, the preceding decomposition shows that sharp control of some projection of $\hat{U}_{\text{left}} H_U - U$ reduces mainly to controlling the off-subspace component $S = U_\perp^\top \hat{U}_{\text{left}} H_U$. Indeed, $C = (I_r - S^\top S)^{1/2}$ and $\|C - I_r\|_{\text{op}} \leq \|S\|_{\text{op}}^2$, so the term involving $C - I_r$ is second order. A direct application of Wedin's theorem (Chen et al., 2021, Section 2.4) would control only the global subspace error $\|S\|_{\text{op}} = \|U_\perp^\top \hat{U}_{\text{left}}\|_{\text{op}} = \|\sin \Theta(\hat{U}_{\text{left}}, U)\|_{\text{op}} \lesssim \|E_{\text{left}}^p\|_{\text{op}} / \sigma_r(M_{\text{left}}^p)$, which is governed by an ambient noise norm and hence scales with the full row dimension N . When combined with the triangle inequality $\|\Pi_N^\top (\hat{U}_{\text{left}} H_U - U)\|_{\text{op}} \leq \|\Pi_N^\top U_\perp\|_{\text{op}} \|S\|_{\text{op}} + \|\Pi_N^\top U\|_{\text{op}} \|S\|_{\text{op}}^2$, this would not exploit the fact that Π_N has only p columns. Instead, we apply the Haar–Stiefel compression bounds from Appendix S6, which allow us to use the fixed projection $\Pi_N^\top U_\perp$ to reduce the random off-subspace component by a factor of order $\sqrt{(p + r + \zeta_T)/N}$, as shown in (S14). This is the key idea that turns an ambient subspace perturbation estimate into the projected bound needed here.

• **Conditioning of $\hat{\Lambda}_c$.** The eigenvalues of $\hat{\Lambda}_c$ are the top r eigenvalues of $\hat{S}$. By the Courant–

Fischer formula restricted to $\text{col}(U)$, we have $\lambda_r(\hat{\Lambda}_c) = \lambda_r(\hat{S}) \geq \lambda_r(U^\top \hat{S} U) = \lambda_r(\Lambda + U^\top \Xi U)$.

Weyl's inequality (Lemma S21) then gives

$$\lambda_r(\hat{\Lambda}_c) \geq \lambda_r(\Lambda) - \|U^\top \Xi U\|_{\text{op}}. \quad (\text{S7})$$

Define the events

$$\begin{aligned} \|G_1 W_{\text{left}}\|_{\text{op}} &\leq c_1 \sigma \|W_{\text{left}}\|_{\text{op}} \sqrt{r + \zeta_T}, \\ \|G_1 G_1^\top - \sigma^2 T_{1,p} I_r\|_{\text{op}} &\leq c_1 \sigma^2 \left\{ \sqrt{T_{1,p}(r + \zeta_T)} + r + \zeta_T \right\}. \end{aligned}$$

By Lemma S19 in Appendix S6 and a standard Wishart concentration bound (e.g. Vershynin, 2019, Theorem 4.6.1), the probability that at least one of the two displayed events fails is at most $\mathcal{O}(p_T^{-10})$. If these bounds hold, the first display in (S5) implies $\|U^\top \Xi U\|_{\text{op}} \leq c_1 \sigma \|W_{\text{left}}\|_{\text{op}} \sqrt{r + \zeta_T} + c_1 \sigma^2 \left\{ \sqrt{T_{1,p}(r + \zeta_T)} + r + \zeta_T \right\}$. By Lemma S3 we have $\lambda_r(\Lambda) \geq c_\ell \gamma_{\min}^2 \rho_T$ and $\|W_{\text{left}}\|_{\text{op}} = \sigma_1(M_{\text{left}}^p) \leq c_u^{1/2} \gamma_{\max} \rho_T^{1/2}$, which yield

$$\begin{aligned} \frac{\|U^\top \Xi U\|_{\text{op}}}{\lambda_r(\Lambda)} &\leq c_1 \kappa \frac{\sigma}{\gamma_{\min}} \sqrt{\frac{T(r + \zeta_T)}{T_{1,p}}} + c_1 \frac{\sigma^2 T}{\gamma_{\min}^2} \left(\sqrt{\frac{r + \zeta_T}{T_{1,p}}} + \frac{r + \zeta_T}{T_{1,p}} \right) \\ &\leq c_1 \frac{\sigma}{\gamma_{\min}} \sqrt{\frac{NT}{T_{1,p}}} + c_1 \frac{\sigma^2 T}{\gamma_{\min}^2} \left(\sqrt{\frac{N}{T_{1,p}}} + \frac{N}{T_{1,p}} \right) \leq c_1 \frac{\sigma}{\gamma_{\min}} \sqrt{\frac{NT}{T_{1,p}}} \leq c_1 \theta \leq \frac{1}{4}, \end{aligned} \quad (\text{S8})$$

where the last inequality follows from Assumptions (A2) and (A3), provided that the absolute constants $c_0 > 0$ and $c_{\text{dim}} > 0$ are sufficiently small. Combining this with (S7) proves (i).

• **Reduction to the range of G_2 .** Using (S4) and $\hat{U}_{\text{left}} H_U = UC + U_\perp S$, and then left-multiplying by $U_\perp^\top$, we obtain

$$G_2 K C + (G_2 G_2^\top - \sigma^2 T_{1,p} I_{N-r}) S = S \hat{\Lambda}_c. \quad (\text{S9})$$

Let $P_2 := \text{Proj}_{\text{col}(G_2)}$ and $P_2^\perp := I_{N-r} - P_2$. Since $P_2^\perp G_2 K = 0$, multiplying (S9) by $P_2^\perp$ gives $P_2^\perp S(\hat{\Lambda}_c + \sigma^2 T_{1,p} I_r) = 0$. On the event where $\hat{\Lambda}_c \succ 0$, we also have that $\hat{\Lambda}_c + \sigma^2 T_{1,p} I_r$ is invertible, hence $P_2^\perp S(\hat{\Lambda}_c + \sigma^2 T_{1,p} I_r) = 0$ implies $P_2^\perp S = 0$, and therefore $S = P_2 S$. Defining $D_2 := (G_2 G_2^\top - \sigma^2 T_{1,p} I_{N-r}) P_2$, we may rewrite (S9) as

$$G_2 K C + D_2 S = S \hat{\Lambda}_c. \quad (\text{S10})$$

The inclusion of the projection P_2 in the definition of D_2 is crucial: although $G_2 G_2^\top - \sigma^2 T_{1,p} I_{N-r}$ may be large on $\text{col}(G_2)^\perp$, the identity $S = P_2 S$ ensures that only its restriction to $\text{col}(G_2)$ is relevant.

• **Restricted centered-Wishart bound for D_2 .** Let $q := \text{rank}(G_2) = \min(N - r, T_{1,p})$ almost surely. Since D_2 acts on $\text{col}(G_2)$, we have $\|D_2\|_{\text{op}} = \max_{1 \leq i \leq q} |\sigma_i(G_2)^2 - \sigma^2 T_{1,p}|$. By the standard two-sided singular value bound for Gaussian matrices (e.g. Vershynin, 2019, Theorem 4.6.1) applied to $G_2^\top / \sigma$, and using (A2) to absorb the terms involving ζ_T into the right-hand side, we have $\|D_2\|_{\text{op}} \leq c_1 \sigma^2 (\sqrt{N T_{1,p}} + N + \zeta_T)$ with probability at least $1 - \mathcal{O}(p_T^{-10})$. This also implies

$$\|D_2\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \leq c_1 \frac{\sigma^2 T}{\gamma_{\min}^2} \left(\sqrt{\frac{N}{T_{1,p}}} + \frac{N + \zeta_T}{T_{1,p}} \right) \leq c_1 \theta^2 \leq \frac{1}{4}, \quad (\text{S11})$$

where the last inequality follows from (A3).

• **Control of $G_2 K$.** We now control $G_2 K \hat{\Lambda}_c^{-1}$ both globally and after projection by $\Pi_N^\top U_\perp$. Conditional on G_1 , the matrix $K = W_{\text{left}} + G_1^\top$ is deterministic, independent of G_2 , and $\text{rank}(K) \leq r$. Hence Lemma S19 gives

$$\begin{aligned} \|\Pi_N^\top U_\perp G_2 K\|_{\text{op}} &\leq c_1 \sigma \|K\|_{\text{op}} \sqrt{p + r + \zeta_T} \\ \|G_2 K\|_{\text{op}} &\leq c_1 \sigma \|K\|_{\text{op}} \sqrt{N + r + \zeta_T} \end{aligned} \quad (\text{S12})$$

with probability at least $1 - \mathcal{O}(p_T^{-10})$. The same bounds hold unconditionally with the same probability. Moreover, on the event $\|G_1\|_{\text{op}} \leq c_1 \sigma(\sqrt{T_{1,p}} + \sqrt{r + \zeta_T})$, which has probability at least $1 - \mathcal{O}(p_T^{-10})$ again by Lemma S19, we have $\|K\|_{\text{op}} \leq \|W_{\text{left}}\|_{\text{op}} + \|G_1\|_{\text{op}} \leq c_1 (\gamma_{\max} \rho_T^{1/2} + \sigma\sqrt{T_{1,p}} + \sigma\sqrt{r + \zeta_T})$, where the second inequality follows from Lemma S3. Combining this bound with (S12) and (i) gives, on an event of probability at least $1 - \mathcal{O}(p_T^{-10})$, we have

$$\begin{aligned}
& \|\Pi_N^\top U_\perp G_2 K\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \\
& \leq c_1 \frac{\sigma\sqrt{p+r+\zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + c_1 \frac{\sigma^2 T}{\gamma_{\min}^2} \left(\sqrt{\frac{p+r+\zeta_T}{T_{1,p}}} + \frac{\sqrt{(p+r+\zeta_T)(r+\zeta_T)}}{T_{1,p}} \right) \\
& \leq c_1 \frac{\sigma\sqrt{p+r+\zeta_T}}{\gamma_{\min} \rho_T^{1/2}}, \\
& \|G_2 K\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \leq c_1 \frac{\sigma\sqrt{N+r+\zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + c_1 \frac{\sigma^2 T}{\gamma_{\min}^2} \left(\sqrt{\frac{N+r+\zeta_T}{T_{1,p}}} + \frac{\sqrt{(N+r+\zeta_T)(r+\zeta_T)}}{T_{1,p}} \right) \\
& \leq c_1 \frac{\sigma\sqrt{N}}{\gamma_{\min} \rho_T^{1/2}},
\end{aligned} \tag{S13}$$

where the final inequalities in both displays follow from (A2), (A3).

• **Haar-measure step.** We now aim to control $\|\Pi_N^\top U_\perp S\|_{\text{op}}$ using Lemma S23. This is achieved through a Haar-measure argument; see Appendix S6 for the statement of the lemma and the relevant background material on the Stiefel manifold. The first step is to rewrite (S10) in coordinates adapted to $\text{col}(G_2)$. In this regard, let $q = \text{rank}(G_2)$. Choose (V_2, Σ_2) measurably from the eigendecomposition of $G_2^\top G_2 = V_2 \Sigma_2^2 V_2^\top$, with the positive eigenvalues sorted in decreasing order, and define $Q := G_2 V_2 \Sigma_2^{-1} \in \text{St}(N-r, q)$, so that $G_2 = Q \Sigma_2 V_2^\top$. Since P_2 is the orthogonal projector onto $\text{col}(G_2)$, we have $P_2 = Q Q^\top$. Thus $S = P_2 S$ implies $S = Q R$, where $R := Q^\top S \in \mathbb{R}^{q \times r}$. Substituting $G_2 = Q \Sigma_2 V_2^\top$ and $S = Q R$

into (S10), and using $D_2 = Q(\Sigma_2^2 - \sigma^2 T_{1,p} I_q) Q^\top$, gives $\Sigma_2 V_2^\top K C + (\Sigma_2^2 - \sigma^2 T_{1,p} I_q) R = R \hat{\Lambda}_c$ after multiplying by $Q^\top$.

Now, let $Q_\perp$ be chosen measurably so that $[Q \ Q_\perp] \in \mathbb{O}(N - r)$, with the last block omitted if $q = N - r$, and set $O := [U \ U_\perp Q \ U_\perp Q_\perp]$. The following calculations allow us to show that $\hat{S}$ has a simple block form with respect to the basis given by O . Using (S5), $\hat{S} = U \Lambda U^\top + \Xi$ and $G_2 = Q \Sigma_2 V_2^\top$, we get

$$U^\top \hat{S} (U_\perp Q) = (Q^\top U_\perp^\top \hat{S} U)^\top = (Q^\top G_2 K)^\top = (\Sigma_2 V_2^\top K)^\top = K^\top V_2 \Sigma_2,$$

$$(U_\perp Q)^\top \hat{S} U = Q^\top U_\perp^\top \hat{S} U = Q^\top G_2 K = \Sigma_2 V_2^\top K,$$

$$(U_\perp Q)^\top \hat{S} (U_\perp Q) = Q^\top (G_2 G_2^\top - \sigma^2 T_{1,p} I_{N-r}) Q = \Sigma_2^2 - \sigma^2 T_{1,p} I_q.$$

Moreover, since $\text{col}(G_2) = \text{col}(Q)$, we have $Q_\perp^\top G_2 = 0$. Hence $U^\top \hat{S} (U_\perp Q_\perp) = 0$, $(U_\perp Q)^\top \hat{S} (U_\perp Q_\perp) = 0$, $(U_\perp Q_\perp)^\top \hat{S} U = 0$, $(U_\perp Q_\perp)^\top \hat{S} (U_\perp Q) = 0$, and $(U_\perp Q_\perp)^\top \hat{S} (U_\perp Q_\perp) = Q_\perp^\top (G_2 G_2^\top - \sigma^2 T_{1,p} I_{N-r}) Q_\perp = -\sigma^2 T_{1,p} I_{N-r-q}$. We therefore get

$$O^\top \hat{S} O = \begin{pmatrix} \Lambda + U^\top \Xi U & K^\top V_2 \Sigma_2 & 0 \\ \Sigma_2 V_2^\top K & \Sigma_2^2 - \sigma^2 T_{1,p} I_q & 0 \\ 0 & 0 & -\sigma^2 T_{1,p} I_{N-r-q} \end{pmatrix}.$$

This shows that, on the event that $\hat{S}$ has at least r positive eigenvalues, the top- r eigenspace of $\hat{S}$ is contained in the column space of $(U, U_\perp Q)$. Formally, using $S = QR$, we have $\hat{U}_{\text{left}} H_U = UC + U_\perp QR = O(C^\top, R^\top, 0)^\top$, and since $\hat{U}_{\text{left}} H_U$ is the aligned top- r eigenspace of $\hat{S}$, we can write

$$(C^\top, R^\top, 0)^\top \hat{\Lambda}_c = O^\top \hat{U}_{\text{left}} H_U \hat{\Lambda}_c = O^\top \hat{S} \hat{U}_{\text{left}} H_U = O^\top \hat{S} O (C^\top, R^\top, 0)^\top.$$

Comparing the first two block rows gives

$$\begin{pmatrix} \Lambda + U^\top \Xi U & K^\top V_2 \Sigma_2 \\ \Sigma_2 V_2^\top K & \Sigma_2^2 - \sigma^2 T_{1,p} I_q \end{pmatrix} \begin{pmatrix} C \\ R \end{pmatrix} = \begin{pmatrix} C \\ R \end{pmatrix} \hat{\Lambda}_c.$$

On the event $\lambda_r(\hat{\Lambda}_c) > 0$, the r largest eigenvalues of $\hat{S}$ are positive and therefore cannot arise from the negative block $-\sigma^2 T_{1,p} I_{N-r-q}$, hence they correspond precisely to the r largest eigenvalues of the reduced block above. We thus get that $(C^\top, R^\top)^\top$ is the top- r eigenspace of the reduced block, and $\hat{\Lambda}_c$ is the associated eigenvalue matrix.

Now, this reduced block depends on G_2 only through (Σ_2, V_2) , and not through Q . Moreover, since $K = W_{\text{left}} + G_1^\top$ and $U^\top \Xi U$ is a function of G_1 , the reduced block is measurable with respect to $\mathcal{F} := \sigma(G_1, \Sigma_2, V_2)$, hence, after fixing deterministic measurable choices of eigenspaces and of the Procrustes alignment, $(C, R, \hat{\Lambda}_c)$ is $\mathcal{F}$ -measurable. Furthermore, writing $R = H_R \Omega_R J_R^\top$ for the compact singular value decomposition of R and $\ell = \text{rank}(R) \leq r$, also H_R is $\mathcal{F}$ -measurable. For completeness, observe that we may assume $\ell \geq 1$; if $\ell = 0$ the desired bound is immediate.

On the other hand, since G_2 has i.i.d. Gaussian entries, its law is left-orthogonally invariant, in the sense that for every deterministic $O_0 \in \mathbb{O}(N-r)$ we have $O_0 G_2 \stackrel{d}{=} G_2$. Moreover, $(O_0 G_2)^\top (O_0 G_2) = G_2^\top G_2$, so left multiplication changes only the left singular subspace, from Q to $O_0 Q$, while leaving (Σ_2, V_2) unchanged. It follows that the conditional law of Q given (Σ_2, V_2) is left-invariant on $\text{St}(N-r, q)$, and by uniqueness of the left-orthogonally invariant probability measure on the Stiefel manifold (e.g. [Chikuse, 2003](#), Theorem 1.2.2 and Section 1.3.1), it is Haar. Furthermore, since this conditional law does not depend on the value of (Σ_2, V_2) , then Q is independent of $\sigma(\Sigma_2, V_2)$, and, since G_1 is independent of G_2 , we also have $Q \perp \mathcal{F}$.

We have therefore shown that, conditional on $\mathcal{F}$, the matrix H_R is fixed, while Q is

Haar-distributed on $\text{St}(N - r, q)$ and independent of $\mathcal{F}$. Hence, by Lemma S23, QH_R is Haar-distributed on $\text{St}(N - r, \ell)$ conditionally on $\mathcal{F}$. We also note here that the first condition in Assumption (A2) implies $r \leq c_{\text{dim}}(N - r)$, and therefore $N - r \geq N/(1 + c_{\text{dim}})$. Moreover, since $\ell \leq r$, $t^2 \asymp \zeta_T$, and $r + \zeta_T \leq c_{\text{dim}}(N - r)$, choosing $c_{\text{dim}} > 0$ sufficiently small ensures $\sqrt{\ell} + t \leq \sqrt{N - r}/2$ as required for the simplified bound in Lemma S23. We then obtain $\|\Pi_N^\top U_\perp QH_R\|_{\text{op}} \leq c_1 \sqrt{(p + r + \zeta_T)/(N - r)} \leq c_1 \sqrt{(p + r + \zeta_T)/N}$ with conditional probability at least $1 - \mathcal{O}(p_T^{-10})$. The same bound also holds unconditionally. Since $S = QR = QH_R \Omega_R J_R^\top$ and $\|\Omega_R\|_{\text{op}} = \|R\|_{\text{op}} = \|QR\|_{\text{op}} = \|S\|_{\text{op}}$, we obtain

$$\|\Pi_N^\top U_\perp S\|_{\text{op}} \leq \|\Pi_N^\top U_\perp QH_R\|_{\text{op}} \|\Omega_R\|_{\text{op}} \|J_R^\top\|_{\text{op}} \leq c_1 \sqrt{\frac{p + r + \zeta_T}{N}} \|S\|_{\text{op}}. \quad (\text{S14})$$

The prefactor $\sqrt{(p + r + \zeta_T)/N}$ is precisely the projection factor that we aimed to obtain, as discussed at the beginning of the proof.

• **Conclusion.** It remains to bound $\|S\|_{\text{op}}$. From (S10) we get $S = G_2 K C \hat{\Lambda}_c^{-1} + D_2 S \hat{\Lambda}_c^{-1}$, and, since $\|C\|_{\text{op}} \leq 1$, the triangle inequality gives $\|S\|_{\text{op}} \leq \|G_2 K\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} + \|D_2\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \|S\|_{\text{op}}$. The preceding Wishart bound for D_2 in (S11) allows reordering this inequality, and gives

$$\|S\|_{\text{op}} \leq c_1 \|G_2 K\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \leq c_1 \frac{\sigma \sqrt{N}}{\gamma_{\min} \rho_T^{1/2}} \quad (\text{S15})$$

by (S13). This, together with (S14), yields $\|\Pi_N^\top U_\perp S\|_{\text{op}} \leq c_1 \sigma \gamma_{\min}^{-1} \sqrt{(p + r + \zeta_T)/\rho_T}$. Finally, since $C = (I_r - S^\top S)^{1/2}$ and $\|C - I_r\|_{\text{op}} \leq \|S\|_{\text{op}}^2$, we have $\|\Pi_N^\top U(C - I_r)\|_{\text{op}} \leq \|\Pi_N^\top U\|_{\text{op}} \|S\|_{\text{op}}^2 \leq c_1 \frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} \|\Pi_N^\top U\|_{\text{op}}$. This proves (S3), and concludes the proof. $\square$

Many useful corollaries can be derived from (S3), yielding bounds that hold with probability

at least $1 - \mathcal{O}(p_T^{-10})$. In particular, for any fixed $x \in \mathcal{S}^{N-1}$, by choosing $\Pi_N = x$ we have

$$\|(\hat{U}_{\text{left}} H_U - U)^\top x\|_2 \leq c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + c_1 \frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} \|U^\top x\|_2 \quad (\text{S16})$$

with high probability, which gives an estimation bound directly for the action of the error $\hat{U}_{\text{left}} H_U - U$ along any fixed unit direction. Furthermore, for fixed $k \in [K]$ the same bounds hold blockwise, in the sense that, for every fixed $1 \leq p \leq N_{1k}$ and $\Pi_N \in \mathbb{R}^{N_{1k} \times p}$ with $\Pi_N^\top \Pi_N = I_p$, we have

$$\|\Pi_N^\top (\hat{U}_{1k} H_U - U_{1k})\|_{\text{op}} \leq c_1 \frac{\sigma \sqrt{p + r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + c_1 \frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} \|\Pi_N^\top U_{1k}\|_{\text{op}}. \quad (\text{S17})$$

As a result, choosing $\Pi_N = I_{N_{1k}}$ and using (A1), (A2) give $\|\hat{U}_{1k} H_U - U_{1k}\|_{\text{op}} \leq c_1 \sigma \gamma_{\min}^{-1} \sqrt{N_{1k}/\rho_T}$ and, as a byproduct, $\hat{U}_{1k}^\top \hat{U}_{1k}$ is invertible, satisfying $\frac{c_\ell}{2} \frac{N_{1k}}{N} I_r \preceq \hat{U}_{1k}^\top \hat{U}_{1k} \preceq 2c_u \frac{N_{1k}}{N} I_r$. This follows from Weyl's inequality (Lemma S21), along similar lines to the proof of (S2). This is particularly useful in our setting, as it ensures that the matrix $\hat{H}_{k,\tau}^{\text{inv}}$ used in Algorithm 1 coincides with $(\hat{U}_{1k}^\top \hat{U}_{1k})^{-1}$ with high probability whenever the algorithm is run with $\tau \leq \frac{c_\ell N_{1k}}{2N}$. Indeed, in this case we have

$$\begin{aligned} \left\{ \frac{c_\ell}{2} \frac{N_{1k}}{N} I_r \preceq \hat{H}_k \preceq 2c_u \frac{N_{1k}}{N} I_r \right\} &= \left\{ \frac{c_\ell}{2} \frac{N_{1k}}{N} I_r \preceq \hat{H}_k \right\} \cap \left\{ \hat{H}_k \preceq 2c_u \frac{N_{1k}}{N} I_r \right\} \\ &\subseteq \left\{ \frac{c_\ell}{2} \frac{N_{1k}}{N} I_r \preceq \hat{H}_k \right\} \subseteq \left\{ \tau I_r \preceq \hat{H}_k \right\}, \end{aligned} \quad (\text{S18})$$

which further implies that $\mathbb{P}(\hat{H}_k \succeq \tau I_r) \geq 1 - \mathcal{O}(p_T^{-10})$. On this event, all eigenvalues of $\hat{H}_k$ are at least τ , hence $\lambda_i \vee \tau = \lambda_i$ for every $i \in [r]$, and $\hat{H}_{k,\tau}^{\text{inv}} = \hat{H}_k^{-1} = (\hat{U}_{1k}^\top \hat{U}_{1k})^{-1}$.

Lemma S5. *Let $\Lambda, \hat{\Lambda}_c, C$ be as in Lemma S4, and suppose that the assumptions of Lemma S4 hold. There exists a constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$ such that, with probability at*

least $1 - \mathcal{O}(p_T^{-10})$, we have

$$\|C\hat{\Lambda}_c^{-1} - \Lambda^{-1}\|_{\text{op}} \leq c_1 \sigma \gamma_{\min}^{-3} \sqrt{N} \rho_T^{-3/2}.$$

Proof. All the bounds below will hold on the events from Lemma S4, which have probability at least $1 - \mathcal{O}(p_T^{-10})$. Using $U_{\perp}^{\top} \Xi U = G_2 K$, $C\hat{\Lambda}_c - \Lambda = \Lambda(C - I_r) + U^{\top} \Xi U C + U^{\top} \Xi U_{\perp} S$, $\Lambda - \hat{\Lambda}_c = \Lambda - C\hat{\Lambda}_c + (C - I_r)\hat{\Lambda}_c$, and $C\hat{\Lambda}_c^{-1} - \Lambda^{-1} = (C - I_r)\hat{\Lambda}_c^{-1} + \hat{\Lambda}_c^{-1}(\Lambda - \hat{\Lambda}_c)\Lambda^{-1}$ from Lemma S4, we can write

$$\begin{aligned} \|C\hat{\Lambda}_c^{-1} - \Lambda^{-1}\|_{\text{op}} &\leq \|C - I_r\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} + \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \|\Lambda - \hat{\Lambda}_c\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}} \\ &\leq \|C - I_r\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} + \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \|\Lambda - C\hat{\Lambda}_c\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}} + \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \|C - I_r\|_{\text{op}} \|\hat{\Lambda}_c\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}} \\ &\leq \|C - I_r\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} + \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \|\Lambda\|_{\text{op}} \|C - I_r\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}} \\ &\quad + \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \|U^{\top} \Xi U\|_{\text{op}} \|C\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}} + \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \|U^{\top} \Xi U_{\perp} S\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}} \\ &\quad + \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \|C - I_r\|_{\text{op}} \|\hat{\Lambda}_c\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}}. \end{aligned} \tag{S19}$$

We now bound the five terms on the right-hand side individually. For the first one, combining

$\|C - I_r\|_{\text{op}} \leq \|S\|_{\text{op}}^2$, $\|\Lambda^{-1}\|_{\text{op}} \lesssim \gamma_{\min}^{-2} \rho_T^{-1}$, (S15) and (S2) gives

$$\|C - I_r\|_{\text{op}} \|\hat{\Lambda}_c^{-1}\|_{\text{op}} \leq c_1 \frac{\sigma^2 N}{\gamma_{\min}^4 \rho_T^2} = c_1 \frac{\sigma \sqrt{N}}{\gamma_{\min} \rho_T^{1/2}} \frac{\sigma \sqrt{N}}{\gamma_{\min}^3 \rho_T^{3/2}} \leq c_1 \frac{\sigma \sqrt{N}}{\gamma_{\min}^3 \rho_T^{3/2}},$$

where the last inequality follows from (A3). The second term can be controlled by combining the previous bound with $\|\Lambda\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}} \leq c_1 \gamma_{\max}^2 \rho_T \gamma_{\min}^{-2} \rho_T^{-1} = \kappa^2 c_1$ and incorporating the constant κ^2 into c_1 . A similar argument applies to the fifth term, using the bound $\|\hat{\Lambda}_c\|_{\text{op}} \lesssim \gamma_{\max}^2 \rho_T$.

For the third term in (S19), we can use $\|C\|_{\text{op}} \leq 1$, $\|\Lambda^{-1}\|_{\text{op}} \lesssim \gamma_{\min}^{-2} \rho_T^{-1}$, (S2) and the second

line in (S8) to write

$$\|\hat{\Lambda}_c^{-1}\|_{\text{op}}\|U^\top \Xi U\|_{\text{op}}\|C\|_{\text{op}}\|\Lambda^{-1}\|_{\text{op}} \leq c_1 \frac{\|U^\top \Xi U\|_{\text{op}}}{\lambda_r(\Lambda)} \frac{1}{\lambda_r(\Lambda)} \leq c_1 \frac{\sigma\sqrt{N}}{\gamma_{\min}\rho_T^{1/2}} \frac{1}{\gamma_{\min}^2\rho_T^2} = c_1 \frac{\sigma\sqrt{N}}{\gamma_{\min}^3\rho_T^{3/2}}.$$

For the fourth piece, (S5) gives $U_\perp^\top \Xi U = G_2 K$ hence $U^\top \Xi U_\perp S = K^\top G_2^\top S$. As a result, combining this with (S15) and (S13) therefore implies

$$\begin{aligned} \|\hat{\Lambda}_c^{-1}\|_{\text{op}}\|U^\top \Xi U_\perp S\|_{\text{op}}\|\Lambda^{-1}\|_{\text{op}} &\leq \|\hat{\Lambda}_c^{-1}\|_{\text{op}}\|G_2 K\|_{\text{op}}\|S\|_{\text{op}}\|\Lambda^{-1}\|_{\text{op}} \leq c_1 \frac{\sigma\sqrt{N}}{\gamma_{\min}\rho_T^{1/2}} \|S\|_{\text{op}}\|\Lambda^{-1}\|_{\text{op}} \\ &\leq c_1 \frac{\sigma^2 N}{\gamma_{\min}^2\rho_T^2} \frac{1}{\gamma_{\min}^2\rho_T^2} = c_1 \frac{\sigma\sqrt{N}}{\gamma_{\min}\rho_T^{1/2}} \frac{\sigma\sqrt{N}}{\gamma_{\min}^3\rho_T^{3/2}} \leq c_1 \frac{\sigma\sqrt{N}}{\gamma_{\min}^3\rho_T^{3/2}}, \end{aligned}$$

where the last inequality follows from (A3). Combining the five bounds concludes the proof. $\square$

We next provide a representation of $\hat{U}_{\text{left}} H_U - U$ in terms of a Gaussian term and a remainder.

Corollary S6. *Use the assumptions and notation of Lemma S4. Set $\Psi_U := \hat{U}_{\text{left}} H_U - U - E_{\text{left}}^p W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1}$, and for fixed $k \in [K]$ define $\Psi_{U,2k} := (\Psi_U)_{\mathcal{I}_k, \bullet}$, where $\mathcal{I}_k = \{N_{1k} + 1, \dots, N\}$. For every fixed $x \in \mathcal{S}^{N_{2k}-1}$, there exists a constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$ such that, with probability at least $1 - \mathcal{O}(p_T^{-10})$, we have*

$$\|(\Psi_{U,2k})^\top x\|_2 \leq c_1 \frac{\sigma^2 \sqrt{(N + T_{1,p})(r + \zeta_T)}}{\gamma_{\min}^2 \rho_T^2} + c_1 \left(\frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T^2} + \frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} \right) \|U_{2k}^\top x\|_2. \quad (\text{S20})$$

Proof. Let $\bar{x} \in \mathcal{S}^{N-1}$ be the zero extension of x to the coordinates $\mathcal{I}_k$, i.e. $\bar{x}_i = x_{i-N_{1k}} \mathbb{1}\{i \in \mathcal{I}_k\}$ for all $i \in [N]$. Then $\|(\Psi_{U,2k})^\top x\|_2 = \|\Psi_U^\top \bar{x}\|_2$ and $\|U^\top \bar{x}\|_2 = \|U_{2k}^\top x\|_2$.

We now recall some important facts that were already stated and justified in the proof of Lemma S4. For readability write $E := E_{\text{left}}^p$, $Y := Y_{\text{left}}^p$, $W := W_{\text{left}}$, $M := M_{\text{left}}^p = UW^\top$, and $\Lambda := W^\top W$. Let $U_\perp \in \mathbb{R}^{N \times (N-r)}$ be such that $[U \ U_\perp] \in \mathcal{O}(N)$, and define $G_1 := U^\top E$

and $G_2 := U_\perp^\top E$. Since E has independent $\mathcal{N}(0, \sigma^2)$ entries, rotational invariance gives that G_1 and G_2 are independent Gaussian matrices with independent $\mathcal{N}(0, \sigma^2)$ entries. Set $\hat{S} := YY^\top - \sigma^2 T_{1,p} I_N$ and $\hat{\Lambda}_c := H_U^\top (\hat{\Sigma}_{\text{left}}^2 - \sigma^2 T_{1,p} I_r) H_U$. Since subtracting $\sigma^2 T_{1,p} I_N$ does not change eigenvectors, $\hat{U}_{\text{left}} H_U$ satisfies $\hat{S} \hat{U}_{\text{left}} H_U = \hat{U}_{\text{left}} H_U \hat{\Lambda}_c$. Write $\hat{U}_{\text{left}} H_U = UC + U_\perp S$, where $C := U^\top \hat{U}_{\text{left}} H_U$ and $S := U_\perp^\top \hat{U}_{\text{left}} H_U$. Because H_U is the Procrustes alignment, C is symmetric positive semidefinite and $C = (I_r - S^\top S)^{1/2}$. Expanding $\hat{S} - U\Lambda U^\top$ gives $\Xi := \hat{S} - U\Lambda U^\top = UW^\top E^\top + EWU^\top + (EE^\top - \sigma^2 T_{1,p} I_N)$. Therefore $U_\perp^\top \Xi U = G_2 W + G_2 G_1^\top = G_2(W + G_1^\top)$, and projecting onto $U_\perp$ gives

$$S = G_2(W + G_1^\top)C\hat{\Lambda}_c^{-1} + D_2 S \hat{\Lambda}_c^{-1}, \quad D_2 := (G_2 G_2^\top - \sigma^2 T_{1,p} I_{N-r})P_2, \quad (\text{S21})$$

where P_2 denotes the orthogonal projector onto the column space of G_2 . In particular, this follows from $S = P_2 S$, which holds under $\hat{\Lambda}_c \succ 0$. Also, since $EW = UG_1 W + U_\perp G_2 W$, the definition of Ψ_U gives $\Psi_U = U_\perp \{S - G_2 W \Lambda^{-1}\} + U \{C - I_r - G_1 W \Lambda^{-1}\}$, consequently we have

$$\|\Psi_U^\top \bar{x}\|_2 \leq \|\{S - G_2 W \Lambda^{-1}\}^\top U_\perp^\top \bar{x}\|_2 + \|(C - I_r)U^\top \bar{x}\|_2 + \|\{G_1 W \Lambda^{-1}\}^\top U^\top \bar{x}\|_2. \quad (\text{S22})$$

We now bound these three terms separately. First, subtracting $G_2 W \Lambda^{-1}$ from the first equation in (S21) yields

$$S - G_2 W \Lambda^{-1} = G_2 W (C\hat{\Lambda}_c^{-1} - \Lambda^{-1}) + G_2 G_1^\top C\hat{\Lambda}_c^{-1} + D_2 S \hat{\Lambda}_c^{-1}. \quad (\text{S23})$$

We recall that on the high-probability event from Lemma S4 we have $\lambda_r(\hat{\Lambda}_c) \geq 3\lambda_r(\Lambda)/4$, $\|\hat{\Lambda}_c^{-1}\|_{\text{op}} \leq c_1 \lambda_r(\Lambda)^{-1}$, $\|S\|_{\text{op}} \leq c_1 \sigma \gamma_{\min}^{-1} \sqrt{N/\rho_T}$, and $\|C - I_r\|_{\text{op}} \leq \|S\|_{\text{op}}^2$. Moreover, arguing as in the paragraph after (S7) and simplifying some bounds using (A2), the same event also gives $\|G_2 W\|_{\text{op}} \leq c_1 \sigma \|W\|_{\text{op}} \sqrt{N}$, $\|G_1 W\|_{\text{op}} \leq c_1 \sigma \|W\|_{\text{op}} \sqrt{r + \zeta_T}$, and $\|G_2 G_1^\top\|_{\text{op}} \leq$

$c_1\sigma^2\sqrt{NT_{1,p}}$. Now, using $\lambda_r(\Lambda) \geq c_1\gamma_{\min}^2\rho_T$, the middle term of (S23) above satisfies

$$\|G_2G_1^\top C\hat{\Lambda}_c^{-1}\|_{\text{op}} \leq c_1 \frac{\sigma^2\sqrt{NT_{1,p}}}{\gamma_{\min}^2\rho_T}.$$

We now deal with the other two terms. First, combining $\|C\hat{\Lambda}_c^{-1} - \Lambda^{-1}\|_{\text{op}} \leq c_1\sigma\gamma_{\min}^{-3}\sqrt{N}\rho_T^{-3/2}$ from Lemma S5 with the above bound for $\|G_2W\|_{\text{op}}$ gives $\|G_2W(C\hat{\Lambda}_c^{-1} - \Lambda^{-1})\|_{\text{op}} \leq c_1\sigma^2\gamma_{\min}^{-2}N/\rho_T$. Similarly, the restricted Wishart bound (S11) for D_2 and the bound on $\|S\|_{\text{op}}$ in (S15) give $\|D_2S\hat{\Lambda}_c^{-1}\|_{\text{op}} \leq c_1\sigma^2\gamma_{\min}^{-2}N/\rho_T$, thereby implying

$$\|S - G_2W\Lambda^{-1}\|_{\text{op}} \leq c_1 \frac{\sigma^2\sqrt{T_{1,p}N}}{\gamma_{\min}^2\rho_T} + c_1 \frac{\sigma^2N}{\gamma_{\min}^2\rho_T}.$$

It remains to convert this operator bound into a Euclidean norm bound. Let $G_2 = Q\Sigma_2V_2^\top$ be its singular value decomposition. Arguing as in the proof of Lemma S4, conditional on $\sigma(G_1, \Sigma_2, V_2)$, the factor Q is Haar-distributed on the appropriate Stiefel manifold. Since $S = QR$ for an $\sigma(G_1, \Sigma_2, V_2)$ -measurable matrix R and $G_2W\Lambda^{-1} = Q\Sigma_2V_2^\top W\Lambda^{-1}$, we can write $S - G_2W\Lambda^{-1} = Q\tilde{R}$, where $\tilde{R}$ is $\sigma(G_1, \Sigma_2, V_2)$ -measurable and has rank at most r . Now, let $\ell := \text{rank}(\tilde{R}) \leq r$, and consider the compact SVD of $\tilde{R} = L_{\tilde{R}}D_{\tilde{R}}M_{\tilde{R}}^\top$ with $L_{\tilde{R}}^\top L_{\tilde{R}} = I_\ell$, where the factors may be chosen $\sigma(G_1, \Sigma_2, V_2)$ -measurable. Conditional on $\sigma(G_1, \Sigma_2, V_2)$, the matrix $L_{\tilde{R}}$ is fixed and Q is Haar-distributed on $\text{St}(N - r, q)$. As a result, Lemma S23, applied with $d = N - r$, $H = L_{\tilde{R}}$, $A = \bar{x}^\top U_\perp$, and $t^2 \asymp \zeta_T$, gives $\|L_{\tilde{R}}^\top Q^\top U_\perp^\top \bar{x}\|_2 \leq c_1 \|\bar{x}^\top U_\perp\|_2 \sqrt{(r + \zeta_T)/(N - r)} \leq c_1 \|\bar{x}^\top U_\perp\|_2 \sqrt{(r + \zeta_T)/N}$, where the last inequality follows from (A2). Finally, using $\|\bar{x}^\top U_\perp\|_2 \leq \|\bar{x}\|_2 = 1$, we obtain

$$\begin{aligned} \|\tilde{R}^\top Q^\top U_\perp^\top \bar{x}\|_2 &= \|M_{\tilde{R}}^\top D_{\tilde{R}}^\top L_{\tilde{R}}^\top Q^\top U_\perp^\top \bar{x}\|_2 \leq \|D_{\tilde{R}}\|_{\text{op}} \|L_{\tilde{R}}^\top Q^\top U_\perp^\top \bar{x}\|_2 \\ &= \|\tilde{R}\|_{\text{op}} \|L_{\tilde{R}}^\top Q^\top U_\perp^\top \bar{x}\|_2 \leq c_1 \sqrt{\frac{r + \zeta_T}{N}} \|\tilde{R}\|_{\text{op}} \end{aligned}$$

with probability at least $1 - \mathcal{O}(p_T^{-10})$, which therefore implies

$$\|\{S - G_2 W \Lambda^{-1}\}^\top U_\perp^\top \bar{x}\|_2 \leq c_1 \frac{\sigma^2 \sqrt{(N + T_{1,p})(r + \zeta_T)}}{\gamma_{\min}^2 \rho_T}.$$

This concludes the analysis for the first and more problematic term in (S22). For the second one, using $\|C - I_r\|_{\text{op}} \leq \|S\|_{\text{op}}^2$ and $\|S\|_{\text{op}} \leq c_1 \sigma \gamma_{\min}^{-1} \sqrt{N/\rho_T}$ from the proof of Lemma S4, we get $\|(C - I_r)U^\top \bar{x}\|_2 \leq c_1 \sigma^2 \gamma_{\min}^{-2} N \rho_T^{-1} \|U^\top \bar{x}\|_2 = c_1 \sigma^2 \gamma_{\min}^{-2} N \rho_T^{-1} \|U_{2k}^\top x\|_2$. For the third term, Lemma S19 gives $\|G_1 W\|_{\text{op}} \leq c_1 \sigma \|W\|_{\text{op}} \sqrt{r + \zeta_T}$ with probability at least $1 - \mathcal{O}(p_T^{-10})$. Also, Lemma S3 gives $\|W\|_{\text{op}} \leq c_u^{1/2} \gamma_{\max} \rho_T^{1/2}$ and $\|\Lambda^{-1}\|_{\text{op}} = \lambda_r(\Lambda)^{-1} = \sigma_r(W)^{-2} \leq c_\ell^{-1} \gamma_{\min}^{-2} \rho_T^{-1}$. As a result, we have $\|G_1 W \Lambda^{-1}\|_{\text{op}} \leq \|G_1 W\|_{\text{op}} \|\Lambda^{-1}\|_{\text{op}} \leq c_1 \sigma \sqrt{r + \zeta_T} \gamma_{\max} \rho_T^{1/2} (\gamma_{\min}^2 \rho_T)^{-1} \leq c_1 \sigma \gamma_{\min}^{-1} \sqrt{(r + \zeta_T)/\rho_T}$, which further implies $\|\{G_1 W \Lambda^{-1}\}^\top U^\top \bar{x}\|_2 \leq c_1 \sigma \gamma_{\min}^{-1} \sqrt{(r + \zeta_T)/\rho_T} \|U_{2k}^\top x\|_2$.

Combining the three bounds concludes the proof. $\square$

As a sanity check, combining (S20) and $\hat{U}_{\text{left}} H_U - U = \Psi_U + E_{\text{left}}^p W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1}$ with Lemmas S3 and S19 and Assumptions (A2), (A3), allows proving a bound for $\|(\hat{U}_{\text{left}} H_U - U)^\top x\|_2$ that agrees with (S16).

Moreover, by applying Lemma S4 and Corollary S6 to $Y_{\text{up}}^\top$ we get the following corollary.

Corollary S7. *Grant Assumptions (A1) with fixed constants $0 < c_\ell \leq c_u$, (A2) and (A3). Suppose further that $0 < \gamma_{\min} \leq \sigma_{\min}(\mathcal{C}_{\bullet, \bullet, j}) \leq \sigma_{\max}(\mathcal{C}_{\bullet, \bullet, j}) \leq \gamma_{\max} < \infty$ for all $j \in [K]$, and let $\kappa := \gamma_{\max}/\gamma_{\min}$. Write $Y_{\text{up}}^p = M_{\text{up}}^p + E_{\text{up}}^p$, with $M_{\text{up}}^p = W_{\text{up}} V^\top$. Let $(\hat{U}_{\text{up}}, \hat{\Sigma}_{\text{up}}, \hat{V}_{\text{up}}) := \text{SVD}_r(Y_{\text{up}}^p)$ and $H_V := \text{sgn}(\hat{V}_{\text{up}}^\top V)$, and define $\Psi_V := \hat{V}_{\text{up}} H_V - V - (E_{\text{up}}^p)^\top W_{\text{up}} (W_{\text{up}}^\top W_{\text{up}})^{-1}$. For fixed $k \in [K]$ we also set $\Psi_{V, 2k} := (\Psi_V)_{\mathcal{J}_k, \bullet}$, where $\mathcal{J}_k = \{T_{1k} + 1, \dots, T\}$. Fix also $\bar{y} \in \mathcal{S}^{T-1}$ and $y \in \mathcal{S}^{T_{2k}-1}$. There exists a constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$ such that, with probability at least $1 - \mathcal{O}(p_N^{-10})$, the following statements hold:*

(i) We have

$$\|(\hat{V}_{\text{up}}H_V - V)^\top \bar{y}\|_2 \leq c_1 \frac{\sigma\sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + c_1 \frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} \|V^\top \bar{y}\|_2. \quad (\text{S24})$$

(ii) We can bound the operator norm as

$$\|\hat{V}_{\text{up}}H_V - V\|_{\text{op}} \leq c_1 \frac{\sigma\sqrt{T}}{\gamma_{\min} \rho_N^{1/2}}. \quad (\text{S25})$$

(iii) We have

$$\|\Psi_{V,2k}^\top y\|_2 \leq c_1 \frac{\sigma^2 \sqrt{(T + N_{1,\text{p}})(r + \zeta_N)}}{\gamma_{\min}^2 \rho_N} + c_1 \left(\frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} + \frac{\sigma\sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} \right) \|V_{2k}^\top y\|_2. \quad (\text{S26})$$

Proof. The proof follows by applying Lemmas S4 and S6 to $(Y_{\text{up}}^{\text{p}})^\top = VW_{\text{up}}^\top + (E_{\text{up}}^{\text{p}})^\top$. In particular, (S24) follows from (S16), (S25) from (S3) with $\Pi_T = I_T$ and (A3), and (S26) from (S20). $\square$

Returning to theoretical guarantees for quantities obtained from the SVD of $Y_{\text{left}}^{\text{p}}$, the following result controls the estimation error for $(U_{1k}^\top U_{1k})^{-1} U_{1k}^\top$, a key object for predicting the c -block from the a -block.

Lemma S8. *Adopt the assumptions and notation of Lemma S4. Fix $k \in [K]$, write $H_k := U_{1k}^\top U_{1k}$ and $\widehat{H}_k := (\hat{U}_{1k} H_U)^\top \hat{U}_{1k} H_U$, and set $D_k := \widehat{H}_k^{-1} (\hat{U}_{1k} H_U)^\top - H_k^{-1} U_{1k}^\top$. There exists $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$ such that, with probability at least $1 - \mathcal{O}(p_T^{-10})$, we have*

$$\|D_k\|_{\text{op}} \leq c_1 \frac{\sigma\sqrt{N}}{\gamma_{\min} \rho_T^{1/2}} \sqrt{\frac{N}{N_{1k}}}. \quad (\text{S27})$$

Proof. We showed in (S17) and the discussion thereafter that $\hat{U}_{1k}H_U$ has full column rank on the high-probability event of Lemma S4. On this event, letting $\Delta_k := \hat{U}_{1k}H_U - U_{1k}$ and using $D_k = \widehat{H}_k^{-1}(\hat{U}_{1k}H_U)^\top - H_k^{-1}U_{1k}^\top = (\hat{U}_{1k}H_U)^\dagger - U_{1k}^\dagger$, Lemma S22 gives

$$D_k = -U_{1k}^\dagger \Delta_k (\hat{U}_{1k}H_U)^\dagger + U_{1k}^\dagger (U_{1k}^\dagger)^\top \Delta_k^\top \{I_{N_{1k}} - (\hat{U}_{1k}H_U)(\hat{U}_{1k}H_U)^\dagger\}.$$

As a result, we have

$$\begin{aligned} \|D_k\|_{\text{op}} &\leq \|U_{1k}^\dagger\|_{\text{op}} \|\Delta_k\|_{\text{op}} \|(\hat{U}_{1k}H_U)^\dagger\|_{\text{op}} + \|U_{1k}^\dagger\|_{\text{op}}^2 \|\Delta_k\|_{\text{op}} \\ &\leq (\sqrt{2} + 1)c_\ell^{-1} \frac{N}{N_{1k}} \|\Delta_k\|_{\text{op}} \leq c_1 \frac{\sigma\sqrt{N}}{\gamma_{\min}\rho_T^{1/2}} \sqrt{\frac{N}{N_{1k}}}, \end{aligned} \quad (\text{S28})$$

where the penultimate inequality follows from (A1) and $\frac{c_\ell}{2} \frac{N_{1k}}{N} I_r \preceq \hat{U}_{1k}^\top \hat{U}_{1k} \preceq 2c_u \frac{N_{1k}}{N} I_r$, while the last one follows from the discussion right after (S17). This completes the proof. $\square$

Lemma S9. *Grant the assumptions of Lemma S4. Fix $k \in [K]$, and recall $H_k = U_{1k}^\top U_{1k}$, $\hat{H}_k = H_U^\top \hat{U}_{1k}^\top \hat{U}_{1k} H_U$. On the high probability event where $\hat{H}_k$ is invertible, define $L := \hat{U}_{2k} H_U \hat{H}_k^{-1} H_U^\top \hat{U}_{1k}^\top - U_{2k} H_k^{-1} U_{1k}^\top$. Writing*

$$\Delta_L := L U_{1k} \mathcal{C}_{\bullet, \bullet, k} - (E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \mathcal{C}_{\bullet, \bullet, k}, \quad (\text{S29})$$

for every fixed $x \in \mathcal{S}^{N_{2k}-1}$ there exists $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$ such that, with probability at least $1 - \mathcal{O}(p_T^{-10})$, we have

$$\|x^\top \Delta_L\|_2 \leq c_1 \frac{\sigma^2 \sqrt{(N + T_{1,\text{p}})(r + \zeta_T)}}{\gamma_{\min}\rho_T} + c_1 \frac{\sigma\sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2. \quad (\text{S30})$$

Proof. Using $H_k^{-1}U_{1k}^\top U_{1k} = I_r$, by definition of L , we have

$$L U_{1k} \mathcal{C}_{\bullet, \bullet, k} = \hat{U}_{2k} H_U \hat{H}_k^{-1} H_U^\top \hat{U}_{1k}^\top U_{1k} \mathcal{C}_{\bullet, \bullet, k} - U_{2k} H_k^{-1} U_{1k}^\top U_{1k} \mathcal{C}_{\bullet, \bullet, k}$$

$$= (\hat{U}_{2k} H_U - U_{2k}) \mathcal{C}_{\bullet, \bullet, k} + \hat{U}_{2k} H_U (\hat{H}_k^{-1} H_U^\top \hat{U}_{1k}^\top - H_k^{-1} U_{1k}^\top) U_{1k} \mathcal{C}_{\bullet, \bullet, k},$$

Also, the definition of $\Psi_{U,2k}$ in Corollary S6 gives

$$\hat{U}_{2k} H_U - U_{2k} = (E_{\text{left}}^p)_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} + \Psi_{U,2k}.$$

Substituting this identity into the above display and using the definition of Δ_L in (S29)

$$\text{give } \Delta_L = \Psi_{U,2k} \mathcal{C}_{\bullet, \bullet, k} + \hat{U}_{2k} H_U (\hat{H}_k^{-1} H_U^\top \hat{U}_{1k}^\top - H_k^{-1} U_{1k}^\top) U_{1k} \mathcal{C}_{\bullet, \bullet, k}.$$

We will use this to control the Euclidean norm of $x^\top \Delta_L$ by bounding the norm of each of the two terms separately. For the first one, combining (S20) with $\|\mathcal{C}_{\bullet, \bullet, k}\|_{\text{op}} \leq \gamma_{\max} = \kappa \gamma_{\min}$ gives

$$\begin{aligned} \|x^\top \Psi_{U,2k} \mathcal{C}_{\bullet, \bullet, k}\|_2 &\leq \|\mathcal{C}_{\bullet, \bullet, k}\|_{\text{op}} \|\Psi_{U,2k}^\top x\|_2 \\ &\leq c_1 \frac{\sigma^2 \sqrt{(N + T_{1,p})(r + \zeta_T)}}{\gamma_{\min} \rho_T} + c_1 \left(\frac{\sigma^2 N}{\gamma_{\min} \rho_T} + \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \right) \|U_{2k}^\top x\|_2. \end{aligned}$$

Second, recalling $D_k = \hat{H}_k^{-1} (\hat{U}_{1k} H_U)^\top - H_k^{-1} U_{1k}^\top$ from Lemma S8, we have

$$\begin{aligned} \|x^\top \hat{U}_{2k} H_U D_k U_{1k} \mathcal{C}_{\bullet, \bullet, k}\|_2 &\leq \|H_U^\top \hat{U}_{2k}^\top x\|_2 \|D_k U_{1k} \mathcal{C}_{\bullet, \bullet, k}\|_{\text{op}} \\ &\leq \left(\|(\hat{U}_{2k} H_U - U_{2k})^\top x\|_2 + \|U_{2k}^\top x\|_2 \right) \|D_k U_{1k} \mathcal{C}_{\bullet, \bullet, k}\|_{\text{op}} \\ &\leq c_1 \left(\frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + \frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} \|U_{2k}^\top x\|_2 + \|U_{2k}^\top x\|_2 \right) \|D_k U_{1k} \mathcal{C}_{\bullet, \bullet, k}\|_{\text{op}} \\ &\leq c_1 \left(\frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + \|U_{2k}^\top x\|_2 \right) \|D_k U_{1k} \mathcal{C}_{\bullet, \bullet, k}\|_{\text{op}} \\ &\leq c_1 \left(\frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + \|U_{2k}^\top x\|_2 \right) \|D_k\|_{\text{op}} \|U_{1k}\|_{\text{op}} \|\mathcal{C}_{\bullet, \bullet, k}\|_{\text{op}} \\ &\leq c_1 \frac{\sigma^2 \sqrt{N(r + \zeta_T)}}{\gamma_{\min} \rho_T} + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2, \end{aligned}$$

where the third inequality follows from (S16), and the last one from (S27), (A1) and

$\|\mathcal{C}_{\bullet,\bullet,k}\|_{\text{op}} \leq \gamma_{\max} = \kappa\gamma_{\min}$. Combining the two displays and using (A2), (A3) concludes the proof. $\square$

Lemma S10. *Grant Assumptions (A1) with fixed constants $0 < c_\ell \leq c_u$, (A2) and (A3). Suppose that $0 < \gamma_{\min} \leq \sigma_{\min}(\mathcal{C}_{\bullet,\bullet,j}) \leq \sigma_{\max}(\mathcal{C}_{\bullet,\bullet,j}) \leq \gamma_{\max} < \infty$ for all $j \in [K]$. Write $Y_{\text{up}}^{\text{p}} = M_{\text{up}}^{\text{p}} + E_{\text{up}}^{\text{p}}$, where $M_{\text{up}}^{\text{p}} = W_{\text{up}}V^{\top}$. Also recall $(\hat{U}_{\text{up}}, \hat{\Sigma}_{\text{up}}, \hat{V}_{\text{up}}) = \text{SVD}_r(Y_{\text{up}}^{\text{p}})$, $H_V = \text{sgn}(\hat{V}_{\text{up}}^{\top}V)$. Writing*

$$\Phi_{\text{up}} := \hat{U}_{\text{up}}\hat{\Sigma}_{\text{up}}\hat{V}_{\text{up}}^{\top} - M_{\text{up}}^{\text{p}} - E_{\text{up}}^{\text{p}}VV^{\top} - W_{\text{up}}(W_{\text{up}}^{\top}W_{\text{up}})^{-1}W_{\text{up}}^{\top}E_{\text{up}}^{\text{p}}, \quad (\text{S31})$$

for every fixed $g \in \mathbb{R}^{N_{1,\text{p}}}$ and $y \in \mathcal{S}^{T-1}$ there exists a constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$ such that

$$\begin{aligned} |g^{\top}\Phi_{\text{up}}y| &\leq c_1 \frac{\sigma^2 \sqrt{T(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2}} \|g\|_2 + c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} \|W_{\text{up}}^{\top}g\|_2 + c_1 \frac{\sigma^2 T}{\gamma_{\min} \rho_N^{1/2}} \|g\|_2 \|V^{\top}y\|_2 \\ &\quad + c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \|W_{\text{up}}^{\top}g\|_2 \|V^{\top}y\|_2 \end{aligned} \quad (\text{S32})$$

with probability at least $1 - \mathcal{O}(p_N^{-10})$.

Proof. Write $\hat{M}_{\text{up}}^{\text{p}} := \hat{U}_{\text{up}}\hat{\Sigma}_{\text{up}}\hat{V}_{\text{up}}^{\top}$, $\delta_V := \hat{V}_{\text{up}}H_V - V$, and $\Psi_V := \delta_V - (E_{\text{up}}^{\text{p}})^{\top}W_{\text{up}}(W_{\text{up}}^{\top}W_{\text{up}})^{-1}$.

Combining this with $\hat{U}_{\text{up}}\hat{\Sigma}_{\text{up}}H_V = Y_{\text{up}}^{\text{p}}(V + \delta_V)$ gives

$$\begin{aligned} \hat{M}_{\text{up}}^{\text{p}} - M_{\text{up}}^{\text{p}} &= Y_{\text{up}}^{\text{p}}(V + \delta_V)(V + \delta_V)^{\top} - W_{\text{up}}V^{\top} \\ &= (W_{\text{up}}V^{\top} + E_{\text{up}}^{\text{p}})(V + \delta_V)(V + \delta_V)^{\top} - W_{\text{up}}V^{\top} \\ &= E_{\text{up}}^{\text{p}}VV^{\top} + E_{\text{up}}^{\text{p}}V\delta_V^{\top} + Y_{\text{up}}^{\text{p}}\delta_V V^{\top} + Y_{\text{up}}^{\text{p}}\delta_V \delta_V^{\top} + W_{\text{up}}\delta_V^{\top} \\ &= E_{\text{up}}^{\text{p}}VV^{\top} + W_{\text{up}}(W_{\text{up}}^{\top}W_{\text{up}})^{-1}W_{\text{up}}^{\top}E_{\text{up}}^{\text{p}} + E_{\text{up}}^{\text{p}}V\delta_V^{\top} + Y_{\text{up}}^{\text{p}}\delta_V V^{\top} + Y_{\text{up}}^{\text{p}}\delta_V \delta_V^{\top} + W_{\text{up}}\Psi_V^{\top}. \end{aligned}$$

This, together with the definition of Φ_{up} in (S31), gives $g^{\top}\Phi_{\text{up}}y = g^{\top}E_{\text{up}}^{\text{p}}V\delta_V^{\top}y +$

$$g^\top Y_{\text{up}}^{\text{p}} \delta_V V^\top y + g^\top Y_{\text{up}}^{\text{p}} \delta_V \delta_V^\top y + g^\top W_{\text{up}} \Psi_V^\top y.$$

We now bound each of the four terms in $g^\top \Phi_{\text{up}} y$ individually. By (S24) and (S25) in Corollary S7, with probability at least $1 - \mathcal{O}(p_N^{-10})$ we have

$$\|\delta_V^\top y\|_2 \leq c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + c_1 \frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} \|V^\top y\|_2, \quad \|\delta_V\|_{\text{op}} \leq c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}}. \quad (\text{S33})$$

Combining the first bound with Lemma S19 and (A2) gives

$$\begin{aligned} |g^\top E_{\text{up}}^{\text{p}} V \delta_V^\top y| &\leq \|(E_{\text{up}}^{\text{p}})^\top g\|_2 \|\delta_V^\top y\|_2 \leq c_1 \sigma \sqrt{T + \zeta_N} \|g\|_2 \|\delta_V^\top y\|_2 \\ &\leq c_1 \frac{\sigma^2 \sqrt{T(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2}} \|g\|_2 + c_1 \frac{\sigma^3 T^{3/2}}{\gamma_{\min}^2 \rho_N} \|g\|_2 \|V^\top y\|_2. \end{aligned}$$

Next, using $\|(Y_{\text{up}}^{\text{p}})^\top g\|_2 \leq \|(M_{\text{up}}^{\text{p}})^\top g\|_2 + \|(E_{\text{up}}^{\text{p}})^\top g\|_2 \leq \|W_{\text{up}}^\top g\|_2 + c_1 \sigma \sqrt{T} \|g\|_2$, we also get

$$\begin{aligned} |g^\top Y_{\text{up}}^{\text{p}} \delta_V V^\top y| &\leq \|(Y_{\text{up}}^{\text{p}})^\top g\|_2 \|\delta_V\|_{\text{op}} \|V^\top y\|_2 \leq c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \|(Y_{\text{up}}^{\text{p}})^\top g\|_2 \|V^\top y\|_2 \\ &\leq c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \left(\|W_{\text{up}}^\top g\|_2 + \sigma \sqrt{T} \|g\|_2 \right) \|V^\top y\|_2 \\ &\leq c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \|W_{\text{up}}^\top g\|_2 \|V^\top y\|_2 + c_1 \frac{\sigma^2 T}{\gamma_{\min} \rho_N^{1/2}} \|g\|_2 \|V^\top y\|_2. \end{aligned}$$

Similarly, for the third term we have

$$\begin{aligned} |g^\top Y_{\text{up}}^{\text{p}} \delta_V \delta_V^\top y| &\leq \|(Y_{\text{up}}^{\text{p}})^\top g\|_2 \|\delta_V\|_{\text{op}} \|\delta_V^\top y\|_2 \leq c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \|(Y_{\text{up}}^{\text{p}})^\top g\|_2 \|\delta_V^\top y\|_2 \\ &\leq c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \left(\|W_{\text{up}}^\top g\|_2 + \sigma \sqrt{T} \|g\|_2 \right) \|\delta_V^\top y\|_2 \\ &\leq c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \left(\|W_{\text{up}}^\top g\|_2 + \sigma \sqrt{T} \|g\|_2 \right) \left(\frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + \frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} \|V^\top y\|_2 \right) \\ &\leq c_1 \frac{\sigma^2 \sqrt{T(r + \zeta_N)}}{\gamma_{\min}^2 \rho_N} \|W_{\text{up}}^\top g\|_2 + c_1 \frac{\sigma^3 T \sqrt{r + \zeta_N}}{\gamma_{\min}^2 \rho_N} \|g\|_2 \\ &\quad + c_1 \frac{\sigma^3 T^{3/2}}{\gamma_{\min}^3 \rho_N^{3/2}} \|W_{\text{up}}^\top g\|_2 \|V^\top y\|_2 + c_1 \frac{\sigma^4 T^2}{\gamma_{\min}^3 \rho_N^{3/2}} \|g\|_2 \|V^\top y\|_2. \end{aligned}$$

It remains to control the term involving Ψ_V . Using the definition of Ψ_V and Lemmas S3 and S19 we can write

$$\begin{aligned}
|g^\top W_{\text{up}} \Psi_V^\top y| &\leq |g^\top W_{\text{up}} \delta_V^\top y| + |g^\top W_{\text{up}} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top E_{\text{up}}^{\text{p}} y| \\
&\leq \|W_{\text{up}}^\top g\|_2 \left(\|\delta_V^\top y\|_2 + \|(W_{\text{up}}^\top W_{\text{up}})^{-1}\|_{\text{op}} \|W_{\text{up}}^\top E_{\text{up}}^{\text{p}} y\|_2 \right) \\
&\leq c_1 \|W_{\text{up}}^\top g\|_2 \left(\frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + c_1 \frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} \|V^\top y\|_2 + \frac{1}{\gamma_{\min}^2 \rho_N} \sigma \|W_{\text{up}}\|_{\text{op}} \sqrt{r + \zeta_N} \right) \\
&\leq c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} \|W_{\text{up}}^\top g\|_2 + c_1 \frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} \|W_{\text{up}}^\top g\|_2 \|V^\top y\|_2.
\end{aligned}$$

Combining the four preceding bounds and simplifying them further using (A2), (A3) proves (S32). $\square$

Lemma S10 provides an upper bound on the approximation error of the entire pooled upper matrix $M_{\text{up}}^{\text{p}} = W_{\text{up}} V^\top$. By restricting Φ_{up} in (S31) to the subsets $\mathcal{I}_k^{\text{up}} = \{s_k + 1, \dots, s_k + N_{1k}\}$ and $\mathcal{J}_k = \{T_{1k} + 1, \dots, T\}$, computations similar to those in the previous proof allow us to quantify the approximation error of $\mathcal{M}_{\bullet, \bullet, k}^{(b)}$.

Corollary S11. *Suppose the assumptions of Lemma S10 are satisfied, and use the notation introduced there. Also define $\hat{M}_b^{(k)} := (\hat{U}_{\text{up}} \hat{\Sigma}_{\text{up}} \hat{V}_{\text{up}}^\top)_{\mathcal{I}_k^{\text{up}}, \mathcal{J}_k}$ and recall $\mathcal{M}_{\bullet, \bullet, k}^{(b)} = (M_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \mathcal{J}_k} = U_{1k} \mathcal{C}_{\bullet, \bullet, k} V_{2k}^\top$. Writing*

$$\Phi_k := \hat{M}_b^{(k)} - \mathcal{M}_{\bullet, \bullet, k}^{(b)} - (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top - U_{1k} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{J}_k}, \quad (\text{S34})$$

for fixed $x \in \mathcal{S}^{N_{2k}-1}$, $y \in \mathcal{S}^{T_{2k}-1}$ there exists a constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$ such that

$$|x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y| \leq c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 + c_1 \frac{\sigma \sqrt{T}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2, \quad (\text{S35})$$

$$\|(U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top \Phi_k y\|_2 \leq c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \sqrt{\frac{N_{1k}}{N}} + c_1 \frac{\sigma \sqrt{T}}{\rho_N^{1/2}} \sqrt{\frac{N_{1k}}{N}} \|V_{2k}^\top y\|_2, \quad (\text{S36})$$

$$\begin{aligned} \|(I_{N_{1k}} - U_{1k} \{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top) \Phi_k y\|_2 &\leq c_1 \frac{\sigma^3 T \sqrt{r + \zeta_N}}{\gamma_{\min}^2 \rho_N} + c_1 \frac{\sigma^2 \sqrt{N_{1k}(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2}} \\ &\quad + c_1 \frac{\sigma^2 \sqrt{T(N_{1k} + T)}}{\gamma_{\min} \rho_N^{1/2}} \|V_{2k}^\top y\|_2 \end{aligned} \quad (\text{S37})$$

with probability at least $1 - O(p_N^{-10})$.

Proof. Restricting (S31) to the subsets $\mathcal{I}_k^{\text{up}}$ and $\mathcal{J}_k$ and using the definition of Φ_k in (S34) immediately yield $\Phi_k = (\Phi_{\text{up}})_{\mathcal{I}_k^{\text{up}}, \mathcal{J}_k}$. In order to prove (S35), set $B_k := U_{2k}(U_{1k}^\top U_{1k})^{-1} U_{1k}^\top$ and define $g \in \mathbb{R}^{N_{1,p}}$ and $\bar{y} \in \mathcal{S}^{T-1}$ to be the vectors with entries $g_i = (B_k^\top x)_{i-s_k} \mathbb{1}\{i \in \mathcal{I}_k^{\text{up}}\}$ and $\bar{y}_t = y_{t-T_{1k}} \mathbb{1}\{t \in \mathcal{J}_k\}$, respectively. This ensures that $V^\top \bar{y} = V_{2k}^\top y$ and that $|g^\top \Phi_{\text{up}} \bar{y}|$ is equal to the left-hand side of (S35). Furthermore, from (A1) we have $\|g\|_2 = \|B_k^\top x\|_2 = \|U_{1k}(U_{1k}^\top U_{1k})^{-1} U_{2k}^\top x\|_2 \leq \|U_{1k}(U_{1k}^\top U_{1k})^{-1}\|_{\text{op}} \|U_{2k}^\top x\|_2 \leq c_\ell^{-1/2} \sqrt{N/N_{1k}} \|U_{2k}^\top x\|_2$, and $\|W_{\text{up}}^\top g\|_2 = \|\mathcal{C}_{\bullet, \bullet, k}^\top U_{1k}^\top B_k^\top x\|_2 = \|\mathcal{C}_{\bullet, \bullet, k}^\top U_{2k}^\top x\|_2 \leq \gamma_{\max} \|U_{2k}^\top x\|_2$. Combining these with (S32) and further simplifying the resulting bound using (A2), (A3) proves (S35).

It remains to prove the last two bounds. Using the expression for Φ_{up} from the proof of Lemma S10 we get

$$\Phi_k y = (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V \delta_V^\top \bar{y} + (Y_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} \delta_V^\top V^\top \bar{y} + (Y_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} \delta_V \delta_V^\top \bar{y} + U_{1k} \mathcal{C}_{\bullet, \bullet, k} \Psi_V^\top \bar{y}. \quad (\text{S38})$$

We next bound the norms of the four terms separately under the action of $(U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top$. As for the first one, using Lemma S19 and the bounds for $\|\delta_V^\top \bar{y}\|_2$ and $\|\delta_V\|_{\text{op}}$ in (S33), we get

$$\|(U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V \delta_V^\top \bar{y}\|_2 \leq \|(U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V\|_{\text{op}} \|\delta_V^\top \bar{y}\|_2$$

$$\begin{aligned}
&\leq c_1 \sigma \sqrt{r + \zeta_N} \|\delta_V^\top \bar{y}\|_2 \\
&\leq c_1 \sigma \sqrt{r + \zeta_N} \left(\frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + \frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} \|V_{2k}^\top y\|_2 \right).
\end{aligned}$$

Similarly, for the second and third terms in (S38) we have

$$\begin{aligned}
&\|(U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top (Y_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} \delta_V (V^\top \bar{y} + \delta_V^\top \bar{y})\|_2 \\
&\leq \|(U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top \{(M_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} + (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet}\} \delta_V (V^\top \bar{y} + \delta_V^\top \bar{y})\|_2 \\
&\leq \|(U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top \{(M_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} + (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet}\} \|_{\text{op}} \|\delta_V\|_{\text{op}} \|V^\top \bar{y} + \delta_V^\top \bar{y}\|_2 \\
&= \|(U_{1k}^\top U_{1k})^{1/2} \mathcal{C}_{\bullet, \bullet, k} V^\top + (U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top E_{\text{up}}^{(k)}\|_{\text{op}} \|\delta_V\|_{\text{op}} \|V^\top \bar{y} + \delta_V^\top \bar{y}\|_2 \\
&\leq c_1 \left(\kappa \gamma_{\min} \sqrt{\frac{N_{1k}}{N}} + \sigma \sqrt{T} \right) \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \left(\frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + \|V_{2k}^\top y\|_2 \right).
\end{aligned}$$

Finally, using (S26) in Corollary S7 to bound $\|\Psi_V^\top \bar{y}\|_2 = \|\Psi_{V, 2k}^\top y\|_2$, with probability at least $1 - \mathcal{O}(p_N^{-10})$ we have

$$\begin{aligned}
&\|(U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top U_{1k} \mathcal{C}_{\bullet, \bullet, k} \Psi_V^\top \bar{y}\|_2 \leq c_1 \gamma_{\max} \sqrt{\frac{N_{1k}}{N}} \|\Psi_V^\top \bar{y}\|_2 = c_1 \kappa \gamma_{\min} \sqrt{\frac{N_{1k}}{N}} \|\Psi_{V, 2k}^\top y\|_2 \\
&\leq c_1 \gamma_{\min} \sqrt{\frac{N_{1k}}{N}} \left\{ \frac{\sigma^2 \sqrt{(T + N_{1, \text{p}})(r + \zeta_N)}}{\gamma_{\min}^2 \rho_N} + \left(\frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} + \frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} \right) \|V_{2k}^\top y\|_2 \right\}.
\end{aligned}$$

Combining the last three displays and further simplifying the bound using (A2), (A3) proves (S36).

In order to prove (S37), we will make use of the fact that $I_{N_{1k}} - U_{1k} \{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top$ is the orthogonal projector onto the orthogonal complement of $\text{col}(U_{1k})$. This also implies that $\|I_{N_{1k}} - U_{1k} \{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top\|_{\text{op}} = 1$ and $\text{rank}(I_{N_{1k}} - U_{1k} \{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top) = N_{1k} - r$. This implies that the signal contribution from the second and third terms in (S38) vanishes, and

we are left with the error matrix only. More precisely, we have

$$\begin{aligned}
& \|(I_{N_{1k}} - U_{1k}\{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top) (Y_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} \delta_V (V^\top \bar{y} + \delta_V^\top \bar{y})\|_2 \\
&= \|(I_{N_{1k}} - U_{1k}\{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top) (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} \delta_V (V^\top \bar{y} + \delta_V^\top \bar{y})\|_2 \\
&= \|(I_{N_{1k}} - U_{1k}\{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top) E_{\text{up}}^{(k)} \delta_V (V^\top \bar{y} + \delta_V^\top \bar{y})\|_2 \\
&\leq \|(I_{N_{1k}} - U_{1k}\{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top) E_{\text{up}}^{(k)}\|_{\text{op}} \|\delta_V\|_{\text{op}} \|V^\top \bar{y} + \delta_V^\top \bar{y}\|_2 \\
&\leq c_1 \sigma \sqrt{T + (N_{1k} - r) + \zeta_N} \|\delta_V\|_{\text{op}} \|V^\top \bar{y} + \delta_V^\top \bar{y}\|_2 \\
&\leq c_1 \sigma \sqrt{N_{1k} + T} \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \left(\frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + \|V_{2k}^\top y\|_2 \right).
\end{aligned}$$

The fourth term in (S38) completely vanishes under the action of $I_{N_{1k}} - U_{1k}\{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top$, while the first one gives

$$\begin{aligned}
& \|(I_{N_{1k}} - U_{1k}\{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top) (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V \delta_V^\top \bar{y}\|_2 \\
&\leq \|(I_{N_{1k}} - U_{1k}\{U_{1k}^\top U_{1k}\}^{-1} U_{1k}^\top) (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V\|_{\text{op}} \|\delta_V^\top \bar{y}\|_2 \\
&\leq c_1 \sigma \sqrt{r + (N_{1k} - r) + \zeta_N} \|\delta_V^\top \bar{y}\|_2 \leq c_1 \sigma \sqrt{N_{1k}} \left(\frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + \frac{\sigma^2 T}{\gamma_{\min}^2 \rho_N} \|V_{2k}^\top y\|_2 \right).
\end{aligned}$$

Combining the last three displays and further simplifying the bound using (A2), (A3) proves (S37). This concludes the proof. $\square$

We now present the main results of this section, which give a first-order expansion of $\hat{\mu}_{xy}^{(k)} - \mu_{xy}^{(k)}$ as the sum of Gaussian terms and remainder terms that can be suitably bounded from above. In the special case $K = 1$ with $x = \mathbf{e}_i$ and $y = \mathbf{e}_t$, analogous expansions were proved in Yan and Wainwright (2024). Our results therefore generalize these earlier expansions to arbitrary $K \geq 1$, $x \in \mathcal{S}^{N_{2k}-1}$, and $y \in \mathcal{S}^{T_{2k}-1}$, while also providing remainder bounds obtained via different techniques. In particular, the approach in Yan and Wainwright (2024) relies on a leave-one-block-out argument, which could also be adapted to the present

setting. However, this approach becomes suboptimal when K grows: in particular, the resulting remainder term is negligible only under a signal-to-noise ratio condition that deteriorates with K . We therefore instead rely on the preceding lemmas, which yield analogous results under weaker conditions.

Lemma S12. *Grant Assumption (A1) with fixed constants c_ℓ, c_u satisfying $0 < c_\ell \leq c_u < \infty$, (A2) and (A3). Suppose further that $0 < \gamma_{\min} \leq \sigma_{\min}(\mathcal{C}_{\bullet, \bullet, j}) \leq \sigma_{\max}(\mathcal{C}_{\bullet, \bullet, j}) \leq \gamma_{\max} < \infty$ for all $j \in [K]$, and let $\kappa := \gamma_{\max}/\gamma_{\min}$. Fix $k \in [K]$, unit vectors $x \in \mathcal{S}^{N_{2k}-1}$, $y \in \mathcal{S}^{T_{2k}-1}$, and let $\hat{\mu}_{xy}^{(k)}$ be the output of Algorithm 1 run with $\tau \leq \frac{c_\ell N_{1k}}{2N}$. Also write the decomposition $\hat{\mu}_{xy}^{(k)} - \mu_{xy}^{(k)} = Z_{xy}^{(1)} + Z_{xy}^{(2)} + Z_{xy}^{(3)} + Z_{xy}^{(4)} + \Delta_{xy} =: Z_{xy} + \Delta_{xy}$, where*

$$\begin{aligned} Z_{xy}^{(1)} &:= x^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{J}_k} y \\ Z_{xy}^{(2)} &:= x^\top U_{2k} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{J}_k} y \\ Z_{xy}^{(3)} &:= x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y \\ Z_{xy}^{(4)} &:= x^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \mathcal{C}_{\bullet, \bullet, k} V_{2k}^\top y. \end{aligned} \tag{S39}$$

There exists an event $\mathcal{G}_1$ with $\mathbb{P}(\mathcal{G}_1) \geq 1 - \mathcal{O}(p_N^{-10} + p_T^{-10})$ such that, under $\mathcal{G}_1$, the remainder satisfies

$$\begin{aligned} |\Delta_{xy}| &\leq c_1 \frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} + c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 \\ &\quad + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{T}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 \end{aligned} \tag{S40}$$

for a sufficiently large constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\text{dim}}, c_{\text{blk}}, \kappa) > 0$.

Proof. Let $\mathcal{G}_1$ be the intersection of the high-probability events in Lemmas S4, S8, S9, S10, S19 and Corollaries S6, S7, S11, applied with the specific deterministic choices of projection matrices and vectors used below. By a union bound, we have $\mathbb{P}(\mathcal{G}_1) \geq 1 - \mathcal{O}(p_N^{-10} + p_T^{-10})$. In

particular, arguing as in (S18), we know that, under $\mathcal{G}_1$, the matrix $\hat{H}_{k,\tau}^{\text{inv}}$ used in Algorithm 1 coincides with $(\hat{U}_{1k}^\top \hat{U}_{1k})^{-1}$ whenever the algorithm is run with $\tau \leq \frac{c_\ell N_{1k}}{2N}$. Recalling the notation $H_k = U_{1k}^\top U_{1k}$, $\hat{H}_k = H_U^\top \hat{U}_{1k}^\top \hat{U}_{1k} H_U$, $L = \hat{U}_{2k} H_U \hat{H}_k^{-1} H_U^\top \hat{U}_{1k}^\top - U_{2k} H_k^{-1} U_{1k}^\top$, and using the expression (S29) and (S34) for Δ_L and Φ_k , respectively, we can write

$$\begin{aligned}
\hat{\mu}_{xy}^{(k)} - \mu_{xy}^{(k)} &= x^\top \hat{U}_{2k} (\hat{U}_{1k}^\top \hat{U}_{1k})^{-1} \hat{U}_{1k}^\top \hat{U}_{\text{up}}^{(k)} \hat{\Sigma}_{\text{up}} \hat{V}_{2k}^\top y - x^\top U_{2k} \mathcal{C}_{\bullet,\bullet,k} V_{2k}^\top y \\
&= x^\top \hat{U}_{2k} (\hat{U}_{1k}^\top \hat{U}_{1k})^{-1} \hat{U}_{1k}^\top \widehat{M}_b^{(k)} y - x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \mathcal{M}_{\bullet,\bullet,k}^{(b)} y \\
&= x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top (\widehat{M}_b^{(k)} - \mathcal{M}_{\bullet,\bullet,k}^{(b)}) y + x^\top L \mathcal{M}_{\bullet,\bullet,k}^{(b)} y + x^\top L (\widehat{M}_b^{(k)} - \mathcal{M}_{\bullet,\bullet,k}^{(b)}) y \\
&= x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \left[(E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}},\bullet} V V_{2k}^\top + U_{1k} \mathcal{C}_{\bullet,\bullet,k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet,\mathcal{J}_k} + \Phi_k \right] y \\
&\quad + x^\top L U_{1k} \mathcal{C}_{\bullet,\bullet,k} V_{2k}^\top y + x^\top L \left[(E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}},\bullet} V V_{2k}^\top + U_{1k} \mathcal{C}_{\bullet,\bullet,k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet,\mathcal{J}_k} + \Phi_k \right] y \\
&= x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}},\bullet} V V_{2k}^\top y + x^\top U_{2k} \mathcal{C}_{\bullet,\bullet,k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet,\mathcal{J}_k} y \\
&\quad + x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y + x^\top L U_{1k} \mathcal{C}_{\bullet,\bullet,k} V_{2k}^\top y + x^\top L (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}},\bullet} V V_{2k}^\top y \\
&\quad + x^\top L U_{1k} \mathcal{C}_{\bullet,\bullet,k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet,\mathcal{J}_k} y + x^\top L \Phi_k y \\
&= Z_{xy}^{(3)} + Z_{xy}^{(2)} + x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y \\
&\quad + x^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k,\bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \mathcal{C}_{\bullet,\bullet,k} V_{2k}^\top y + x^\top \Delta_L V_{2k}^\top y \\
&\quad + x^\top L (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}},\bullet} V V_{2k}^\top y + x^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k,\bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \mathcal{C}_{\bullet,\bullet,k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet,\mathcal{J}_k} y \\
&\quad + x^\top \Delta_L (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet,\mathcal{J}_k} y + x^\top L \Phi_k y \\
&= Z_{xy}^{(1)} + Z_{xy}^{(2)} + Z_{xy}^{(3)} + Z_{xy}^{(4)} + x^\top U_{2k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y + x^\top \Delta_L V_{2k}^\top y \\
&\quad + x^\top \Delta_L (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet,\mathcal{J}_k} y + x^\top L (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}},\bullet} V V_{2k}^\top y + x^\top L \Phi_k y. \tag{S41}
\end{aligned}$$

We will now bound each of the remainder terms individually. The first term is controlled directly by (S35), while for the second term it is enough to write $|x^\top \Delta_L V_{2k}^\top y| \leq \|x^\top \Delta_L\|_2 \|V_{2k}^\top y\|_2$, and bound the first factor using (S30). For the third one, start by observing that Lemma S19 gives $\|(W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet,\mathcal{J}_k} y\|_2 \leq c_1 \sigma \gamma_{\min}^{-1} \sqrt{r + \zeta_N} \rho_N^{-1/2}$.

Applying again (S30) from Lemma S9 then gives

$$\begin{aligned}
|x^\top \Delta_L (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{I}_k} y| &\leq \|x^\top \Delta_L\|_2 \|(W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{I}_k} y\|_2 \\
&\leq c_1 \frac{\sigma^2 \sqrt{(N + T_{1,\text{p}})(r + \zeta_T)}}{\gamma_{\min} \rho_T} \frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} \\
&= c_1 \frac{\sigma^3 \sqrt{(N + T_{1,\text{p}})(r + \zeta_N)(r + \zeta_T)}}{\gamma_{\min}^2 \rho_N^{1/2} \rho_T} + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} \|U_{2k}^\top x\|_2 \\
&\leq c_1 \frac{\sigma^2 \sqrt{(r + \zeta_N)(r + \zeta_T)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} + c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2,
\end{aligned}$$

where the last inequality follows from (A3).

We next control the fourth term in (S41). By the definition of L , for any vector $w \in \mathbb{R}^{N_{1k}}$ we have $Lw = (\hat{U}_{2k} H_U - U_{2k}) H_k^{-1} U_{1k}^\top w + \hat{U}_{2k} H_U D_k w$, where $D_k = \hat{H}_k^{-1} (\hat{U}_{1k} H_U)^\top - H_k^{-1} U_{1k}^\top$. Applying this identity with $w = (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y$ and using $\hat{U}_{2k} H_U - U_{2k} = (E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} + \Psi_{U,2k}$ with $\Psi_{U,2k}$ defined in Corollary S6, gives

$$\begin{aligned}
x^\top L(E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y &= x^\top (E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} H_k^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y \\
&\quad + x^\top \Psi_{U,2k} H_k^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y + x^\top \hat{U}_{2k} H_U D_k (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y.
\end{aligned} \tag{S42}$$

We bound these three pieces separately. Since $(E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet}$ and $(E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet}$ are independent, conditionally on the upper-pooled noise $(E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet}$, the first term is a Gaussian random variable with conditional variance $\sigma^2 \|W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} H_k^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y\|_2^2$ and mean zero. Combining this with Lemma S19 and a standard Gaussian tail bound gives

$$\begin{aligned}
|x^\top (E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} H_k^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y| \\
\leq c_1 \sigma \sqrt{\zeta_T} \left\| W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y \right\|_2 \\
\leq c_1 \sigma \sqrt{\zeta_T} \left\| W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top (E_{\text{up}}^{\text{p}})_{\mathcal{I}_k^{\text{up}}, \bullet} V \right\|_{\text{op}} \left\| V_{2k}^\top y \right\|_2
\end{aligned}$$

$$\begin{aligned}
&\leq c_1 \sigma^2 \sqrt{\zeta_T (r + \zeta_N)} \left\| W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \right\|_{\text{op}} \left\| (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \right\|_{\text{op}} \left\| V_{2k}^\top y \right\|_2 \\
&\leq c_1 \frac{\sigma^2}{\gamma_{\min}} \sqrt{\frac{N \zeta_T (r + \zeta_N)}{N_{1k} \rho_T}} \left\| V_{2k}^\top y \right\|_2 \leq c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \left\| V_{2k}^\top y \right\|_2
\end{aligned}$$

with probability at least $1 - \mathcal{O}(p_N^{-10} + p_T^{-10})$, where in the penultimate inequality we used Lemma S3 to get $\|W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1}\|_{\text{op}} \leq \sigma_r^{-1}(M_{\text{left}}^p) \leq c_\ell^{-1/2} \gamma_{\min}^{-1} \rho_T^{-1/2}$, and (A1) to get $\|(U_{1k}^\top U_{1k})^{-1} U_{1k}^\top\|_{\text{op}} \leq c_\ell^{-1/2} \sqrt{N/N_{1k}}$. Similarly, for the second piece in (S42), Lemma S19, (A1), (A2), (A3) and (S20) give

$$\begin{aligned}
&\left| x^\top \Psi_{U,2k} H_k^{-1} U_{1k}^\top (E_{\text{up}}^p)_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y \right| \\
&\leq \|\Psi_{U,2k}^\top x\|_2 \left\| H_k^{-1} U_{1k}^\top (E_{\text{up}}^p)_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y \right\|_2 \leq c_1 \sigma \sqrt{\frac{N(r + \zeta_N)}{N_{1k}}} \|V_{2k}^\top y\|_2 \|\Psi_{U,2k}^\top x\|_2 \\
&\leq c_1 \sigma \sqrt{\frac{N(r + \zeta_N)}{N_{1k}}} \|V_{2k}^\top y\|_2 \left\{ \frac{\sigma^2 \sqrt{(N + T_{1,p})(r + \zeta_T)}}{\gamma_{\min}^2 \rho_T} + \left(\frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} + \frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} \right) \|U_{2k}^\top x\|_2 \right\} \\
&\leq c_1 \frac{\sigma^3 \sqrt{N(r + \zeta_N)(N + T_{1,p})(r + \zeta_T)}}{\gamma_{\min}^2 \rho_T \sqrt{N_{1k}}} \|V_{2k}^\top y\|_2 \\
&\quad + c_1 \sigma \sqrt{\frac{N(r + \zeta_N)}{N_{1k}}} \left(\frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} + \frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} \right) \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 \\
&\leq c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2.
\end{aligned}$$

For the third piece, (S16), (A2), (A3), Lemmas S8 and S19 yield

$$\begin{aligned}
&\left| x^\top \hat{U}_{2k} H_U D_k (E_{\text{up}}^p)_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y \right| \leq \|H_U^\top \hat{U}_{2k}^\top x\|_2 \|D_k\|_{\text{op}} \left\| (E_{\text{up}}^p)_{\mathcal{I}_k^{\text{up}}, \bullet} V V_{2k}^\top y \right\|_2 \\
&\leq c_1 \left(\frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + \|U_{2k}^\top x\|_2 \right) \frac{\sigma \sqrt{N}}{\gamma_{\min} \rho_T^{1/2}} \sqrt{\frac{N}{N_{1k}}} \sigma \sqrt{N_{1k}} \|V_{2k}^\top y\|_2 \\
&\leq c_1 \frac{\sigma^3 N \sqrt{r + \zeta_T}}{\gamma_{\min}^2 \rho_T} \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma^2 N}{\gamma_{\min} \rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 \\
&\leq c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2.
\end{aligned}$$

Combining the last three displays leads to

$$\left| x^\top L(E_{\text{up}}^{\text{p}}) \mathcal{I}_k^{\text{up}}, \bullet V V_{2k}^\top y \right| \leq c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2.$$

Finally, we control the fifth term in (S41). Letting $P_{1k} := U_{1k}(U_{1k}^\top U_{1k})^{-1}U_{1k}^\top$, we can write

$$x^\top L \Phi_k y = x^\top L P_{1k} \Phi_k y + x^\top L (I_{N_{1k}} - P_{1k}) \Phi_k y. \quad (\text{S43})$$

We will now bound the first projected component. Set $z := \mathcal{C}_{\bullet, \bullet, k}^{-1} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y$, so that $P_{1k} \Phi_k y = U_{1k} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y = U_{1k} \mathcal{C}_{\bullet, \bullet, k} z$. Using the definition of Δ_L in (S29) we obtain

$$\begin{aligned} x^\top L P_{1k} \Phi_k y &= x^\top (E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} \mathcal{C}_{\bullet, \bullet, k} z + x^\top \Delta_L z \\ &= x^\top (E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y + x^\top \Delta_L z. \end{aligned}$$

We bound these two pieces separately. For the first one, Lemma S19, (A1) and (S36) give

$$\begin{aligned} &\left| x^\top (E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y \right| \\ &\leq \left\| x^\top (E_{\text{left}}^{\text{p}})_{\{N_{1k}+1, \dots, N\}, \bullet} W_{\text{left}} (W_{\text{left}}^\top W_{\text{left}})^{-1} (U_{1k}^\top U_{1k})^{-1/2} \right\|_2 \left\| (U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top \Phi_k y \right\|_2 \\ &\leq \frac{\sigma}{\gamma_{\min}} \sqrt{\frac{N(r + \zeta_T)}{N_{1k} \rho_T}} \left\{ \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \sqrt{\frac{N_{1k}}{N}} + \frac{\sigma \sqrt{T}}{\rho_N^{1/2}} \sqrt{\frac{N_{1k}}{N}} \|V_{2k}^\top y\|_2 \right\} \\ &\leq c_1 \frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} + c_1 \frac{\sigma^2 \sqrt{T(r + \zeta_T)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} \|V_{2k}^\top y\|_2 \\ &\leq c_1 \frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} + c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2. \end{aligned} \quad (\text{S44})$$

For the second projected piece, using $\sigma_{\min}(\mathcal{C}_{\bullet, \bullet, k}) \geq \gamma_{\min} > 0$, (A1), and (S36), we have

$$\|z\|_2 = \left\| \mathcal{C}_{\bullet, \bullet, k}^{-1} (U_{1k}^\top U_{1k})^{-1} U_{1k}^\top \Phi_k y \right\|_2$$

$$\leq \gamma_{\min}^{-1} \left\| (U_{1k}^\top U_{1k})^{-1/2} \right\|_{\text{op}} \left\| (U_{1k}^\top U_{1k})^{-1/2} U_{1k}^\top \Phi_k y \right\|_2 \leq c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\gamma_{\min} \rho_N^{1/2}} + c_1 \frac{\sigma \sqrt{T}}{\gamma_{\min} \rho_N^{1/2}} \|V_{2k}^\top y\|_2.$$

Combining this with (S30) gives

$$\begin{aligned} |x^\top \Delta_L z| &\leq \|x^\top \Delta_L\|_2 \|z\|_2 \leq c_1 \frac{\sigma^2 \sqrt{(N + T_{1,p})(r + \zeta_T)}}{\gamma_{\min} \rho_T} \|z\|_2 + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|z\|_2 \\ &\leq c_1 \frac{\sigma^3 \sqrt{(N + T_{1,p})(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min}^2 \rho_T \rho_N^{1/2}} + c_1 \frac{\sigma^3 \sqrt{T(N + T_{1,p})(r + \zeta_T)}}{\gamma_{\min}^2 \rho_T \rho_N^{1/2}} \|V_{2k}^\top y\|_2 \\ &\quad + c_1 \frac{\sigma^2 \sqrt{N(r + \zeta_N)}}{\gamma_{\min} \rho_T^{1/2} \rho_N^{1/2}} \|U_{2k}^\top x\|_2 + c_1 \frac{\sigma^2 \sqrt{NT}}{\gamma_{\min} \rho_T^{1/2} \rho_N^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 \\ &\leq c_1 \frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} + c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 \quad (\text{S45}) \end{aligned}$$

$$+ c_1 \frac{\sigma \sqrt{T}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2, \quad (\text{S46})$$

and concludes the analysis of the first term in (S43). It remains to control the component orthogonal to U_{1k} . In particular, since $U_{1k}^\top (I_{N_{1k}} - P_{1k}) \Phi_k y = 0$, the population part in the decomposition of L vanishes, hence

$$\begin{aligned} L(I_{N_{1k}} - P_{1k}) \Phi_k y &= \hat{U}_{2k} H_U \hat{H}_k^{-1} H_U^\top \hat{U}_{1k}^\top (I_{N_{1k}} - P_{1k}) \Phi_k y - U_{2k} H_k^{-1} U_{1k}^\top (I_{N_{1k}} - P_{1k}) \Phi_k y \\ &= \hat{U}_{2k} H_U \left(\hat{H}_k^{-1} H_U^\top \hat{U}_{1k}^\top - H_k^{-1} U_{1k}^\top \right) (I_{N_{1k}} - P_{1k}) \Phi_k y \\ &= \hat{U}_{2k} H_U D_k (I_{N_{1k}} - P_{1k}) \Phi_k y. \end{aligned}$$

Therefore, (S16), (S27), (S37) and (A3) give

$$\begin{aligned} |x^\top L(I_{N_{1k}} - P_{1k}) \Phi_k y| &\leq \|H_U^\top \hat{U}_{2k}^\top x\|_2 \|D_k\|_{\text{op}} \|(I_{N_{1k}} - P_{1k}) \Phi_k y\|_2 \\ &\leq c_1 \left(\frac{\sigma \sqrt{r + \zeta_T}}{\gamma_{\min} \rho_T^{1/2}} + \|U_{2k}^\top x\|_2 \right) \frac{\sigma \sqrt{N}}{\gamma_{\min} \rho_T^{1/2}} \sqrt{\frac{N}{N_{1k}}} \\ &\quad \times \left(\frac{\sigma^3 T \sqrt{r + \zeta_N}}{\gamma_{\min}^2 \rho_N} + \frac{\sigma^2 \sqrt{N_{1k}(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2}} + \frac{\sigma^2 \sqrt{T(N_{1k} + T)}}{\gamma_{\min} \rho_N^{1/2}} \|V_{2k}^\top y\|_2 \right) \end{aligned}$$

$$\begin{aligned}
&= c_1 \frac{\sigma^5 NT \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min}^4 \rho_T \rho_N \sqrt{N_{1k}}} + c_1 \frac{\sigma^4 N \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min}^3 \rho_T \rho_N^{1/2}} \\
&\quad + c_1 \frac{\sigma^4 N \sqrt{(r + \zeta_T)T(N_{1k} + T)}}{\gamma_{\min}^3 \rho_T \rho_N^{1/2} \sqrt{N_{1k}}} \|V_{2k}^\top y\|_2 \\
&\quad + c_1 \frac{\sigma^4 NT \sqrt{r + \zeta_N}}{\gamma_{\min}^3 \rho_T^{1/2} \rho_N \sqrt{N_{1k}}} \|U_{2k}^\top x\|_2 + c_1 \frac{\sigma^3 N \sqrt{r + \zeta_N}}{\gamma_{\min}^2 \rho_T^{1/2} \rho_N^{1/2}} \|U_{2k}^\top x\|_2 \\
&\quad + c_1 \frac{\sigma^3 N \sqrt{T(N_{1k} + T)}}{\gamma_{\min}^2 \rho_T^{1/2} \rho_N^{1/2} \sqrt{N_{1k}}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 \\
&\leq c_1 \frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} + c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 \\
&\quad + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2. \tag{S47}
\end{aligned}$$

Combining (S43), (S44), (S46), and (S47), we obtain

$$\begin{aligned}
|x^\top L \Phi_k y| &\leq c_1 \frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} + c_1 \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 \\
&\quad + c_1 \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 + c_1 \frac{\sigma \sqrt{T}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2.
\end{aligned}$$

Combining all the previous inequalities gives a bound on $|\Delta_{xy}|$ and completes the proof. $\square$

Lemma S13. *Consider the setting of Lemma S12, and further assume (A4) with $\nu_x \neq 0, \nu_y \neq 0$. Also let $\mathcal{G}_1$ be the event such that $\mathbb{P}(\mathcal{G}_1) \geq 1 - \mathcal{O}(p_N^{-10} + p_T^{-10})$ under which (S40) holds. Define*

$$\Upsilon_{xy} := \frac{\sigma^2(r + \zeta_N)}{\rho_N} \|U_{2k}^\top x\|_2^2 + \frac{\sigma^2(r + \zeta_T)}{\rho_T} \|V_{2k}^\top y\|_2^2 + \frac{\sigma^2 N}{N_{1k}} \|U_{2k}^\top x\|_2^2 \|V_{2k}^\top y\|_2^2.$$

We have $\mathbb{E}[Z_{xy}^2] \leq c_1 \Upsilon_{xy}$ for a sufficiently large constant $c_1 \equiv c_1(c_\ell, c_u, c_0, c_{\dim}, c_{\text{blk}}, \kappa, \nu_x, \nu_y) > 0$.

Furthermore, under $\mathcal{G}_1$, the remainder satisfies $\Delta_{xy}^2 \leq c_1 \Upsilon_{xy}$.

Proof. We will use $\mathbb{E}[Z_{xy}^2] \leq 4 \sum_{i=1}^4 \mathbb{E}[(Z_{xy}^{(i)})^2]$, and bound the second moment of each $Z_{xy}^{(i)}$

in (S39) separately. First, from $Z_{xy}^{(2)} \sim \mathcal{N}(0, \sigma^2 \|x^\top U_{2k} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top\|_2^2)$ we get

$$\begin{aligned} \mathbb{E}[(Z_{xy}^{(2)})^2] &= \text{Var}(Z_{xy}^{(2)}) = \sigma^2 \|x^\top U_{2k} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top\|_2^2 \\ &\leq \sigma^2 \|U_{2k}^\top x\|_2^2 \|\mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1} W_{\text{up}}^\top\|_{\text{op}}^2 \leq \frac{\kappa^2 \sigma^2}{c_\ell \rho_N} \|U_{2k}^\top x\|_2^2, \end{aligned}$$

where the second inequality follows from Lemma S3. Furthermore, similar computations allow showing that $\mathbb{E}[(Z_{xy}^{(4)})^2] = \text{Var}(Z_{xy}^{(4)}) \leq \kappa^2 \sigma^2 c_\ell^{-1} \|V_{2k}^\top y\|_2^2 \rho_T^{-1}$ and $\mathbb{E}[(Z_{xy}^{(3)})^2] = \text{Var}(Z_{xy}^{(3)}) \leq \sigma^2 c_\ell^{-1} (N/N_{1k}) \|U_{2k}^\top x\|_2^2 \|V_{2k}^\top y\|_2^2$. Finally, for the first term we have

$$\begin{aligned} \mathbb{E}[(Z_{xy}^{(1)})^2] &\leq \|(W_{\text{left}}^\top W_{\text{left}})^{-1/2} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1/2}\|_{\text{op}}^2 \\ &\quad \times \mathbb{E} \left[\left\| (W_{\text{left}}^\top W_{\text{left}})^{-1/2} W_{\text{left}}^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k, \bullet}^\top x \right\|_2^2 \left\| (W_{\text{up}}^\top W_{\text{up}})^{-1/2} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{J}_k} y \right\|_2^2 \right] \\ &\leq \|(W_{\text{left}}^\top W_{\text{left}})^{-1/2} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1/2}\|_{\text{op}}^2 \\ &\quad \times \left\{ \mathbb{E} \left\| (W_{\text{left}}^\top W_{\text{left}})^{-1/2} W_{\text{left}}^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k, \bullet}^\top x \right\|_2^4 \right\}^{1/2} \left\{ \mathbb{E} \left\| (W_{\text{up}}^\top W_{\text{up}})^{-1/2} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{J}_k} y \right\|_2^4 \right\}^{1/2} \\ &= \sigma^4 r(r+2) \|(W_{\text{left}}^\top W_{\text{left}})^{-1/2} \mathcal{C}_{\bullet, \bullet, k} (W_{\text{up}}^\top W_{\text{up}})^{-1/2}\|_{\text{op}}^2 \\ &\leq \frac{\gamma_{\max}^2}{c_\ell^2 \gamma_{\min}^4} \frac{\sigma^4 r(r+2)}{\rho_T \rho_N} \leq c_1 \frac{\sigma^4 r^2}{\gamma_{\min}^2 \rho_T \rho_N}. \end{aligned}$$

The second bound follows from the Cauchy–Schwarz inequality, the first and only equality uses the fact that $\sigma^{-1} (W_{\text{left}}^\top W_{\text{left}})^{-1/2} W_{\text{left}}^\top (E_{\text{left}}^{\text{p}})_{\mathcal{I}_k, \bullet}^\top x$ and $\sigma^{-1} (W_{\text{up}}^\top W_{\text{up}})^{-1/2} W_{\text{up}}^\top (E_{\text{up}}^{\text{p}})_{\bullet, \mathcal{J}_k} y$ are standard normal vectors in $\mathbb{R}^r$, and hence have fourth moment $r(r+2)$ in squared Euclidean norm, while the penultimate inequality uses Lemma S3 to obtain $\lambda_{\min}(W_{\text{left}}^\top W_{\text{left}}) \geq c_\ell \gamma_{\min}^2 \rho_T$ and $\lambda_{\min}(W_{\text{up}}^\top W_{\text{up}}) \geq c_\ell \gamma_{\min}^2 \rho_N$. Furthermore, we can show that this term is dominated by either one of the first two terms in Υ_{xy} . Indeed, we have

$$\frac{\sigma^4 r^2}{\gamma_{\min}^2 \rho_T \rho_N} = \frac{\sigma^4 r^2}{\gamma_{\min}^2 \rho_T \rho_N} \frac{N}{\nu_x^2 r} \|U_{2k}^\top x\|_2^2 = \frac{\sigma^2 N}{\gamma_{\min}^2 \rho_T} \frac{1}{\nu_x^2} \frac{\sigma^2 r}{\rho_N} \|U_{2k}^\top x\|_2^2 \leq \frac{c_0^2}{\nu_x^2} \frac{\sigma^2 (r + \zeta_N)}{\rho_N} \|U_{2k}^\top x\|_2^2.$$

This, combined with the previous bounds and $\min(r + \zeta_N, r + \zeta_T) \geq 1$, gives $\mathbb{E}[Z_{xy}^2] \leq c_1 \Upsilon_{xy}$.

Coming now to bounding the remainder, we observe that the second and third terms in (S40) appear in the definition of Υ_{xy} . It thus remains to control the other three. Using the definition of ν_x in (A4), for the first one we get

$$\begin{aligned} \frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} &= \frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} \frac{\sqrt{N}}{\nu_x \sqrt{r}} \|U_{2k}^\top x\|_2 \\ &= \frac{\sigma \sqrt{N}}{\gamma_{\min} \rho_T^{1/2}} \frac{\sigma \sqrt{(r + \zeta_N)(1 + \zeta_T/r)}}{\nu_x \rho_N^{1/2}} \|U_{2k}^\top x\|_2 \\ &\leq \frac{c_0}{\nu_x} \frac{\sigma \sqrt{(r + \zeta_N)(1 + \zeta_T/r)}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2. \end{aligned}$$

Arguing by symmetry and using $c_{\text{blk}} \min(\zeta_N, \zeta_T) \leq r$, we can thus conclude

$$\frac{\sigma^2 \sqrt{(r + \zeta_T)(r + \zeta_N)}}{\gamma_{\min} \rho_N^{1/2} \rho_T^{1/2}} \leq c_0 \sqrt{1 + c_{\text{blk}}^{-1}} \max \left\{ \frac{1}{\nu_x} \frac{\sigma \sqrt{r + \zeta_N}}{\rho_N^{1/2}} \|U_{2k}^\top x\|_2, \frac{1}{\nu_y} \frac{\sigma \sqrt{r + \zeta_T}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2 \right\}.$$

Similarly, we can bound the fourth term in (S40) by

$$\frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \|U_{2k}^\top x\|_2 \|V_{2k}^\top y\|_2 = \frac{\sigma \sqrt{N}}{\rho_T^{1/2}} \frac{\nu_x \sqrt{r}}{\sqrt{N}} \|V_{2k}^\top y\|_2 = \nu_x \frac{\sigma \sqrt{r}}{\rho_T^{1/2}} \|V_{2k}^\top y\|_2.$$

An analogous bound holds for the fifth term, thereby showing that $|\Delta_{xy}| \leq c_1 \sqrt{\Upsilon_{xy}}$ on the event where (S40) holds. This completes the proof. $\square$

S5 Additional results

S5.1 Sufficient conditions for (A1)

We comment further on (A1) by providing two sufficient conditions under which it holds, either deterministically or with high probability.

Lemma S14. Let $U \in \mathbb{R}^{N \times r}$ have orthonormal columns and assume there exists $\mu \geq 2$ such that $\|U^\top \mathbf{e}_j\|_2^2 \leq \mu r/N$ for all $j \in [N]$. Let U_1 be any submatrix formed by selecting N_1 rows of U . If $N_1 \geq \frac{2\mu r}{2\mu r + 1} N$ then assumption (A1) holds with $c_\ell = \frac{1}{2}$ and $c_u = \frac{5}{4}$.

Proof. Write the row vectors $u_j := U^\top \mathbf{e}_j \in \mathbb{R}^r$, so that $\sum_{j=1}^N u_j u_j^\top = U^\top U = I_r$. Let $S \subset [N]$ index the N_1 selected rows, and let S^c be the complement with $N_2 := |S^c| = N - N_1$. Then

$$U_1^\top U_1 = \sum_{j \in S} u_j u_j^\top = I_r - \sum_{j \in S^c} u_j u_j^\top = I_r - U_2^\top U_2,$$

where U_2 is the submatrix of the remaining N_2 rows. Since $U_2^\top U_2 \succeq 0$ we immediately get $U_1^\top U_1 \preceq I_r$.

For the lower bound, combining $\lambda_{\max}(U_2^\top U_2) \leq \text{tr}(U_2^\top U_2) = \|U_2\|_F^2$ with incoherence gives

$$\|U_2\|_{\text{op}}^2 \leq \|U_2\|_F^2 = \sum_{j \in S^c} \|u_j\|_2^2 \leq N_2 \frac{\mu r}{N},$$

which leads to

$$U_1^\top U_1 = I_r - U_2^\top U_2 \succeq \left(1 - \frac{\mu r N_2}{N}\right) I_r \succeq \frac{1}{2} I_r \succeq \frac{1}{2} \frac{N_1}{N} I_r.$$

In particular, the last step follows from $N_1/N \leq 1$, while in the penultimate inequality we used $N_1 \geq \frac{2\mu r}{2\mu r + 1} N$, which implies $N_2/N \leq \frac{1}{2\mu r + 1}$, and thus $1 - \frac{\mu r N_2}{N} \geq 1 - \frac{\mu r}{2\mu r + 1} = \frac{\mu r + 1}{2\mu r + 1} \geq \frac{1}{2}$.

For the upper bound, it is useful to notice that $\frac{5}{4} \frac{N_1}{N} \geq \frac{5}{4} \frac{2\mu r}{2\mu r + 1} \geq 1$, where the middle inequality follows from $\mu r \geq 2$. This, together with $U_1^\top U_1 \preceq I_r$, shows that $U_1^\top U_1 \preceq I_r \preceq \frac{5}{4} \frac{N_1}{N} I_r$, and concludes the proof. $\square$

Lemma S15. Let $U \in \mathbb{R}^{N \times r}$ have orthonormal columns and satisfy $\|U^\top \mathbf{e}_j\|_2^2 \leq \mu r/N$ for all $j \in [N]$ for some $\mu \geq 1$. Let U_1 be formed by selecting N_1 rows from U uniformly at

random without replacement. For all $\varepsilon \in (0, 1)$ we have

$$\mathbb{P} \left\{ (1 - \varepsilon) \frac{N_1}{N} I_r \preceq U_1^\top U_1 \preceq (1 + \varepsilon) \frac{N_1}{N} I_r \right\} \geq 1 - 2r \exp \left\{ -\frac{N_1 \varepsilon^2}{3\mu r} \right\}.$$

Proof. For each $j \in [N]$, define $u_j := U^\top e_j \in \mathbb{R}^r$. Let S denote the random set of N_1 row indices sampled uniformly at random without replacement from $[N]$, so that $U_1^\top U_1 = \sum_{j \in S} u_j u_j^\top$. Equivalently, we may write $U_1^\top U_1 = \sum_{i=1}^{N_1} X_i$, where $X_1, \dots, X_{N_1}$ are sampled uniformly without replacement from $\{u_j u_j^\top : j \in [N]\}$.

Each $u_j u_j^\top$ is positive semidefinite, and by the incoherence assumption we have $\lambda_{\max}(u_j u_j^\top) = \|u_j\|_2^2 \leq \mu r / N =: B$ for all $j \in [N]$. Moreover, each X_i has marginal distribution uniform on $\{u_j u_j^\top : j \in [N]\}$, so

$$\mathbb{E} X_i = \frac{1}{N} \sum_{j=1}^N u_j u_j^\top = \frac{1}{N} U^\top U = \frac{1}{N} I_r.$$

Writing $\lambda_- := \lambda_{\min}(\sum_{i=1}^{N_1} \mathbb{E} X_i)$ and $\lambda_+ := \lambda_{\max}(\sum_{i=1}^{N_1} \mathbb{E} X_i)$, we thus have $\lambda_- = \lambda_+ = N_1/N$.

Although the matrices X_i are dependent, the trace-moment argument in [Gross and Nesme \(2010\)](#) allows the usual matrix Chernoff bounds to be applied to sampling without replacement from a finite collection. Hence, using [Tropp \(2012, Theorem 1.1\)](#) with dimension r , norm bound $B = \mu r / N$, and mean eigenvalues $\lambda_- = \lambda_+ = N_1/N$, gives, for $\varepsilon \in (0, 1)$,

$$\begin{aligned} \mathbb{P} \left\{ \lambda_{\min}(U_1^\top U_1) \leq (1 - \varepsilon) \frac{N_1}{N} \right\} &\leq r \left\{ \frac{e^{-\varepsilon}}{(1 - \varepsilon)^{1-\varepsilon}} \right\}^{\lambda_-/B} \\ &= r \left\{ \frac{e^{-\varepsilon}}{(1 - \varepsilon)^{1-\varepsilon}} \right\}^{N_1/(\mu r)} \leq r \exp \left\{ -\frac{N_1 \varepsilon^2}{3\mu r} \right\}, \end{aligned}$$

$$\mathbb{P} \left\{ \lambda_{\max}(U_1^\top U_1) \geq (1 + \varepsilon) \frac{N_1}{N} \right\} \leq r \left\{ \frac{e^{\varepsilon}}{(1 + \varepsilon)^{1+\varepsilon}} \right\}^{\lambda_+/B}$$

$$= r \left\{ \frac{e^\varepsilon}{(1+\varepsilon)^{1+\varepsilon}} \right\}^{N_1/(\mu r)} \leq r \exp \left\{ -\frac{N_1 \varepsilon^2}{3\mu r} \right\}.$$

In the last inequalities we used the standard bounds

$$\frac{e^{-\varepsilon}}{(1-\varepsilon)^{1-\varepsilon}} \leq \exp \left\{ -\frac{\varepsilon^2}{2} \right\} \leq \exp \left\{ -\frac{\varepsilon^2}{3} \right\}, \quad \frac{e^\varepsilon}{(1+\varepsilon)^{1+\varepsilon}} \leq \exp \left\{ -\frac{\varepsilon^2}{3} \right\},$$

both of which hold for $\varepsilon \in (0, 1)$. A union bound over the lower- and upper-tail events concludes the proof. $\square$

Note that both lemmas become uninformative as soon as the rank r is of the same order as the sampled dimension N_1 . In Lemma S14, the sufficient condition $N_1 \geq \frac{2\mu r}{2\mu r + 1} N$ forces $N_1/N \approx 1$ when r is large. Interpreted in the four-block setting used in the main body, this means that the fraction of rows containing missing entries must be exceptionally small. In Lemma S15, while $U_1^\top U_1$ concentrates around $(N_1/N)I_r$ when N_1 is much larger than $\mu r \log r$, a Marchenko–Pastur-type heuristic suggests that $\lambda_{\min}(U_1^\top U_1) \approx \frac{N_1}{N}(1 - \sqrt{\gamma})^2$ as $r/N_1 \rightarrow \gamma \in (0, 1)$, so the lower constant c_ℓ in (A1) deteriorates and can be arbitrarily small when r is too large relative to N_1 .

Returning to Lemma S14, the requirement that the missing block be small has close analogues in the MNAR causal-panel matrix-completion literature. For instance, horizontal regression (see, e.g., Athey et al., 2021, for a discussion) in the unconfoundedness literature is most appropriate when there are many control units relative to the number of periods, whereas vertical regression in the synthetic-control literature is most appropriate when there are many pre-treatment periods relative to the number of donor units. More precisely, horizontal regression is essentially feasible only when $N \gg T$, whereas vertical regression is viable when $T \gg N$. In either case, estimation is reliable only if the corresponding regression design matrices and the implied factor structure are sufficiently well-conditioned.

Moreover, settings with limited missing values are also a central building block of [Choi and Yuan \(2026\)](#). They first show that nuclear-norm regularization can accurately estimate the low-rank signal under MNAR when the total number of missing entries is sufficiently small (Assumption (iii) in Theorem 2.1). They then extend to more general MNAR patterns by partitioning the missing set into small groups so that each subproblem has few missing entries.

S5.2 Hardness results for $c_\ell = 0$

We illustrate why restricting our attention to $c_\ell > 0$ in (A1) is essential for $\mu_{xy}^{(k)}$ to be identifiable. Indeed, when $c_\ell = 0$, the restricted Gram matrices $U_{1j}^\top U_{1j}$ and $V_{1j}^\top V_{1j}$ are allowed to be singular. This creates directions in which the slice-specific core $\mathcal{C}_{\bullet,\bullet,k}$ can be perturbed so that only the unobserved d -block of $\mathcal{M}_{\bullet,\bullet,k}$ changes, while all observed entries across all slices remain the same. Therefore, two elements of the class can induce the same distribution while having different values of $\mu_{xy}^{(k)}$, so consistent estimation is impossible.

We recall $\mu_{xy}^{(k)}(\mathcal{M}) = x^\top \mathcal{M}_{\bullet,\bullet,k}^{(d)} y$ and $Z_\Omega = \{ \mathcal{M}_{itj} + \mathcal{E}_{itj} : \Omega_{i,t,j} = 1, (i, t, j) \in [N] \times [T] \times [K] \}$, with the mask Ω fixed and known, and write $\mathbb{P}_\mathcal{M}$ and $\mathbb{E}_\mathcal{M}$ for probability and expectation under the law of Z_Ω .

Proposition S16. *Fix an index $k \in [K]$, constants $\gamma_{\max} > \gamma_{\min} > 0$, and unit vectors $x \in \mathcal{S}^{N_{2k}-1}$, $y \in \mathcal{S}^{T_{2k}-1}$. Let $\bar{c}_u := \max(N / \min_{j \in [K]} N_{1j}, T / \min_{j \in [K]} T_{1j})$. Then*

$$\inf_{\phi} \sup_{\mathcal{M} \in \mathcal{F}(0, \bar{c}_u)} \mathbb{E}_\mathcal{M} \left[\left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}) \right\}^2 \right] \geq \frac{(\gamma_{\max} - \gamma_{\min})^2}{4},$$

where the infimum is over all Borel-measurable functions ϕ of the observed entries Z_Ω .

Proof. Define $\bar{x} \in \mathcal{S}^{N-1}$ and $\bar{y} \in \mathcal{S}^{T-1}$ to be the vectors with entries $\bar{x}_i := x_{i-N_{1k}} \mathbb{1}\{i > N_{1k}\}$ and $\bar{y}_t := y_{t-T_{1k}} \mathbb{1}\{t > T_{1k}\}$, respectively. Choose vectors $\{u_2, \dots, u_r\} \subset \mathbb{R}^N$ and

$\{v_2, \dots, v_r\} \subset \mathbb{R}^T$ such that $\{\bar{x}, u_2, \dots, u_r\}$ and $\{\bar{y}, v_2, \dots, v_r\}$ are orthonormal sets. In particular, this is possible since $r \leq \min(N, T)$. Define $U := (\bar{x} \mid u_2 \mid \dots \mid u_r) \in \mathbb{R}^{N \times r}$, $V := (\bar{y} \mid v_2 \mid \dots \mid v_r) \in \mathbb{R}^{T \times r}$ so that $U^\top U = V^\top V = I_r$. For every slice $j \neq k$ we take $C_{\bullet, \bullet, j} = \gamma_{\min} I_r$. For the specific slice k , writing $\mathbf{e}_1 = (1, 0, \dots, 0)^\top \in \mathbb{R}^r$ for the first vector of the canonical basis in $\mathbb{R}^r$, we set $\mathcal{C}_{\bullet, \bullet, k}^- = \gamma_{\min} I_r$ and $\mathcal{C}_{\bullet, \bullet, k}^+ = \gamma_{\min} I_r + (\gamma_{\max} - \gamma_{\min}) \mathbf{e}_1 \mathbf{e}_1^\top$. Finally, we define $\mathcal{M}^\pm := \mathcal{C}^\pm \times_1 U \times_2 V \times_3 I_K$.

We first verify that $\mathcal{M}^+, \mathcal{M}^- \in \mathcal{F}(0, \bar{c}_u)$. The orthonormality constraints on U and V hold by construction. Moreover, for every $j \in [K]$,

$$0 \preceq U_{1j}^\top U_{1j} \preceq I_r \preceq \bar{c}_u \frac{N_{1j}}{N} I_r, \quad 0 \preceq V_{1j}^\top V_{1j} \preceq I_r \preceq \bar{c}_u \frac{T_{1j}}{T} I_r,$$

by the definition of $\bar{c}_u$, hence Assumption (A1) holds with $c_\ell = 0$ and $c_u = \bar{c}_u$. Finally, for $j \neq k$, all singular values of $\mathcal{C}_{\bullet, \bullet, j}$ are equal to $\gamma_{\min}$; for $j = k$, the singular values of $\mathcal{C}_{\bullet, \bullet, k}^-$ are all equal to $\gamma_{\min}$ and those of $\mathcal{C}_{\bullet, \bullet, k}^+$ lie in $[\gamma_{\min}, \gamma_{\max}]$. Therefore both tensors belong to $\mathcal{F}(0, \bar{c}_u)$.

Next, since $\bar{x}$ and $\bar{y}$ are supported on the row and column indices of the missing d -block of slice k , the rank-one perturbation $(\gamma_{\max} - \gamma_{\min}) \bar{x} \bar{y}^\top$ is supported entirely on that missing block. Hence $P_{\Omega_{\bullet, \bullet, j}}(\mathcal{M}_{\bullet, \bullet, j}^+) = P_{\Omega_{\bullet, \bullet, j}}(\mathcal{M}_{\bullet, \bullet, j}^-)$ for all $j \in [K]$. Since the noise distribution is the same under $\mathcal{M}^+$ and $\mathcal{M}^-$, it follows that the induced laws of the observed data coincide, that is $\mathbb{P}_{\mathcal{M}^+} = \mathbb{P}_{\mathcal{M}^-}$. On the other hand, the corresponding target parameters are separated. Indeed,

$$\mu_{xy}^{(k)}(\mathcal{M}^+) - \mu_{xy}^{(k)}(\mathcal{M}^-) = x^\top (\mathcal{M}_{\bullet, \bullet, k}^{+, (d)} - \mathcal{M}_{\bullet, \bullet, k}^{-, (d)}) y = (\gamma_{\max} - \gamma_{\min}) x^\top x y^\top y = \gamma_{\max} - \gamma_{\min}.$$

Now, since $\mathbb{P}_{\mathcal{M}^+} = \mathbb{P}_{\mathcal{M}^-}$, expectations under the two laws are identical for every measurable

function ϕ of Z_Ω . We can therefore bound

$$\begin{aligned} & \mathbb{E}_{\mathcal{M}^+} \left[\left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}^+) \right\}^2 \right] + \mathbb{E}_{\mathcal{M}^-} \left[\left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}^-) \right\}^2 \right] \\ &= \mathbb{E}_{\mathcal{M}^+} \left[\left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}^+) \right\}^2 + \left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}^-) \right\}^2 \right] \\ &\geq \frac{1}{2} \left\{ \mu_{xy}^{(k)}(\mathcal{M}^+) - \mu_{xy}^{(k)}(\mathcal{M}^-) \right\}^2 = \frac{1}{2} (\gamma_{\max} - \gamma_{\min})^2, \end{aligned}$$

where we used the elementary inequality $(a - b)^2 + (a - c)^2 \geq (b - c)^2/2$. Consequently, we have

$$\begin{aligned} & \sup_{\mathcal{M} \in \mathcal{F}(0, \bar{c}_u)} \mathbb{E}_{\mathcal{M}} \left[\left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}) \right\}^2 \right] \\ &\geq \frac{1}{2} \left(\mathbb{E}_{\mathcal{M}^+} \left[\left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}^+) \right\}^2 \right] + \mathbb{E}_{\mathcal{M}^-} \left[\left\{ \phi(Z_\Omega) - \mu_{xy}^{(k)}(\mathcal{M}^-) \right\}^2 \right] \right) \geq \frac{(\gamma_{\max} - \gamma_{\min})^2}{4}. \end{aligned}$$

Taking the infimum over all Borel-measurable functions ϕ gives the claimed lower bound. $\square$

The choice of $\bar{c}_u$ is to make the construction feasible over arbitrary missingness patterns.

The same lower bound can be proved for each fixed $c_u > 0$ when $N_{2k} \leq N_{2j}$ and $T_{2k} \leq T_{2j}$ for all $j \neq k$, using a similar construction.

The second result shows that this issue is not merely an artifact of allowing slice-specific heterogeneity in $\mathcal{C}_{\bullet, \bullet, j}$. Even if we impose the strongest possible homogeneity assumption, i.e. $\mathcal{C}_{\bullet, \bullet, 1} = \dots = \mathcal{C}_{\bullet, \bullet, K}$, which is equivalent to $\mathcal{M}_{\bullet, \bullet, 1} = \dots = \mathcal{M}_{\bullet, \bullet, K}$, setting $c_\ell = 0$ can still make the problem information-theoretically hard unless auxiliary layers provide enough complementary information on the missing structure relevant to the target functional. We write $\mathcal{F}_{\text{id}}(c_\ell, c_u) := \{\mathcal{M} \in \mathcal{F}(c_\ell, c_u) : \mathcal{C}_{\bullet, \bullet, 1} = \dots = \mathcal{C}_{\bullet, \bullet, K}\}$.

Proposition S17. *Fix an index $k \in [K]$, constants $\gamma_{\max} \geq \gamma_{\min} > 0$, and unit vectors $x \in \mathcal{S}^{N_{2k}-1}$, $y \in \mathcal{S}^{T_{2k}-1}$. Also let $\bar{x} \in \mathcal{S}^{N-1}$ and $\bar{y} \in \mathcal{S}^{T-1}$ to be the vectors with entries*

$\bar{x}_i := x_{i-N_{1k}} \mathbb{1}\{i > N_{1k}\}$ and $\bar{y}_t := y_{t-T_{1k}} \mathbb{1}\{t > T_{1k}\}$, respectively. Define $S_k(x, y) := \sum_{j=1}^K \|P_{\Omega_{\bullet, \bullet, j}}(\bar{x}\bar{y}^\top)\|_F^2$, and $\bar{c}_u := \max(N/\min_{j \in [K]} N_{1j}, T/\min_{j \in [K]} T_{1j})$. Then

$$\inf_{\phi} \sup_{\mathcal{M} \in \mathcal{F}_{\text{id}}(0, \bar{c}_u)} \mathbb{E}_{\mathcal{M}} \left[\left\{ \phi(Z_{\Omega}) - \mu_{xy}^{(k)}(\mathcal{M}) \right\}^2 \right] \geq \max_{\gamma \in [\gamma_{\min}, \gamma_{\max}]} \gamma^2 \left[2 - 2\Phi \left(\frac{\gamma \sqrt{S_k(x, y)}}{\sigma} \right) \right],$$

where the infimum is over all Borel-measurable functions ϕ of the observed entries Z_{Ω} .

Proof. Choose vectors $\{u_2, \dots, u_r\} \subset \mathbb{R}^N$ and $\{v_2, \dots, v_r\} \subset \mathbb{R}^T$ such that $\{\bar{x}, u_2, \dots, u_r\}$ and $\{\bar{y}, v_2, \dots, v_r\}$ are orthonormal sets. In particular, this is possible since $r \leq \min(N, T)$. Define $U^{\pm} := (\pm \bar{x} \mid u_2 \mid \dots \mid u_r) \in \mathbb{R}^{N \times r}$, $V := (\bar{y} \mid v_2 \mid \dots \mid v_r) \in \mathbb{R}^{T \times r}$ so that $(U^{\pm})^\top U^{\pm} = V^\top V = I_r$. Also, for all $j \in [K]$ set $\mathcal{C}_{\bullet, \bullet, j} = \gamma I_r$, where $\gamma \in [\gamma_{\min}, \gamma_{\max}]$.

These choices induce $\mathcal{M}^{\pm} \in \mathcal{F}_{\text{id}}(0, \bar{c}_u)$ with $\mathcal{M}_{\bullet, \bullet, j}^{\pm} = \gamma U^{\pm} V^\top$. The precise computation follows an argument similar to the construction used in the proof of Proposition S16. Furthermore, we have

$$\mu_{xy}^{(k)}(\mathcal{M}^{\pm}) = x^\top \mathcal{M}_{\bullet, \bullet, k}^{\pm, (d)} y = \bar{x}^\top \mathcal{M}_{\bullet, \bullet, k}^{\pm} \bar{y} = \gamma \bar{x}^\top U^{\pm} V^\top \bar{y} = \pm \gamma.$$

hence $\{\mu_{xy}^{(k)}(\mathcal{M}^+) - \mu_{xy}^{(k)}(\mathcal{M}^-)\}^2 = 4\gamma^2$. By Le Cam's two-point method (Tsybakov, 2009, Theorem 2.2), for any measurable function ϕ , the minimax risk is lower bounded by $\gamma^2\{1 - \text{TV}(\mathbb{P}_{\mathcal{M}^+}, \mathbb{P}_{\mathcal{M}^-})\}$, where $\mathbb{P}_{\mathcal{M}^{\pm}}$ denotes the law of the observed entries Z_{Ω} under the signal $\mathcal{M}^{\pm}$. In particular, under Gaussian noise with common variance σ^2 , independence of the errors across slices implies that the joint law of all observed entries across all slices is multivariate Gaussian with mean vector equal to the vectorization of $\{P_{\Omega_{\bullet, \bullet, j}}(\mathcal{M}_{\bullet, \bullet, j}^{\pm})\}_{j \in [K]}$ and covariance $\sigma^2 I$. This, combined with $\mathcal{M}_{\bullet, \bullet, j}^+ - \mathcal{M}_{\bullet, \bullet, j}^- = \gamma(U^+ - U^-)V^\top = 2\gamma \bar{x}\bar{y}^\top$, gives

$$\text{TV}(\mathbb{P}_{\mathcal{M}^+}, \mathbb{P}_{\mathcal{M}^-}) = 2\Phi \left(\frac{\sqrt{\sum_{j=1}^K \|P_{\Omega_{\bullet, \bullet, j}}(2\gamma \bar{x}\bar{y}^\top)\|_F^2}}{2\sigma} \right) - 1 = 2\Phi \left(\frac{\gamma \sqrt{S_k(x, y)}}{\sigma} \right) - 1,$$

and concludes the proof upon taking the maximum over $\gamma \in [\gamma_{\min}, \gamma_{\max}]$. $\square$

Proposition S17 gives a lower bound on the minimax risk over $\mathcal{F}_{\text{id}}(0, \bar{c}_u)$, where all slices are identical. In particular, the lower bound depends on $S_k(x, y) = \sum_{j=1}^K \|P_{\Omega_{\bullet, \bullet, j}}(\bar{x}\bar{y}^\top)\|_F^2$, which quantifies how often the rank-one pattern $\bar{x}\bar{y}^\top$, supported on slice k 's missing block, is observed across other layers. For example, when $x = N_{2k}^{-1/2} \mathbf{1}_{N_{2k}}$ and $y = T_{2k}^{-1/2} \mathbf{1}_{T_{2k}}$ we have

$$S_k(x, y) = \frac{1}{N_{2k}T_{2k}} \sum_{j=1}^K \|(1 - \Omega_{\bullet, \bullet, k}) \odot \Omega_{\bullet, \bullet, j}\|_0,$$

so $S_k(x, y)$ is the fraction of slice k 's missing block that is observed elsewhere. When a slice $j \neq k$ has a much smaller missing block, this overlap increases and drives the lower bound to zero. On the other hand, in general, if $S_k(x, y) \lesssim \sigma^2/\gamma^2$, the two alternatives in the proof remain statistically close, and the minimax risk is of constant order. The extreme case corresponds to $S_k(x, y) = 0$, which occurs when the entire d -block missing under slice k is also missing under every other slice; for example, this holds when $N_{2k} \leq N_{2j}$ and $T_{2k} \leq T_{2j}$ for all $j \neq k$.

Taken together, Propositions S16 and S17 highlight two distinct failure modes when $c_\ell = 0$. Proposition S16 shows that with slice-specific cores, the target functional can be non-identifiable, as different parameter values can induce the same distribution of observed entries while yielding different $\mu_{xy}^{(k)}$. Proposition S17 shows that even if the slices share a common core, the target may still be hard to estimate, as the minimax risk can remain bounded away from zero unless the missing entries of slice k are sufficiently observed in other slices.

S5.3 Background on tensors

We now provide a brief background on tensors to familiarize the reader with the notation used in our model $\mathcal{M} = \mathcal{C} \times_1 U \times_2 V \times_3 I_K$. Although the main body of the paper considers only order-3 tensors, we keep this section fairly general at first. We then state and prove a result connecting the Tucker2 model with standard low-rank matrix factorizations.

A tensor is a multidimensional array $\mathcal{X} \in \mathbb{R}^{n_1 \times \dots \times n_d}$ with entries $x_{i_1 \dots i_d}$, where $1 \leq i_j \leq n_j$ for $j \in [d]$; order-1 tensors are vectors and order-2 tensors are matrices. Fixing all indices except i_j yields a mode- j fiber, which is the higher-order analogue of rows and columns. Fixing all but two indices yields a slice, which forms a two-dimensional subarray, e.g. for an order-3 tensor $\mathcal{X}$ the frontal slice $\mathcal{X}_{\bullet, \bullet, j}$ fixes the third index.

To connect tensor algebra to matrix algebra, it is convenient to use matricization and define the mode- j unfolding $X_{(j)} \in \mathbb{R}^{n_j \times \prod_{k \neq j} n_k}$ as a rearrangement of $\mathcal{X}$ so that the mode- j fibers become the columns of a matrix. Under this representation, the j -mode product (Kolda and Bader, 2009, Section 2.5) reduces to ordinary matrix multiplication. Specifically, for $A \in \mathbb{R}^{n' \times n_j}$, the tensor $\mathcal{Y} = \mathcal{X} \times_j A$ has dimensions $n_1 \times \dots \times n_{j-1} \times n' \times n_{j+1} \times \dots \times n_d$ and entries

$$(\mathcal{X} \times_j A)_{i_1 \dots i_{j-1} k i_{j+1} \dots i_d} = \sum_{i_j=1}^{n_j} x_{i_1 \dots i_d} a_{k i_j}.$$

Furthermore, its unfolding satisfies $(\mathcal{X} \times_j A)_{(j)} = A X_{(j)}$.

A core model in multilinear algebra is the Tucker decomposition, which represents a tensor as a low-dimensional core transformed along each mode and is of the form $\mathcal{X} = \mathcal{G} \times_1 A^{(1)} \times_2 \dots \times_d A^{(d)}$, where the core $\mathcal{G}$ encodes interactions among latent components and the factor matrices map these components to the ambient spaces (Kolda and Bader, 2009, Equations 4.1–4.2). Setting one factor matrix to the identity yields the Tucker2 model, introduced in Section 2; in our notation this gives $\mathcal{M} = \mathcal{C} \times_1 U \times_2 V \times_3 I_K$, so mode 3 is

left unchanged, i.e. slices are not mixed across k , while U and V act along modes 1 and 2, respectively.

We next provide an equivalent characterization for this model.

Proposition S18. *The following are equivalent:*

1. $M^{(j)} = UR_jV^\top$ for all $j \in [K]$, with common orthonormal $U \in \mathbb{R}^{N \times r}$, $V \in \mathbb{R}^{T \times r}$ and $R_j \in \mathbb{R}^{r \times r}$;
2. The column (resp. row) spaces of all $M^{(j)}$ are contained in a common subspace $\mathcal{U}$ (resp. $\mathcal{V}$) of dimension at most r ;
3. Stacking the matrices $M^{(j)}$ yields a tensor $\mathcal{M} \in \mathbb{R}^{N \times T \times K}$ that admits a Tucker2 decomposition $\mathcal{M} = \mathcal{C} \times_1 U \times_2 V \times_3 I_K$, with core $\mathcal{C} \in \mathbb{R}^{r \times r \times K}$ and shared mode-1/mode-2 orthonormal factor matrices U and V .

Proof. (1) $\Rightarrow$ (2): If $M^{(j)} = UR_jV^\top$ with $U \in \mathbb{R}^{N \times r}$, $V \in \mathbb{R}^{T \times r}$ column-orthonormal, then $\text{col}(M^{(j)}) \subseteq \text{col}(U) =: \mathcal{U}$ and $\text{row}(M^{(j)}) \subseteq \text{col}(V) =: \mathcal{V}$ for all $j \in [K]$, so (2) holds.

(2) $\Rightarrow$ (1): Let $\mathcal{U}, \mathcal{V}$ be subspaces containing all column and row spaces, with $\dim(\mathcal{U}) \leq r$ and $\dim(\mathcal{V}) \leq r$, and let U, V be orthonormal bases of $\mathcal{U}, \mathcal{V}$. Denote the orthogonal projections by $P_U := UU^\top$ and $P_V := VV^\top$. For each $j \in [K]$, the assumptions imply $P_U M^{(j)} = M^{(j)}$ and $M^{(j)} P_V = M^{(j)}$, hence

$$M^{(j)} = P_U M^{(j)} P_V = U(U^\top M^{(j)} V)V^\top.$$

Setting $R_j := U^\top M^{(j)} V$ gives (1).

(1) $\Rightarrow$ (3): Stack the matrices as a tensor $\mathcal{M} \in \mathbb{R}^{N \times T \times K}$ with frontal slices $\mathcal{M}_{\bullet, \bullet, j} = M^{(j)}$.

Define a core tensor $\mathcal{C} \in \mathbb{R}^{r \times r \times K}$ by $\mathcal{C}_{\bullet, \bullet, j} := R_j$. Then $\mathcal{M} = \mathcal{C} \times_1 U \times_2 V \times_3 I_K$, which is a

Tucker2 decomposition with shared mode-1/mode-2 orthonormal factors U, V .

(3) $\Rightarrow$ (1): Conversely, suppose $\mathcal{M}$ has a Tucker decomposition $\mathcal{M} = \mathcal{C} \times_1 U \times_2 V \times_3 I_K$ with $U \in \mathbb{R}^{N \times r}$, $V \in \mathbb{R}^{T \times r}$ column-orthonormal and $\mathcal{C} \in \mathbb{R}^{r \times r \times K}$. Writing $\mathcal{C}_{\bullet, \bullet, j}$ for the j -th frontal slice of $\mathcal{C}$, the j -th slice of $\mathcal{M}$ is

$$\mathcal{M}_{\bullet, \bullet, j} = \sum_{k=1}^K \delta_{jk} U \mathcal{C}_{\bullet, \bullet, k} V^\top = U \left(\sum_{k=1}^K \delta_{jk} \mathcal{C}_{\bullet, \bullet, k} \right) V^\top = U \mathcal{C}_{\bullet, \bullet, j} V^\top.$$

This concludes the proof. $\square$

S6 Auxiliary probabilistic results

In this appendix we collect some useful results that are used in the proofs of our main results.

We begin with two tail probability bounds. For $\sigma > 0$, a random variable X with mean $\mu = \mathbb{E}[X]$ is said to be σ -subgaussian if $\mathbb{E}[e^{\lambda(X-\mu)}] \leq e^{\sigma^2 \lambda^2 / 2}$ for all $\lambda \in \mathbb{R}$.

Lemma S19. *Let $E \in \mathbb{R}^{n_1 \times n_2}$ be a random matrix with mean-zero independent σ -subgaussian entries. For any fixed matrices $X \in \mathbb{R}^{n_1 \times p_1}$ and $Y \in \mathbb{R}^{n_2 \times p_2}$, for all $\delta \in (0, 1)$ there exists an absolute constant $c_1 > 0$ such that*

$$\|X^\top E Y\|_{\text{op}} \leq c_1 \sigma \|X\|_{\text{op}} \|Y\|_{\text{op}} \sqrt{\text{rank}(X) + \text{rank}(Y) + \log(\delta^{-1})}$$

with probability at least $1 - \delta$.

Proof. Write the compact singular value decompositions $X = U_X \Sigma_X V_X^\top$ and $Y = U_Y \Sigma_Y V_Y^\top$, where $U_X \in \mathbb{R}^{n_1 \times r_X}$, $V_X \in \mathbb{R}^{p_1 \times r_X}$, $U_Y \in \mathbb{R}^{n_2 \times r_Y}$, $V_Y \in \mathbb{R}^{p_2 \times r_Y}$ have orthonormal columns,

$r_X := \text{rank}(X)$ and $r_Y := \text{rank}(Y)$, and $\|\Sigma_X\|_{\text{op}} = \|X\|_{\text{op}}$, $\|\Sigma_Y\|_{\text{op}} = \|Y\|_{\text{op}}$. Then, we can write $X^\top EY = V_X \Sigma_X (U_X^\top E U_Y) \Sigma_Y V_Y^\top$, and by the submultiplicativity and orthonormal invariance of the spectral norm, we have $\|X^\top EY\|_{\text{op}} \leq \|X\|_{\text{op}} \|Y\|_{\text{op}} \|U_X^\top E U_Y\|_{\text{op}}$. It remains to bound $\|U_X^\top E U_Y\|_{\text{op}}$. For any $x \in \mathcal{S}^{r_X-1}$ and $y \in \mathcal{S}^{r_Y-1}$, define

$$Z(x, y) := x^\top U_X^\top E U_Y y = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} E_{ij} (U_X x)_i (U_Y y)_j,$$

so that $\|U_X^\top E U_Y\|_{\text{op}} = \sup_{x \in \mathcal{S}^{r_X-1}, y \in \mathcal{S}^{r_Y-1}} |Z(x, y)|$. For every fixed pair (x, y) and any $\delta \in (0, 1)$, Hoeffding's inequality for sums of subgaussian random variables and the fact that $\|U_X x\|_2 = \|U_Y y\|_2 = 1$ imply that there exists an absolute constant $c_1 > 0$ such that

$$\mathbb{P}\left\{|Z(x, y)| > c_1 \sigma \sqrt{\log(\delta^{-1})}\right\} \leq \delta. \quad (\text{S48})$$

In order to deal with the supremum, we will combine the above display with a standard netting argument [Wainwright \(2019, Chapter 5\)](#). In particular, let $\mathcal{N}_X$ and $\mathcal{N}_Y$ be $\frac{1}{4}$ -nets of $\mathcal{S}^{r_X-1}$ and $\mathcal{S}^{r_Y-1}$, respectively. Their cardinalities satisfy $|\mathcal{N}_X| \leq 9^{r_X}$ and $|\mathcal{N}_Y| \leq 9^{r_Y}$ ([Vershynin, 2019, Equation 4.20](#)). Now, for all $x_1 \in \mathcal{S}^{r_X-1}$ and $y_1 \in \mathcal{S}^{r_Y-1}$, we can find $x_2 \in \mathcal{N}_X$, $y_2 \in \mathcal{N}_Y$ such that $\|x_1 - x_2\|_2 \leq 1/4$ and $\|y_1 - y_2\|_2 \leq 1/4$. This, combined with $Z(x_1, y_1) = \{Z(x_1, y_1) - Z(x_2, y_1)\} + \{Z(x_2, y_1) - Z(x_2, y_2)\} + Z(x_2, y_2)$, allows showing that $\sup_{x \in \mathcal{S}^{r_X-1}, y \in \mathcal{S}^{r_Y-1}} |Z(x, y)| \leq 2 \max_{x \in \mathcal{N}_X, y \in \mathcal{N}_Y} |Z(x, y)|$ after taking the supremum on both sides. We thus get

$$\begin{aligned} \|U_X^\top E U_Y\|_{\text{op}} &= \sup_{x \in \mathcal{S}^{r_X-1}, y \in \mathcal{S}^{r_Y-1}} |Z(x, y)| \leq 2 \max_{x \in \mathcal{N}_X, y \in \mathcal{N}_Y} |Z(x, y)| \\ &\leq c_1 \sigma \sqrt{\log(|\mathcal{N}_X| |\mathcal{N}_Y| / \delta)} \leq c_1 \sigma \sqrt{r_X + r_Y + \log(\delta^{-1})} \end{aligned}$$

with probability at least $1 - \delta$, where the second inequality follows from an application

of (S48) with $\delta/(|\mathcal{N}_X||\mathcal{N}_Y|)$ in place of δ , and a union bound over $\mathcal{N}_X \times \mathcal{N}_Y$. This concludes the proof. $\square$

Lemma S20. *Let $E \in \mathbb{R}^{n_1 \times n_2}$ have independent entries distributed as $\mathcal{N}(0, \sigma^2)$. For every $p \geq 1$ we have*

$$\left(\mathbb{E}\|E\|_{\text{op}}^p\right)^{1/p} \leq \sigma\left(\sqrt{n_1} + \sqrt{n_2} + c_1\sqrt{p}\right),$$

where $c_1 > 0$ is an absolute constant.

Proof. By homogeneity, it suffices to consider the case $\sigma^2 = 1$. Indeed, writing $E = \sigma H$ with independent $H_{ij} \sim \mathcal{N}(0, 1)$, we have $\|E\|_{\text{op}} = \sigma\|H\|_{\text{op}}$, and therefore $\left(\mathbb{E}\|E\|_{\text{op}}^p\right)^{1/p} = \sigma\left(\mathbb{E}\|H\|_{\text{op}}^p\right)^{1/p}$. Thus it is enough to prove that $\left(\mathbb{E}\|H\|_{\text{op}}^p\right)^{1/p} \leq \sqrt{n_1} + \sqrt{n_2} + c_1\sqrt{p}$. In this regard, writing $c > 0$ for an absolute constant, we recall from Vershynin (2019, Theorem 7.3.1 and Corollary 7.3.2) that $\mathbb{E}\|H\|_{\text{op}} \leq \sqrt{n_1} + \sqrt{n_2}$, and $\mathbb{P}\left(\|H\|_{\text{op}} \geq \sqrt{n_1} + \sqrt{n_2} + t\right) \leq 2e^{-ct^2}$ for all $t \geq 0$. Furthermore, letting $Y := (\|H\|_{\text{op}} - \sqrt{n_1} - \sqrt{n_2})_+$, we also have $\|H\|_{\text{op}} \leq \sqrt{n_1} + \sqrt{n_2} + Y$ and $(\mathbb{E}\|H\|_{\text{op}}^p)^{1/p} \leq \sqrt{n_1} + \sqrt{n_2} + (\mathbb{E}Y^p)^{1/p}$, where the latter bound follows from Minkowski's inequality.

It remains to bound $(\mathbb{E}Y^p)^{1/p}$. Using the layer-cake formula, the change of variables $u = ct^2$, and Stirling's approximation, we obtain

$$\begin{aligned} \mathbb{E}Y^p &= \int_0^\infty p t^{p-1} \mathbb{P}(Y > t) dt = \int_0^\infty p t^{p-1} \mathbb{P}\left(\|H\|_{\text{op}} > \sqrt{n_1} + \sqrt{n_2} + t\right) dt \\ &\leq 2p \int_0^\infty t^{p-1} e^{-ct^2} dt = p c^{-p/2} \int_0^\infty u^{p/2-1} e^{-u} du \\ &= p c^{-p/2} \Gamma(p/2) = 2 c^{-p/2} \Gamma(p/2 + 1) \leq (c_1\sqrt{p})^p, \end{aligned}$$

for some constant $c_1 > 0$ depending only on c . Combining the above bounds gives

$$(\mathbb{E}\|H\|_{\text{op}}^p)^{1/p} \leq \sqrt{n_1} + \sqrt{n_2} + (\mathbb{E}Y^p)^{1/p} \leq \sqrt{n_1} + \sqrt{n_2} + c_1\sqrt{p},$$

thereby completing the proof. $\square$

We next recall Weyl's inequality for singular values and eigenvalues (Chen et al., 2021, Lemmas 2.2–2.3).

Lemma S21. *Let $A, E \in \mathbb{R}^{n \times m}$. Then, for every $1 \leq i \leq \min(n, m)$, the i -th largest singular values of A and $A + E$ satisfy*

$$|\sigma_i(A + E) - \sigma_i(A)| \leq \|E\|_{\text{op}}.$$

Moreover, if $n = m$ and $A, E \in \mathbb{R}^{n \times n}$ are symmetric, then, for every $1 \leq i \leq n$, the i -th largest eigenvalues of A and $A + E$ satisfy

$$|\lambda_i(A + E) - \lambda_i(A)| \leq \|E\|_{\text{op}}.$$

We recall that the Moore–Penrose pseudoinverse of $A = U \text{diag}(\sigma_1, \dots, \sigma_r) V^\top$, with column orthonormal $U \in \mathbb{R}^{n_1 \times r}$, $V \in \mathbb{R}^{n_2 \times r}$ and $\sigma_i > 0$, is $A^\dagger = V \text{diag}(\sigma_1^{-1}, \dots, \sigma_r^{-1}) U^\top$. The following lemma gives an exact identity for how the Moore–Penrose inverse changes when a full-column-rank matrix A is perturbed to $B = A + \Delta$, as well as a simple operator-norm bound and a useful formula for the action of $B^\dagger - A^\dagger$ on A .

Lemma S22. *Let $A, B \in \mathbb{R}^{n_1 \times n_2}$ have full column rank, and let $\Delta := B - A$. Then $B^\dagger - A^\dagger = -A^\dagger \Delta B^\dagger + A^\dagger (A^\dagger)^\top \Delta^\top (I_{n_1} - B B^\dagger)$. Consequently,*

$$\|B^\dagger - A^\dagger\|_{\text{op}} \leq \|A^\dagger\|_{\text{op}} \|\Delta\|_{\text{op}} \|B^\dagger\|_{\text{op}} + \|A^\dagger\|_{\text{op}}^2 \|\Delta\|_{\text{op}}.$$

Moreover, we have $(B^\dagger - A^\dagger)A = -B^\dagger \Delta$.

Proof. Since A and B have full column rank, $A^\dagger A = I_{n_2}$, $B^\dagger B = I_{n_2}$, and $A^\dagger = (A^\top A)^{-1} A^\top$. Therefore, $B^\dagger - A^\dagger = (B^\dagger B - A^\dagger B)B^\dagger - A^\dagger(I_{n_1} - BB^\dagger) = -A^\dagger \Delta B^\dagger - A^\dagger(I_{n_1} - BB^\dagger)$. It remains to rewrite the last term. Since $BB^\dagger$ is the orthogonal projector onto $\text{col}(B)$, we have $B^\top(I_{n_1} - BB^\dagger) = 0$. As $B = A + \Delta$, this gives $A^\top(I_{n_1} - BB^\dagger) = -\Delta^\top(I_{n_1} - BB^\dagger)$, and hence $-A^\dagger(I_{n_1} - BB^\dagger) = (A^\top A)^{-1} \Delta^\top(I_{n_1} - BB^\dagger)$. Using $A^\dagger(A^\dagger)^\top = (A^\top A)^{-1}$ proves the first identity.

The norm bound follows from this identity and $\|I_{n_1} - BB^\dagger\|_{\text{op}} \leq 1$. Finally, $(B^\dagger - A^\dagger)A = B^\dagger A - I_{n_2} = B^\dagger A - B^\dagger B = -B^\dagger \Delta$. $\square$

Finally, we present the auxiliary results on the Stiefel manifold used in the proofs. These include standard facts on the Haar measure and its generation via Gaussian QR decompositions; see, for example, [Stewart \(1980\)](#); [Mezzadri \(2007\)](#); [Chikuse \(2003\)](#). For integers $1 \leq q \leq d$, the Stiefel manifold is

$$\text{St}(d, q) := \{Q \in \mathbb{R}^{d \times q} : Q^\top Q = I_q\}.$$

Thus, $\text{St}(d, q)$ is the set of $d \times q$ matrices whose columns are orthonormal. The special cases $q = 1$ and $q = d$ reduce to the unit sphere $\mathcal{S}^{d-1}$ and the orthogonal group $\mathbb{O}(d)$, respectively.

Although $\text{St}(d, q)$ is not a group when $q < d$, it carries a natural probability measure that is invariant under left multiplication by orthogonal matrices. A random matrix $Q \in \text{St}(d, q)$ is said to be Haar-distributed or uniformly distributed on the Stiefel manifold if $OQ \stackrel{d}{=} Q$ for every deterministic $O \in \mathbb{O}(d)$. Such a left-orthogonally invariant probability measure on $\text{St}(d, q)$ is unique (e.g. [Chikuse, 2003](#), Theorem 1.2.2 and Section 1.3.1). In words, multiplying Q by any deterministic rotation or reflection does not change its law, hence a Haar-distributed element of $\text{St}(d, q)$ may be viewed as a uniformly random matrix with orthonormal columns. A standard construction of such matrices is obtained from a Gaussian

matrix. If $G \in \mathbb{R}^{d \times q}$ has i.i.d. $\mathcal{N}(0, 1)$ entries and $G = QR$ is its thin QR decomposition with the diagonal entries of R taken positive, then $Q \in \text{St}(d, q)$ is Haar-distributed.

The following lemma formalizes a stability property of Haar-distributed Stiefel matrices. If Q is uniformly distributed on $\text{St}(d, q)$, then multiplying it on the right by any fixed $H \in \text{St}(q, \ell)$ produces an ℓ -dimensional orthonormal system which is itself uniformly distributed on $\text{St}(d, \ell)$. The lemma also gives a high-probability bound for the size of this random orthonormal system after applying a fixed linear map A . In particular, when ℓ is small relative to d , the operator norm $\|AQH\|_{\text{op}}$ is at most of the order

$$\|A\|_{\text{op}} \sqrt{\frac{\text{rank}(A) + \ell}{d}},$$

with high probability, up to universal constants.

Lemma S23. *Let $d, q, p \in \mathbb{N}$, and let $1 \leq \ell \leq q \leq d$. Let $Q \in \mathbb{R}^{d \times q}$ be Haar-distributed on $\text{St}(d, q)$, and let $\mathcal{F}$ be a sigma-field independent of Q . Let $H \in \text{St}(q, \ell)$ be $\mathcal{F}$ -measurable, and let $A \in \mathbb{R}^{p \times d}$ be deterministic. Then, conditional on $\mathcal{F}$, the matrix QH is Haar-distributed on $\text{St}(d, \ell)$. Moreover, there is a universal constant $c_1 > 0$ such that, for every $t \geq 0$ with $\sqrt{\ell} + t < \sqrt{d}$, we have*

$$\mathbb{P} \left\{ \|AQH\|_{\text{op}} \leq c_1 \|A\|_{\text{op}} \frac{\sqrt{\text{rank}(A) + \ell + t^2}}{\sqrt{d} - \sqrt{\ell} - t} \middle| \mathcal{F} \right\} \geq 1 - 2e^{-t^2/2}. \quad (\text{S49})$$

In particular, if $\sqrt{\ell} + t \leq \sqrt{d}/2$, then

$$\mathbb{P} \left\{ \|AQH\|_{\text{op}} \leq 2c_1 \|A\|_{\text{op}} \sqrt{\frac{\text{rank}(A) + \ell + t^2}{d}} \middle| \mathcal{F} \right\} \geq 1 - 2e^{-t^2/2}. \quad (\text{S50})$$

Proof. Since $Q^\top Q = I_q$ and $H^\top H = I_\ell$, we have $(QH)^\top (QH) = H^\top Q^\top QH = I_\ell$. Hence $QH \in \text{St}(d, \ell)$.

We now identify the conditional law of QH . Fix $O \in \mathbb{O}(d)$. Since Q is Haar-distributed on $\text{St}(d, q)$ and is independent of $\mathcal{F}$, its conditional law given $\mathcal{F}$ is still Haar. Hence $OQ \stackrel{d}{=} Q$ conditionally on $\mathcal{F}$. Since H is $\mathcal{F}$ -measurable, it is fixed after conditioning on $\mathcal{F}$, and therefore $OQH \stackrel{d}{=} QH$ conditionally on $\mathcal{F}$. Thus the conditional law of QH is invariant under left multiplication by every deterministic orthogonal matrix $O \in \mathbb{O}(d)$. By uniqueness of the left-orthogonally invariant probability measure on $\text{St}(d, \ell)$ (e.g. [Chikuse, 2003](#), Theorem 1.2.2 and Section 1.3.1), this conditional law is the Haar measure on $\text{St}(d, \ell)$.

Let $G \in \mathbb{R}^{d \times \ell}$ have independent $\mathcal{N}(0, 1)$ entries and be independent of $\mathcal{F}$. By the Gaussian representation of the Haar measure on the Stiefel manifold ([Chikuse, 2003](#), Theorem 2.4.3), $QH \stackrel{d}{=} G(G^\top G)^{-1/2}$ conditionally on $\mathcal{F}$. Hence

$$\|AQH\|_{\text{op}} \stackrel{d}{=} \|AG(G^\top G)^{-1/2}\|_{\text{op}} \leq \frac{\|AG\|_{\text{op}}}{\sigma_\ell(G)}$$

conditionally on $\mathcal{F}$. By Lemma [S19](#), applied with $X = A^\top$, $Y = I_\ell$, $\sigma = 1$, and $\delta = e^{-t^2/2}$, we have $\|AG\|_{\text{op}} \leq c_1 \|A\|_{\text{op}} \sqrt{\text{rank}(A) + \ell + t^2}$ with probability at least $1 - e^{-t^2/2}$. Also, the standard lower-tail bound for the smallest singular value of a Gaussian matrix ([Davidson and Szarek, 2001](#), Theorem II.13) gives $\sigma_\ell(G) \geq \sqrt{d} - \sqrt{\ell} - t$ with probability at least $1 - e^{-t^2/2}$. On the intersection of these two events, which has probability at least $1 - 2e^{-t^2/2}$, the bound [\(S49\)](#) follows.

Finally, if $\sqrt{\ell} + t \leq \sqrt{d}/2$, then $\sqrt{d} - \sqrt{\ell} - t \geq \sqrt{d}/2$. Substituting this lower bound into [\(S49\)](#) gives [\(S50\)](#). $\square$

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